DISCRETE-TIME NEURAL NETWORK BASED STATE OBSERVER WITH COMMAND-FILTERED NEURAL NETWORK CONTROL FOR A CLASS OF SYSTEMS WITH UNMATCHED UNCERTAINTIES

by

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ABSTRACT

An observer is a dynamic system that estimates the state variables of another system from noisy measurements, either to reconstruct unmeasured states or to improve the accuracy of measured ones. The Modified State Observer (MSO) is a standard observer structure augmented with an online neural network that estimates both the system states and the system uncertainty. It has been applied to orbit uncertainty estimation, atmospheric reentry uncertainty estimation, and the control of a nonlinear electrohydraulic system with parameter uncertainty.

This work develops the discrete-time extension of the MSO (DMSO) and establishes its stability using Lyapunov theory. The stability argument evaluates the Lyapunov first difference exactly rather than bounding it term by term. Doing so exposes a family of cancellations produced by the adaptation law itself, and yields admissibility conditions that are both satisfiable and physically interpretable: the error-injection matrix must satisfy $\sigma_{\max}(F - K_{\mathrm{MSO}}) < 1$, and the adaptation gain must be bounded above. In particular, small adaptation gains are admissible.

To demonstrate the DMSO, simulation studies are performed on a two-wheeled inverted pendulum (TWIP) robot, an unstable nonlinear system with unmatched uncertainties, and the observer is implemented on hardware. The robot model is tested against the robot's recorded balancing runs: an input-free angular-momentum test, the response to the saturated voltage at switch-on, and the closed loop with the gains the robot ran fix the sign of the motor's reaction torque on the body, although the model is not validated quantitatively.

A command-filtered backstepping controller with neural network uncertainty compensation is then developed for the TWIP. The controller uses the tilt angle as a virtual control for the position subsystem, in coordinates built on the flat output of the linearized robot, so that the voltage enters only at the last step. The formulation (i) retains the coupling term between the position and attitude error subsystems and cancels it explicitly, (ii) assigns damping to every error coordinate rather than only to the two driven coordinates, and (iii) uses command filtering so that the functions approximated by the neural networks depend only on measured states and the reference trajectory, removing any algebraic dependence of each network's target on the other network's output. Uniform ultimate boundedness of all tracking errors and all weight estimation errors follows under a command-filter error assumption, with σ-modification supplying weight boundedness without a persistency-of-excitation requirement. In simulation the controller reduces the RMS position tracking error of an LQR by factors of 4 to 20.

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NOMENCLATURE

x_k	System state vector at time step k, $x_k \in \mathbb{R}^n$
$\hat{x}_k$	State estimate
e_k	State estimation error, $e_k = x_k - \hat{x}_k$
F, G	Discretized system and input matrices
B_{MSO}	Uncertainty selection matrix, $B_{\text{MSO}} \in \mathbb{R}^{n \times p}$
K_{MSO}	Observer error-injection gain
A	Error-injection matrix, $A = F - K_{\text{MSO}}$
$f(\cdot)$	Unknown uncertainty vector
W, $\hat{W}_k$	Ideal and estimated neural network weights
$\tilde{W}_k$	Weight estimation error, $\tilde{W}_k = W - \hat{W}_k$
ϕ_k	Basis (activation) vector, $\phi_k = \phi(x_k)$
ψ_k	Functional approximation error, $\psi_k = \tilde{W}_k^T \phi_k$
ϵ_k	Network reconstruction error, $\|\epsilon_k\| \leq \epsilon_N$
Γ	Observer network adaptation gain
κ	σ-modification (leakage) gain
y, η	Flat output of the linearized robot, $y = x_1 - bx_3$, and its derivative
b	Input-gain ratio B_1/B_2
a_2	Virtual-control gain of the position subsystem, $a_2 = A_2 - bA_4$
z_i	Backstepping error coordinates, $i = 1, \ldots, 4$
χ_i	Command filter errors
c_i	Backstepping damping gains
w	Attitude weight in the controller Lyapunov function
ϖ_i	Command filter bandwidths
θ	Pendulum angle; $\phi = \theta - \pi$ is the tilt from upright
V	Motor terminal voltage (control input)

Chapter 1

INTRODUCTION

1.1 TWO WHEELED INVERTED PENDULUM

The two-wheeled inverted pendulum (TWIP) is a mobile robot consisting of a body mounted on a single wheel axis, with the center of gravity located above that axis. The system is statically unstable and must be actively controlled to remain upright, which makes it a convenient laboratory analogue for a broad class of unstable, underactuated aerospace systems: it has more degrees of freedom than control inputs, it has strongly coupled translational and rotational dynamics, and its linearization is valid only in a neighborhood of the operating point.

The TWIP is attractive as a testbed for three reasons. First, its dynamics are well understood and can be derived from first principles, so a truth model is available against which estimator performance can be assessed. Second, the uncertainty in a physical build is real and not synthetic: mass properties are difficult to measure accurately, the motor constants drift with temperature and load, and the gear train exhibits deadzone and backlash. Third, the system is genuinely underactuated, with a single motor voltage command driving both the translational and rotational accelerations, so the uncertainties entering the two acceleration equations cannot both be cancelled directly. The uncertainty is therefore *unmatched*,

and this is the case of practical interest.

1.2 CONTRIBUTION OF THIS WORK

This work makes three contributions.

(i) **A discrete-time Modified State Observer with a complete stability proof.** The MSO of Rajagopal et al. is extended to discrete time. Every digital implementation is a discrete-time system, so a discrete-time result is required before the technique can be claimed for digital flight control, rover autonomy, or hardware-in-the-loop use. The proof given in Chapter 3 evaluates the Lyapunov first difference exactly. The resulting admissibility conditions are $\sigma_{\max}(F - K_{\mathrm{MSO}}) < 1$ together with an upper bound on the adaptation gain, both of which are satisfiable over an open set of designs.

(ii) **A command-filtered neural control law for unmatched uncertainty.** Chapter 6 develops an additive control signal that uses the tilt angle as a virtual control for the position subsystem. The design is a four-step backstepping procedure with command filtering, carried out on the flat output of the linearized robot. It assigns damping to every error coordinate, retains and cancels the coupling between the position and attitude error subsystems, and makes each neural network's approximation target a function of measured state and reference trajectory alone.

(iii) **Hardware validation.** The observer is implemented on a physical TWIP robot, and the model of the robot is tested against its recorded balancing runs (Section 5.3). Chapter 4 documents the platform and the practical obstacles encountered; Appendix A gives a staged validation procedure for reproducing and extending the results.

1.3 ORGANIZATION

Chapter 2 reviews the relevant literature on two-wheeled inverted pendulum control and on modified state observers. Chapter 3 develops the DMSO and its stability proof. Chapter 4 describes the physical robot. Chapter 5 derives the TWIP equations of motion from the DC motor model upward and tests them against the robot's recorded data. Chapter 6 presents the LQR baseline and the command-filtered neural controller. Chapter 7 describes the simulation environment, including the discretization, the noise models, and the actuator nonlinearities. Chapter 8 presents results. Chapter 9 concludes. Appendix A gives implementation and test guidance.

Chapter 2

LITERATURE REVIEW

2.1 TWO WHEELED INVERTED PENDULUM

Research on two wheeled inverted pendulums began in the 1990s. Ha and Yuta at the University of Tsukuba created the Yamabico Kurara and used a linear quadratic regulator design for trajectory tracking control [1]. Shiroma et al. built a wheeled inverted pendulum and analyzed its stability in the presence of applied forces, using an observer to estimate the unknown force [2]. Ozaki et al. implemented decoupling control by adding an additional state variable, creating a square system, and proposed algorithms to cope with the singularity introduced by the feedback [3]. Grasser et al. at the Industrial Electronics Laboratory of the Swiss Federal Institute of Technology created JOE, a mobile inverted pendulum, decoupling the robot dynamics and applying pole placement together with a filter to suppress the backlash effects of the DC motors [4]. Ooi at the University of Western Australia built a balancing robot to investigate Kalman filtering for sensor fusion, implementing both pole placement and linear quadratic regulator controllers [5]. Various forms of PID control have been investigated by a number of researchers [6, 7, 8, 9]. Kim and Kwon developed a feedforward controller based on the state dependent Riccati equation [10].

Two observations about this body of work bear on the present development. First, every

one of these designs depends on the physical parameters being known accurately, since the gains are computed from them; the sensitivity of the closed loop to parameter error is the practical limitation rather than the linearization itself. Second, Grasser et al. treat backlash as something to be filtered out of the measurement rather than compensated in the control, which is a reasonable engineering choice but leaves the nonlinearity in the loop. Both points recur in Chapter 4.

The work above is based on linearization of the equations of motion. Nonlinear control of two wheeled inverted pendulums is a current topic of research. Kim et al. investigated the exact dynamics of a two wheeled inverted pendulum robot, examining stability on an inclined plane and in turning motion [11]. Pathak et al. used the full nonlinear equations obtained by the Euler-Lagrange method and applied partial feedback linearization to control velocity and position [12]. This last is worth singling out: the partial feedback linearization decomposition makes explicit that the actuated and unactuated subsystems must be treated differently, and that the tilt state rather than the motor voltage is the effective control authority available to the position loop. That observation is the structural basis for the virtual control used in Chapter 6. Askari et al. implemented a form of model predictive control and studied performance under input disturbances [13]. Shibayama et al. implemented an observer-based robust controller to counteract unknown disturbances, permitting the use of a simplified system model [14]. Ja and Lee implemented sliding mode control, improving on linear controllers when far from equilibrium [15]. Tsai and Ju implemented backstepping sliding mode control for trajectory tracking and stabilization by first decoupling the kinematics and dynamics [16]. Do and Seet combined partial feedback linearization with p-times differentiable saturation and backstepping; the resulting controller has a large domain of attraction and is simple to tune [17]. Rudra and Barai developed a robust adaptive backstepping technique that estimates the system parameters and removes the need for prior knowledge of them [18]. Durdevic and Yang proposed a hybrid switching controller to

overcome the backlash nonlinearity of DC motors and demonstrated its effectiveness experimentally [19]. Yue, Wei, and Li developed an adaptive sliding mode technique based on zero dynamics theory, allowing parameter uncertainties to be estimated and providing robustness in the presence of nonlinearities [20].

The general backstepping machinery underlying several of these designs is due to Krstić et al. [21], and the standard stability tools used throughout are collected in Khalil [22] and in Slotine and Li [23].

Neural networks have been used by a number of researchers to control the two wheeled inverted pendulum. Noh, Lee, and Jung used radial basis function networks to control the unstable system, deriving an online training law from the back-propagation algorithm [24, 25]. Tsai, Huang, and Lin proposed an adaptive technique using radial basis function networks, employing backstepping and Lyapunov analysis to synthesize stable adaptive control laws [26]. Li and Yang formulated an output feedback adaptive neural network controller with a linear dynamic compensator and showed that it outperforms a model based controller when parametric uncertainties are present [27]. Li, Yang, and Fan devote a chapter of their book on advanced nonlinear control of wheeled inverted pendulums to neural network control using radial basis functions [28]. Optimal control remains a smaller area of research for these systems, though Gomez et al. developed a strategy based on Control Adjoining Cell Mapping Reinforcement Learning and tested it on a robot built from LEGO NXT components [29].

Applications of the wheeled inverted pendulum configuration extend beyond the laboratory; Neugebauer et al. describe mobile systems of this type used for machining large work pieces [30].

The common difficulty across all of the nonlinear and adaptive work above, for the TWIP specifically, is underactuation. A single input drives both acceleration equations, so an uncertainty appearing in the tilt acceleration cannot be cancelled by the same signal that cancels an uncertainty in the translational acceleration. Two of the works cited treat the

actuator nonlinearity directly rather than absorbing it into a lumped uncertainty [4, 19], and the broader adaptive treatment of actuator nonlinearities is due to Tao and Kokotović [31]. This matters here because backlash is not smooth and therefore falls outside the function class covered by the universal approximation property. A network can approximate such a nonlinearity only in an averaged sense, and the residual appears as a floor on the achievable tracking error rather than as something further adaptation can remove; see Remark 7.3.

2.1.1 Command Filtering

Classical backstepping requires analytic differentiation of each virtual control law, which produces an expanding chain of terms and, when the virtual control contains an online function estimate, requires differentiating the weight update law itself. Command filtering, in the form of Farrell et al. [32], and the closely related dynamic surface control of Swaroop et al. [33], replace analytic differentiation with a bandwidth-limited filter and compensate for the resulting filter error within the stability argument. This device is used in Chapter 6, where it serves a second purpose beyond avoiding the explosion of terms: it breaks the algebraic dependence that would otherwise exist between the two neural networks' approximation targets, as detailed in Remark 6.3.

2.2 MODIFIED STATE OBSERVER

Artificial neural networks have emerged as a capable tool in adaptive control. Modeled after the behavior of neurons in biological systems, they have proven effective as function approximators [34, 35]. This approximation capability has been adapted into many methods of adaptive control that use networks to estimate nonlinear system uncertainty [36, 37]. Rajagopal et al. developed the Modified State Observer (MSO) as a method of estimating nonlinear uncertainty by combining a standard Luenberger observer structure with a neural

network [38]. The MSO allows the designer to use large adaptive gains without the high frequency oscillation in the control signal that could otherwise excite unmodeled dynamics.

Although originally applied to control, the MSO has also been used purely for state estimation. Harl et al. applied it to an orbit uncertainty estimation problem and extended the method to a reduced order formulation for cases where the full state cannot be measured [39]. Darling et al. used the MSO and its reduced order form in an atmospheric reentry uncertainty estimation problem, estimating the uncertain aerodynamic acceleration of falling debris [40], and subsequently developed the Sigma Point MSO, which applies sigma point filtering in the manner of the unscented Kalman filter and admits nonlinear measurements with a variable observer gain [41]. Yang et al. formulated a model reference adaptive control design based on the MSO to control a nonlinear electrohydraulic system with parameter uncertainty from a simplified linear model [36]. Pappu et al. formulated an MSO-based adaptive control law and applied it to the Black Kite micro air vehicle [42].

All of this previous work on the MSO is in continuous time. In practice, implementation on a digital system requires integrating the dynamics over time at every step. Discrete time implementations have the advantage of reduced computation in addition to being designed for digital hardware directly. Discrete time neural networks have themselves been a topic of study in adaptive control [37]. To a lesser extent they have been used specifically as observers. Salgado and Chairez used recurrent networks with offline training based on a least mean square method to construct a discrete time neural observer [43]. Alanis et al. proposed a reduced order discrete time neural observer trained offline by an extended Kalman filter using high order recurrent networks [44]. What these works lack is an online training law with guaranteed boundedness. Lewis et al. developed a multilayer discrete time neural network observer using online training [45]; the present work differs in the weight update law and in the observer structure, and requires substantially fewer tunable parameters, which simplifies both the design and the tuning effort.

2.2.1 Method Notes

Three points of technique motivate the proof given in Chapter 3.

First, the weight error recursion in discrete time adaptive schemes has the form $\tilde{W}_{k+1} = (I - \Gamma\phi_k\phi_k^T)\tilde{W}_k + (\cdot)$, in which $I - \Gamma\phi\phi^T$ is a symmetric rank-one perturbation of the identity. Its spectrum is $\{1,\, 1 - \Gamma\left\|\phi\right\|^2\}$, so its induced two-norm is $\max\{1, \left|1 - \Gamma\left\|\phi\right\|^2\right|\}$ and not $\left|1 - \Gamma\left\|\phi\right\|^2\right|$ [46]. Bounding this quantity before the Lyapunov difference is formed discards the structure that the adaptation law was constructed to produce.

Second, the condition $0 < \Gamma\left\|\phi_k\right\|^2 < 2$ familiar from the persistency of excitation literature [35, 45] is recovered directly from the exact first difference, and is not an independent assumption.

Third, weight drift in the absence of persistent excitation is a long-standing concern in adaptive control, and σ-modification was introduced by Ioannou and Kokotović [47, 48] precisely to address it. The device is used twice in this work: on the controller networks in Chapter 6 and on the observer network in Section 3.6.2. Applying it in both places makes the two stability arguments structurally consistent and removes reliance on an excitation condition that regulation about an equilibrium does not supply.

2.3 SUPPORTING MATERIAL

The matrix algebra used in Chapter 3, in particular the properties of semi-orthogonal matrices and of the Frobenius norm, follows Abadir and Magnus [49] and Horn and Johnson [46]. Young's inequality is due to Young [50], and the triangle inequality and related numerical results follow Kress [51]. The rotation sequences used to obtain the tilt angle from the accelerometer in Section 4.3.1 follow Kuipers [52], and the complementary filter used to fuse the accelerometer and gyroscope follows Higgins [53]. The rigid body dynamics used in Chapter 5 follow Hibbeler [54], and the DC motor and mechatronic subsystem models follow

de Silva [55]. The virtual control construction extended in Chapter 6 is due to Huang [56]. The Kalman filter baseline follows Kalman [57] and Simon [58], and the inertial sensor error models used in Section 7.5 follow Gebre-Egziabher [59] and Flenniken [60].

Chapter 3

DISCRETE-TIME MODIFIED STATE OBSERVER

The Discrete-Time Modified State Observer (DMSO) is an extension of the Modified State Observer of Rajagopal et al. [38] to discrete time. This chapter states the problem, gives the observer and its adaptation law, and establishes a mean-square bound on, and, with bounded weights, uniform ultimate boundedness of, the state estimation error and the functional approximation error by Lyapunov analysis.

The derivation evaluates the Lyapunov first difference exactly before any bounding is performed. This is not a stylistic preference. The adaptation law is constructed so that the cross terms coupling the weight error to the state estimation error and to the network reconstruction error cancel identically, and a term-by-term bound applied in advance would destroy those cancellations.

3.1 NOTATION

$\|\cdot\|$ denotes the Euclidean norm on vectors and the induced 2-norm on matrices; $\|\cdot\|_F$ denotes the Frobenius norm, $\|M\|_F^2 = \mathrm{Tr}(M^T M)$. For square M, $\sigma_{\max}(M)$ is the largest singular

value and $\rho(M)$ the spectral radius. Recall $\rho(M) \le \sigma_{\max}(M) = \|M\|$, with equality when M is normal [46]; the converse does not hold, since a non-normal M with a dominant normal block, such as $\operatorname{diag}\big(1, \left[\begin{smallmatrix} 0 & 1/2 \\ 0 & 0 \end{smallmatrix}\right]\big)$, also attains equality. $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ denote extreme eigenvalues of a symmetric matrix. Properties of semi-orthogonal matrices and of the Frobenius norm follow Abadir and Magnus [49].

3.2 PROBLEM STATEMENT

Assumption 3.1 (Plant)**.** The plant evolves as

$$x_{k+1} = Fx_k + Gu_k + B_{\mathrm{MSO}} f(x_k), \qquad y_k = x_k, \tag{3.1}$$

with $x_k \in \mathbb{R}^n$ the state, $u_k \in \mathbb{R}^m$ the control, $F \in \mathbb{R}^{n\times n}$ the discretized system matrix, $G \in \mathbb{R}^{n\times m}$ the discretized input matrix, and $f : \mathbb{R}^n \to \mathbb{R}^p$ an unknown smooth vector of uncertainties, p being the number of states carrying uncertainty.

Assumption 3.2 (Selection matrix)**.** $B_{\mathrm{MSO}} \in \mathbb{R}^{n\times p}$ contains exactly one unit entry per column, all other entries zero, with no two columns equal. Its columns are therefore distinct standard basis vectors of $\mathbb{R}^n$, so

$$B_{\mathrm{MSO}}^T B_{\mathrm{MSO}} = I_p, \qquad \|B_{\mathrm{MSO}}\| = 1, \qquad B_{\mathrm{MSO}} B_{\mathrm{MSO}}^T = \Pi, \tag{3.2}$$

where Π is the orthogonal projector onto the uncertain coordinates. Only the left identity in (3.2) is used below; note that $B_{\mathrm{MSO}} B_{\mathrm{MSO}}^T \neq I_n$ whenever $p < n$.

Assumption 3.3 (Functional approximation)**.** On the compact operating set there exist ideal weights $W \in \mathbb{R}^{l\times p}$ and a basis $\phi : \mathbb{R}^n \to \mathbb{R}^l$ with

$$f(x_k) = W^T \phi(x_k) + \epsilon_k, \qquad \|\epsilon_k\| \le \epsilon_N, \qquad \|\phi(x_k)\| \le \phi_{\max}, \tag{3.3}$$

ϵ_N and $\phi_{\max}$ known and finite, and the plant trajectory x_k remains in that set for all $k \geq 0$ (for the TWIP, because a controller keeps it there). Write $\phi_k := \phi(x_k)$. The universal approximation property of two-layer networks is due to Hornik et al. [34]; Sadegh [35] extended it to the single-layer functional link form used here, in which one layer suffices provided ϕ is selected as a basis.

Assumption 3.4 (Bounded ideal weights)**.** $\|W\|_F \leq W_M < \infty$. Used only in Section 3.6.2.

3.3 OBSERVER AND ADAPTATION LAW

The observer is the modified Luenberger form

$$\hat{x}_{k+1} = F\hat{x}_k + Gu_k + B_{\mathrm{MSO}}\hat{f}(x_k) + K_{\mathrm{MSO}}\big(y_k - \hat{x}_k\big), \qquad \hat{f}(x_k) = \hat{W}_k^T\phi_k, \tag{3.4}$$

with $K_{\mathrm{MSO}} \in \mathbb{R}^{n\times n}$ a constant design gain and $\hat{W}_k \in \mathbb{R}^{l\times p}$ the weight estimate. Define the error-injection matrix

$$A := F - K_{\mathrm{MSO}}. \tag{3.5}$$

Because $y_k = x_k$, K_{MSO} is entirely at the designer's disposal and A may be assigned arbitrarily, including $A = 0$; see Corollary 3.10.

The adaptation law is

$$\hat{W}_{k+1} = \hat{W}_k + \Gamma\,\phi_k\, e_{k+1}^T B_{\mathrm{MSO}}, \qquad \Gamma > 0, \tag{3.6}$$

where $e_k := x_k - \hat{x}_k$ and $\tilde{W}_k := W - \hat{W}_k$. The law is causal: $\hat{W}_{k+1}$ is formed at step $k+1$, at which time e_{k+1} is available.

Three quantities recur and are named once:

$$\psi_k := \tilde{W}_k^T\phi_k \in \mathbb{R}^p, \qquad \zeta_k := \psi_k + \epsilon_k \in \mathbb{R}^p, \qquad a_k := B_{\mathrm{MSO}}^T A e_k \in \mathbb{R}^p. \tag{3.7}$$

Here ψ_k is the *functional* approximation error, the quantity the network must drive small, and ζ_k the total uncertainty reconstruction error. Since $\|B_{\mathrm{MSO}}\| = 1$, $\|a_k\| \le \|Ae_k\|$.

3.4 ERROR DYNAMICS

Lemma 3.5 (State estimation error)**.** *Under Assumptions 3.1–3.3,*

$$e_{k+1} = Ae_k + B_{\mathrm{MSO}}\zeta_k. \tag{3.8}$$

Proof. Subtracting (3.4) from (3.1) with $y_k = x_k$,

$$e_{k+1} = Fe_k - K_{\mathrm{MSO}}e_k + B_{\mathrm{MSO}}\big(f(x_k) - \hat{f}(x_k)\big) = Ae_k + B_{\mathrm{MSO}}\big(W^T\phi_k + \epsilon_k - \hat{W}_k^T\phi_k\big),$$

and $W^T\phi_k - \hat{W}_k^T\phi_k = \tilde{W}_k^T\phi_k = \psi_k$. □

Lemma 3.6 (Weight error)**.** *With* $\xi_k := a_k + \zeta_k$,

$$\tilde{W}_{k+1} = \tilde{W}_k - \Gamma\phi_k\xi_k^T = \big(I - \Gamma\phi_k\phi_k^T\big)\tilde{W}_k - \Gamma\phi_k\big(a_k + \epsilon_k\big)^T. \tag{3.9}$$

Proof. From (3.6), $\tilde{W}_{k+1} = \tilde{W}_k - \Gamma\phi_k e_{k+1}^T B_{\mathrm{MSO}}$. By Lemma 3.5 and $B_{\mathrm{MSO}}^T B_{\mathrm{MSO}} = I$,

$$e_{k+1}^T B_{\mathrm{MSO}} = \big(Ae_k + B_{\mathrm{MSO}}\zeta_k\big)^T B_{\mathrm{MSO}} = e_k^T A^T B_{\mathrm{MSO}} + \zeta_k^T = a_k^T + \zeta_k^T = \xi_k^T.$$

The second equality follows from $\Gamma\phi_k\zeta_k^T = \Gamma\phi_k\phi_k^T\tilde{W}_k + \Gamma\phi_k\epsilon_k^T$. □

3.5 LYAPUNOV STABILITY PROOF

Take the Lyapunov candidate

$$V_k = e_k^T e_k + \frac{1}{\Gamma}\operatorname{Tr}\big(\tilde{W}_k^T \tilde{W}_k\big) = \|e_k\|^2 + \frac{1}{\Gamma}\left\|\tilde{W}_k\right\|_F^2 \;\geq\; 0. \tag{3.10}$$

The adaptation gain Γ itself serves as the weighting; no independent Lyapunov constant is introduced.

3.5.1 The Exact First Difference

Proposition 3.7 (Exact difference)**.** *Let* $\gamma_k := \Gamma\,\|\phi_k\|^2$*. Then*

$$\Delta V_k = -\,\|e_k\|^2 + \|Ae_k\|^2 + 2a_k^T\epsilon_k + \|\epsilon_k\|^2 - \|\psi_k\|^2 + \gamma_k\,\|a_k + \zeta_k\|^2\,. \tag{3.11}$$

Proof. Weight term. From Lemma 3.6,

$$\operatorname{Tr}\big(\tilde{W}_{k+1}^T\tilde{W}_{k+1}\big) = \operatorname{Tr}\big(\tilde{W}_k^T\tilde{W}_k\big) - 2\Gamma\operatorname{Tr}\big(\tilde{W}_k^T\phi_k\xi_k^T\big) + \Gamma^2\operatorname{Tr}\big(\xi_k\phi_k^T\phi_k\xi_k^T\big).$$

Since $\operatorname{Tr}(\tilde{W}^T\phi\xi^T) = \xi^T\tilde{W}^T\phi = \psi^T\xi$ and $\operatorname{Tr}(\xi\phi^T\phi\xi^T) = \|\phi\|^2\,\|\xi\|^2$,

$$\frac{1}{\Gamma}\Big[\left\|\tilde{W}_{k+1}\right\|_F^2 - \left\|\tilde{W}_k\right\|_F^2\Big] = -2\psi_k^T\xi_k + \gamma_k\,\|\xi_k\|^2\,. \tag{3.12}$$

State term. From Lemma 3.5 and $B_{\mathrm{MSO}}^T B_{\mathrm{MSO}} = I$,

$$\|e_{k+1}\|^2 - \|e_k\|^2 = \|Ae_k\|^2 - \|e_k\|^2 + 2a_k^T\zeta_k + \|\zeta_k\|^2\,, \tag{3.13}$$

using $2e_k^T A^T B_{\mathrm{MSO}}\zeta_k = 2a_k^T\zeta_k$.

Combine. Adding (3.12) and (3.13) with $\xi_k = a_k + \zeta_k$,

$$\Delta V_k = \|Ae_k\|^2 - \|e_k\|^2 + \underbrace{2a_k^T\zeta_k - 2\psi_k^T a_k}_{\text{(i)}} + \underbrace{\|\zeta_k\|^2 - 2\psi_k^T\zeta_k}_{\text{(ii)}} + \gamma_k \|a_k + \zeta_k\|^2 .$$

For (i), $2a_k^T(\zeta_k - \psi_k) = 2a_k^T\epsilon_k$. For (ii), with $\zeta_k = \psi_k + \epsilon_k$,

$$\|\psi_k + \epsilon_k\|^2 - 2\psi_k^T(\psi_k + \epsilon_k) = \|\psi_k\|^2 + 2\psi_k^T\epsilon_k + \|\epsilon_k\|^2 - 2\|\psi_k\|^2 - 2\psi_k^T\epsilon_k = -\|\psi_k\|^2 + \|\epsilon_k\|^2 .$$

Substituting gives (3.11). □

The mechanism is now visible. Outside the adaptation term $\gamma_k \|a_k + \zeta_k\|^2$, every cross term between ψ_k and ϵ_k, and every cross term between ψ_k and a_k, cancels exactly. The negative term $-\|\psi_k\|^2$ is produced by the adaptation law and is not an artifact of any inequality. The only price paid for adaptation is the single term $\gamma_k \|a_k + \zeta_k\|^2$, which is small when Γ is small. Small adaptation gains are therefore admissible on structural grounds, and the conditions below reflect this.

3.5.2 Main Result

Theorem 3.8 (Uniform ultimate boundedness)**.** *Let Assumptions 3.1–3.3 hold. Set* $\bar{\sigma} := \sigma_{\max}(A)$ *and* $\bar{\gamma} := \Gamma\phi_{\max}^2$. *Fix* $\nu > 0$ *and define*

$$c_1 := 1 + \bar{\gamma}\Big(1 + \tfrac{1}{\nu}\Big), \qquad c_2 := 1 - \bar{\gamma}\big(1 + \nu\big), \qquad \mu := \bar{\sigma}\sqrt{c_1}. \tag{3.14}$$

If

$$\text{(C1)} \quad c_2 > 0, \qquad \textit{i.e.} \quad \Gamma\phi_{\max}^2 < \frac{1}{1+\nu}, \tag{3.15}$$

$$\text{(C2)} \quad \mu < 1, \qquad \textit{i.e.} \quad \sigma_{\max}(A)^2\Big[1 + \Gamma\phi_{\max}^2\big(1 + \tfrac{1}{\nu}\big)\Big] < 1, \tag{3.16}$$

then

$$\Delta V_k \;\le\; -\|e_k\|^2 + c_1\big(\bar{\sigma}\,\|e_k\| + \epsilon_N\big)^2 - c_2\,\|\psi_k\|^2\,, \tag{3.17}$$

and $\Delta V_k < 0$ *whenever*

$$\|e_k\| > b_e := \frac{\sqrt{c_1}\,\epsilon_N}{1-\mu} \qquad or \qquad \|\psi_k\| > b_\psi := \epsilon_N\sqrt{\frac{c_1}{c_2\big(1-\mu^2\big)}}. \tag{3.18}$$

Consequently:

(a) without any further assumption, the time averages of the errors satisfy

$$\limsup_{N\to\infty}\frac{1}{N}\sum_{k=0}^{N-1}\Big[\tfrac{1-\mu^2}{2}\,\|e_k\|^2 + c_2\,\|\psi_k\|^2\Big] \le c_1\epsilon_N^2\,\frac{1+\mu^2}{1-\mu^2}; \tag{3.19}$$

(b) if the weight error $\tilde{W}_k$ *is bounded (Section 3.6), the state estimation error* e_k *and the weight error* $\tilde{W}_k$*, hence* $\psi_k = \tilde{W}_k^T\phi_k$*, are uniformly ultimately bounded.*

The thresholds b_e *and* b_ψ *are referred to below as the ultimate bounds. Strictly, the bound that the Lyapunov argument delivers in (b) is the largest value of* $\|e_k\|$ *on the smallest level set of* V *containing the region where* $\Delta V_k \ge 0$*, which can be larger;* b_e *is the level beyond which the error is always driven down.*

Proof. Begin from (3.11). Since $\|B_{\mathrm{MSO}}\| = 1$, $\|a_k\| \le \|Ae_k\|$, so

$$\|Ae_k\|^2 + 2a_k^T\epsilon_k + \|\epsilon_k\|^2 \le \big(\|Ae_k\| + \epsilon_N\big)^2.$$

Write $s_k := \|Ae_k\| + \epsilon_N$ and $w_k := \|\psi_k\|$. Then $\|a_k + \zeta_k\| \le \|Ae_k\| + \|\psi_k\| + \|\epsilon_k\| \le s_k + w_k$, and $\gamma_k \le \bar{\gamma}$, so

$$\Delta V_k \le -\|e_k\|^2 + s_k^2 - w_k^2 + \bar{\gamma}(s_k + w_k)^2.$$

Applying Young's inequality [50] once, in the form $(s+w)^2 \le (1+\frac{1}{\nu})s^2 + (1+\nu)w^2$,

$$\Delta V_k \le -\|e_k\|^2 + c_1 s_k^2 - c_2 w_k^2,$$

and $\|Ae_k\| \le \bar{\sigma}\|e_k\|$ gives (3.17).

For the first branch of (3.18), $c_2 > 0$ by (C1), so it suffices that $c_1(\bar{\sigma}\|e_k\| + \epsilon_N)^2 < \|e_k\|^2$, that is, $\sqrt{c_1}(\bar{\sigma}\|e_k\| + \epsilon_N) < \|e_k\|$, that is, $\|e_k\|(1-\mu) > \sqrt{c_1}\epsilon_N$. This requires $\mu < 1$, which is (C2).

For the second branch, let $g(t) := c_1(\bar{\sigma}t + \epsilon_N)^2 - t^2 = (\mu^2-1)t^2 + 2c_1\bar{\sigma}\epsilon_N t + c_1\epsilon_N^2$ for $t = \|e_k\| \ge 0$. By (C2) the leading coefficient is negative, so g is maximized at $t^\star = c_1\bar{\sigma}\epsilon_N/(1-\mu^2)$ with

$$g(t^\star) = c_1\epsilon_N^2 + \frac{c_1^2\bar{\sigma}^2\epsilon_N^2}{1-\mu^2} = c_1\epsilon_N^2\Big[1 + \frac{\mu^2}{1-\mu^2}\Big] = \frac{c_1\epsilon_N^2}{1-\mu^2}.$$

Hence $\Delta V_k \le g(\|e_k\|) - c_2 w_k^2 < 0$ whenever $c_2 w_k^2 > c_1\epsilon_N^2/(1-\mu^2)$, which is $\|\psi_k\| > b_\psi$.

For (a), split the cross term in (3.17) as $c_1(\bar{\sigma}\|e_k\| + \epsilon_N)^2 \le \frac{1+\mu^2}{2}\|e_k\|^2 + c_1\epsilon_N^2\frac{1+\mu^2}{1-\mu^2}$, which is Young's inequality with weight $(1-\mu^2)/(2\mu^2)$ (for $\mu = 0$ it holds trivially). Then

$$\Delta V_k \le -\tfrac{1-\mu^2}{2}\|e_k\|^2 - c_2\|\psi_k\|^2 + c_1\epsilon_N^2\frac{1+\mu^2}{1-\mu^2},$$

and summing from $k = 0$ to $N-1$, using $V_N \ge 0$ and dividing by N, gives (3.19).

For (b), $\Delta V_k < 0$ outside $\{\|e_k\| \le b_e\} \cap \{\|\psi_k\| \le b_\psi\}$. That set is not bounded in $\tilde{W}_k$, since components of $\tilde{W}_k$ orthogonal to ϕ_k leave ψ_k unchanged, so the standard Lyapunov extension theorem for discrete-time systems [45, 37] does not apply to it directly. With $\tilde{W}_k$ confined to a bounded set its intersection with that set is compact, and the theorem gives uniform ultimate boundedness; see also Khalil [22] for the continuous-time counterpart. □

Corollary 3.9 (Simple sufficient conditions)**.** *Take $\nu = 1$, so $c_1 = 1 + 2\bar{\gamma}$ and $c_2 = 1 - 2\bar{\gamma}$.*

Then Theorem 3.8 applies provided

$$\sigma_{\max}(A) < 1 \quad \textit{and} \quad 0 < \Gamma < \frac{1}{\phi_{\max}^2} \min\left\{ \frac{1}{2},\ \frac{1-\sigma_{\max}(A)^2}{2\,\sigma_{\max}(A)^2} \right\}. \tag{3.20}$$

As $\Gamma \to 0^+$ *the ultimate bounds tend to*

$$b_e \to \frac{\epsilon_N}{1-\sigma_{\max}(A)}, \qquad b_\psi \to \frac{\epsilon_N}{\sqrt{1-\sigma_{\max}(A)^2}}. \tag{3.21}$$

Proof. With $\nu = 1$, (C1) is $\bar{\gamma} < 1/2$ and (C2) is $\bar{\sigma}^2(1+2\bar{\gamma}) < 1$, i.e. $\bar{\gamma} < (1-\bar{\sigma}^2)/(2\bar{\sigma}^2)$, feasible for some $\bar{\gamma} > 0$ if and only if $\bar{\sigma} < 1$. Substituting $\bar{\gamma} = \Gamma\phi_{\max}^2$ gives (3.20); the limits follow by setting $c_1, c_2 \to 1$ and $\mu \to \bar{\sigma}$ in (3.18). □

Corollary 3.10 (Deadbeat error injection)**.** *Since* $y_k = x_k$*, choosing* $K_{\mathrm{MSO}} = F$ *gives* $A = 0$*. Working directly from* (3.11) *with* $a_k = 0$*,*

$$\Delta V_k = -\|e_k\|^2 - \big(1-\gamma_k\big)\|\psi_k\|^2 + 2\gamma_k \psi_k^T \epsilon_k + \big(1+\gamma_k\big)\|\epsilon_k\|^2,$$

so for any $\bar{\gamma} < 1$*, completing the square on the* ψ_k *terms gives*

$$\Delta V_k \le -\|e_k\|^2 + \frac{\epsilon_N^2}{1-\bar{\gamma}},$$

so the time average of $\|e_k\|^2$ *is asymptotically at most* $\epsilon_N^2/(1-\bar{\gamma})$ *and, with* $\tilde{W}_k$ *bounded as in Theorem 3.8(b),* $\|e_k\|$ *is driven down whenever it exceeds* $\epsilon_N/\sqrt{1-\bar{\gamma}}$*. No condition on* A *is required and the restriction on* Γ *relaxes to* $\bar{\gamma} < 1$*.*

Remark 3.11 (Interpretation). Condition (C2) requires the error-injection matrix to be a contraction in the 2-norm; (C1) requires the adaptation gain to be small relative to the basis magnitude. Faster error injection buys tolerance for a larger learning rate, and a slower learning rate always helps. Both are what one would expect on physical grounds. In the

limit $\Gamma \to 0^+$ the conditions collapse to $\sigma_{\max}(A) < 1$ and the bounds (3.21) reduce to the gain of the error dynamics applied to the network reconstruction floor ϵ_N, which is the correct limiting answer.

Remark 3.12 (Non-normal A). If A is Schur but $\sigma_{\max}(A) \geq 1$, Theorem 3.8 does not apply in the Euclidean norm. Since $\rho(A) < 1$, for any $\delta > 0$ there is a nonsingular T with $\|TAT^{-1}\| \leq \rho(A) + \delta < 1$. Repeating the derivation with $V_k = e_k^T T^T T e_k + \Gamma^{-1} \left\|\tilde{W}_k\right\|_F^2$ preserves the cancellations of Proposition 3.7 only if the adaptation law is weighted by the same matrix, $\hat{W}_{k+1} = \hat{W}_k + \Gamma \phi_k e_{k+1}^T T^T T B_{\text{MSO}}$. With that law the exact difference has the form (3.11), with ψ_k and ϵ_k measured in the norm weighted by $M := B_{\text{MSO}}^T T^T T B_{\text{MSO}} \succ 0$ and a_k replaced by $B_{\text{MSO}}^T T^T T A e_k$; no normalization $M = I$ is needed. The conclusions are the same with $\bar{\sigma}$ replaced by $\|TAT^{-1}\|$, at the cost of factors depending on $\kappa(T)$ and on the extreme eigenvalues of M in the bounds. Since K_{MSO} is fully free here, it is simpler in practice to place A so that $\sigma_{\max}(A)$ is directly small. Note that $\lambda_{\max}(A)$ is not an upper bound for $\|A\|$ unless A is normal, and may be complex.

3.6 BOUNDEDNESS OF THE WEIGHT ESTIMATES

Without a bound on $\tilde{W}_k$, Theorem 3.8 gives only the mean-square bound (a) on e_k and $\psi_k = \tilde{W}_k^T \phi_k$, and nothing on $\tilde{W}_k$ itself. This is the correct statement: the negative term in (3.17) acts only along ϕ_k, so components of $\tilde{W}_k$ orthogonal to the regressor direction are unconstrained by (3.17) and may drift. Two standard remedies apply.

3.6.1 Persistency of Excitation

Lemma 3.13 (Sadegh [35]). *Let $\Phi(k_1, k_0)$ denote the state-transition matrix of $\tilde{W}_{k+1} = M(k)\tilde{W}_k$. If $\|\Phi(k_1, k_0)\| < 1$ for all $k_1 > k_0 \geq 0$, the homogeneous system is stable.*

Definition 3.14 (Persistency of excitation)**.** The sequence $\{\phi_k\}$ is persistently exciting if there exist $\beta > 0$ and an integer $L \geq 1$ such that

$$\lambda_{\min}\left(\sum_{j=k}^{k+L-1} \phi_j \phi_j^T\right) \geq \beta, \qquad \forall\, k \geq 0. \tag{3.22}$$

Lemma 3.15. *If $M(k) = I - \Gamma\phi_k\phi_k^T$ with $0 < \Gamma \|\phi_k\|^2 < 2$ and (3.22) holds, then the homogeneous part of (3.9) is uniformly exponentially stable, and since the forcing term $-\Gamma\phi_k(a_k + \epsilon_k)^T$ is bounded whenever e_k is bounded, $\tilde{W}_k$ is bounded.*

Lemma 3.15 bounds $\tilde{W}_k$ given a bound on e_k, while Theorem 3.8(b) bounds e_k given a bound on $\tilde{W}_k$. Under (3.22) the two are combined by a windowed Lyapunov or small-gain argument, which is standard but not reproduced here. σ-modification, below, avoids the circularity by bounding $\tilde{W}_k$ through V_k directly.

The condition $0 < \Gamma \|\phi_k\|^2 < 2$ required here is exactly the condition $\gamma_k < 2$ that renders the ψ_k coefficient in (3.12) negative. In the present derivation it is not an independent assumption but a consequence of the same inequality that drives Theorem 3.8, and Corollary 3.9 implies it.

3.6.2 σ-Modification

Persistency of excitation is not always available, particularly during regulation, where the regressor may be nearly constant. Replacing (3.6) by

$$\hat{W}_{k+1} = \big(1 - \Gamma\kappa\big)\hat{W}_k + \Gamma\phi_k e_{k+1}^T B_{\mathrm{MSO}}, \qquad \kappa > 0, \tag{3.23}$$

gives $\tilde{W}_{k+1} = \tilde{W}_k - \Gamma\phi_k\xi_k^T + \Gamma\kappa\hat{W}_k$. The linear part of the trace difference acquires

$$2\kappa\,\mathrm{Tr}\big(\tilde{W}_k^T W\big) - 2\kappa\left\|\tilde{W}_k\right\|_F^2 \;\leq\; -\kappa\left\|\tilde{W}_k\right\|_F^2 + \kappa W_M^2, \tag{3.24}$$

using Assumption 3.4 and $2ab \le a^2 + b^2$. This supplies negative definiteness in $\tilde{W}_k$ directly. The quadratic contributions of $\Gamma\kappa\hat{W}_k$ are dominated for $\Gamma\kappa$ sufficiently small, and the cost is a bias of order κW_M^2 in the ultimate bound. σ-modification [47, 48] is the same device used for the controller networks in Chapter 6; applying it to the observer network as well makes the two stability arguments structurally consistent and is the recommended implementation.

3.7 OBSERVER OR FILTER

With $y_k = x_k$ the full state is measured, so (3.4) is properly a filter and uncertainty identifier rather than an observer in the unmeasured-state sense. Nothing in the proof depends on this, but it should be stated plainly, and it is the reason K_{MSO} may be chosen freely in Corollary 3.10. Extension to $y_k = Cx_k$ with $\operatorname{rank} C < n$ requires $A = F - K_{\mathrm{MSO}}C$ and a detectability assumption; the structure of Proposition 3.7 carries over unchanged with $a_k = B_{\mathrm{MSO}}^T A e_k$.

Chapter 4

TWO WHEELED INVERTED PENDULUM IMPLEMENTATION

4.1 PHYSICAL DESIGN

The robot consists of two mounting plates separated by threaded rods, with two geared DC motors mounted on the lower plate and the electronics distributed between the plates. The center of gravity sits above the wheel axis, giving an inverted pendulum whose pivot is the wheel contact line. The physical parameters are summarized in Table 4.1.

Table 4.1: TWIP robot physical parameters

Parameter	Value	Units
Pendulum moment of inertia about the center of gravity, I_p	0.0169 [a]	$\mathrm{kg\,m^2}$
Wheel moment of inertia, I_w	1.183×10^{-4} [b]	$\mathrm{kg\,m^2}$
Distance to CG from wheel axis, l	0.075	m
Mass of pendulum, M_p	1.432	kg
Mass of wheels, M_w	0.1168	kg
Wheel radius, r	0.045	m

[a] Computed from a lumped-mass idealization of the plates and components; the same model gives $0.025\,\mathrm{kg\,m^2}$ about the wheel axis, and the equations of motion of Chapter 5 take the value about the center of gravity.

[b] Uniform disk, $M_w r^2/2$.

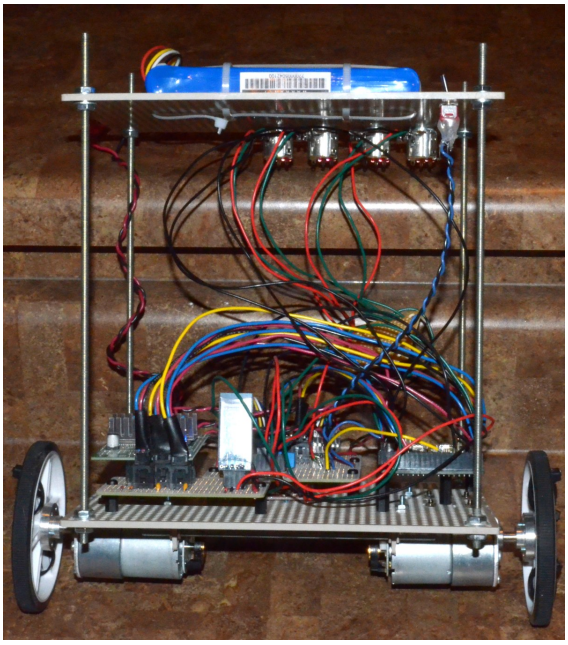


Figure 4.1: Two wheeled inverted pendulum robot.

The moment of inertia and the CG offset are the two parameters least well known on a build of this kind. Both are computed from a lumped-mass idealization of the plates and components rather than measured. The gains that balanced the hardware differ from the LQR gains computed from Table 4.1, but much of that difference is in the position and velocity gains, whose sign the firmware's encoder frame affects (Section 5.3.6), so it does not measure the parameter error. The recorded runs give only indications of that error: at switch-on the response bounds the tilt input gain of the model to within about 40% (Section 5.3.5), and the accelerometer gives an axle coefficient 20 to 50% above the model, which may point to an I_p above the lumped-mass estimate (Section 5.3.4). This is the motivation for the uncertainty estimation problem rather than an incidental nuisance; Appendix A describes a pendulum-swing procedure for measuring I_p and l directly.

4.2 ELECTRONIC DESIGN

Electronics are a major component of any robot, and the two wheeled inverted pendulum is no exception. The robot consists of a main computer, two motors, various sensors, and power and communication devices.

4.2.1 Computer

The robot is controlled using an Arduino Due microcontroller board based on the Atmel SAM3X8E ARM Cortex-M3, a 32-bit 84 MHz CPU. The Due has 96 kB of onboard SRAM and 512 kB of flash available for programming. It provides 54 digital input/output channels, twelve of which can be configured for pulse width modulation output, twelve analog inputs, and two analog outputs, and supports serial communication over USB. The microcontroller is programmed in C through the Arduino interface. The Arduino is a widely used hobbyist platform and a broad range of integrated shields can be purchased to work directly with it.

The Due is adequate for the control law but not generous. As documented in Section 4.4, serial transmission for telemetry dominated the loop budget. Appendix A recommends separating the deterministic control path from the telemetry path so that the achievable loop rate is set by the control computation rather than by the logging.

4.2.2 Motors

Two Pololu 12 V 37 mm brushed DC motors drive the wheels. The motors have a 30:1 gearbox and 64 counts-per-revolution integrated Hall effect quadrature encoders measuring the position of the motor output shaft. The motors are characterized by their free-run speed and current and their stall torque and current. The back-EMF constant follows from the free-run data through the steady armature equation $V = Ri + k_e\omega$ at the free-run speed of

350 rpm (36.65 rad/s),

$$k_e = \frac{12\,\mathrm{V} - (0.3\,\mathrm{A})(2.5\,\Omega)}{36.65\,\mathrm{rad/s}} = 0.307\,\mathrm{V\,s/rad}.$$

The torque constant k_m is the data-sheet value. The stall data give $k_m \approx 0.155\,\mathrm{N\,m/A}$, 35% above it.

The two constants are related but not equal. At the armature they are the same quantity: the electrical power ei taken up by the back-EMF e is the mechanical power $\tau\omega$ developed, so the torque per ampere and the back-EMF per unit speed are equal in SI units. Both constants here are referred to the output shaft of the gearbox, and the gear train treats them differently. The gear ratio N multiplies the back-EMF constant exactly, because the ratio of the two shaft speeds is fixed by the teeth. It multiplies the torque as well, but friction in the motor and the gears takes part of it, so only a fraction $\eta_g < 1$ reaches the output shaft. With k_a the armature constant,

$$k_e = N\,k_a, \qquad k_m = \eta_g\,N\,k_a, \qquad \text{so that} \qquad k_m = \eta_g\,k_e. \tag{4.1}$$

The values of Table 4.2 correspond to $\eta_g = 0.38$, and the stall data to $\eta_g = 0.51$. The fraction is not a constant of the motor. It depends on load and speed, and when the wheel drives the motor the friction adds to the motor torque instead of subtracting from it. A single k_m is therefore an average, and what it leaves out is part of the uncertainty that the observer of Chapter 3 is meant to estimate. The recorded switch-on response bounds the tilt input gain, which is proportional to k_m, to within about 40% of its nominal value (Section 5.3.5), and Stage 0 of Appendix A describes the direct measurement of both constants.

The nominal terminal resistance is measured directly across the output terminals at several shaft positions and averaged. The motor constants are summarized in Table 4.2.

Motor drive is provided by a Pololu Dual VNH5019 Motor Driver Arduino Shield. The

VNH5019 chip allows DC motor operation between 5.5 V and 24 V at currents up to 12 A, provides current sensing, and supports PWM operation at ultrasonic frequencies up to 20 kHz for quieter operation. The shield is designed to work directly with the Arduino platform and a supplied library provides straightforward control of two independent motors.

Table 4.2: DC motor physical parameters

Parameter	Value	Units
Free-run speed	350	RPM
Free-run current	300	mA
Stall torque	110	oz-in
Stall current	5000	mA
Torque motor constant, k_m	0.11541	N m/A
Back-EMF constant, k_e	0.307 [a]	V s/rad
Nominal resistance, R	2.5	Ω

[a] From the free-run speed and current above.

4.2.3 Sensors

Sensors are necessary in any robot in order to measure the state of the system, and the state must be known in order to properly control it. In this system the measurements are obtained from an integrated inertial measurement unit and from Hall effect digital encoders integrated on each motor.

Inertial measurement unit.

Integrated System in Package (SiP) microchips are widely used in consumer electronics owing to their small size and low production cost. The InvenSense MPU-9150 is a low cost, low power SiP motion tracking device marketed for equipment such as smartphones and tablets. It includes a three-axis gyroscope, a three-axis accelerometer, a three-axis magnetometer, and an onboard Digital Motion Processor (DMP) intended to offload algorithms from the main processor. The SparkFun MPU-9150 breakout board is used here, which places the

surface mount device on a board with the communication pins available for direct soldering. The DMP and the magnetometer were not used; only the direct accelerometer and gyroscope readings are required to obtain the necessary states.

The gyroscope has a 16-bit digital output with user-selectable ranges from $\pm 250\,^\circ/\mathrm{s}$ to $\pm 2000\,^\circ/\mathrm{s}$. The lowest range is selected because it has the highest sensitivity, $131\,\mathrm{LSB}/(^\circ/\mathrm{s})$, corresponding to a resolution of 1/131 or approximately $0.007\,63\,^\circ/\mathrm{s}$. Noise is specified by two parameters: a total RMS noise of $0.06\,^\circ/\mathrm{s}$ at a 92 Hz rate, and a noise spectral density of $0.005\,^\circ/\mathrm{s}/\sqrt{\mathrm{Hz}}$ at 10 Hz. These are characterized more carefully by direct test in Section 7.5. The specifications are summarized in Table 4.3.

The accelerometer also has a 16-bit digital output, reporting in multiples of gravity. Calibration was required to obtain a 1 g reading at rest. At the minimum range of $\pm 2\,\mathrm{g}$ the sensitivity is $16\,384\,\mathrm{LSB/g}$, an exact resolution of 1/16384 or approximately 61.0 µg. Noise is specified as 4 mg RMS at a 100 Hz rate and a power spectral density of $400\,\mu\mathrm{g}/\sqrt{\mathrm{Hz}}$ at 10 Hz. The difference in the reference rates between the two sensors necessitates a direct analysis of the noise from the units actually used, so that the simulation is representative. The accelerometer specifications are summarized in Table 4.4.

Table 4.3: Gyroscope specifications

Parameter	Value	Units
Range	±250	°/s
Sensitivity	131	LSB/(°/s)
Resolution	0.00763	°/s
Total RMS noise	0.06	°/s RMS
Power spectral density	0.005	°/s/$\sqrt{\mathrm{Hz}}$

Digital encoders.

Each Pololu metal DC motor has an integrated Hall effect quadrature encoder measuring the relative position of the motor output shaft. The shaft carries a magnetic disk with 16

Table 4.4: Accelerometer specifications

Parameter	Value	Units
Range	±2	g
Sensitivity	16384	LSB/g
Resolution	61.0	µg
Total RMS noise	4	mg RMS
Power spectral density	400	$\mu g/\sqrt{Hz}$

magnetic divisions, and two Hall effect sensors detect the changing field as the shaft rotates. Counting both the rising and falling edges of both signals gives a total resolution of 64 counts per revolution of the motor output shaft. The two signals are a quarter cycle (90°) apart, in quadrature, which is the origin of the term and what allows the direction of rotation to be decoded. In combination with the 30:1 gearbox, the gearbox output has a measurable resolution of 1920 counts per revolution. With 90 mm diameter wheels the encoder measures discrete steps of 1.47×10^{-4} m.

The encoders are mounted on the motor shaft, ahead of the gearbox. Gear backlash is therefore invisible to the encoder: the measured shaft angle advances while the wheel does not. This is a structural measurement limitation, not a resolution limitation, and it cannot be removed by increasing the sample rate. Appendix A discusses load-side encoding as the remedy.

4.2.4 Miscellaneous Electronics

Four potentiometers are connected to the Arduino Due to allow easy adjustment of parameters, in this case the controller gains, and a switch is used to enable and disable the controller. Miscellaneous level-shifting electronics are included to permit communication with the Due, as is an HC-05 Bluetooth module for wireless transmission of serial output data. The robot is powered by a single high-discharge 11.1 V 20C lithium polymer battery.

4.3 PROGRAMMING

On initialization the various pins and connections are configured, enabling communication between the sensors, switches, Bluetooth module, and the microcontroller. The motor controller is initialized and the IMU is enabled. On initialization of the IMU the constant bias is removed from the sensor measurements by averaging a run of samples taken with the robot stationary.

4.3.1 Measurements

The accelerometer measures the specific force vector, which at rest is the gravitational field vector expressed in body axes. Writing the measured vector as $\mathbf{G}_p = [a_x \; a_y \; a_z]^T$ and relating it to the gravitational field through a rotation sequence in roll φ and pitch ϑ_p [52],

$$\frac{\mathbf{G}_p}{\|\mathbf{G}_p\|} = \frac{1}{\sqrt{a_x^2 + a_y^2 + a_z^2}} \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} = \begin{bmatrix} -\sin\vartheta_p \\ \cos\vartheta_p \sin\varphi \\ \cos\vartheta_p \cos\varphi \end{bmatrix}, \tag{4.2}$$

which can be solved for the roll and pitch angles,

$$\varphi = \tan^{-1}\left(\frac{a_y}{a_z}\right), \qquad \vartheta_p = \tan^{-1}\left(\frac{-a_x}{\sqrt{a_y^2 + a_z^2}}\right). \tag{4.3}$$

The relationship in (4.3) is only one way to compute the pitch from the accelerometer. Rotation matrices do not commute, but as only the pitch angle is of concern a different rotation sequence may be used to obtain the same pitch measurement. Computation on the microcontroller must be minimized wherever possible, so by using the rotation sequence R_{yxz} in the same manner an easier relationship for the tilt angle is obtained,

$$\theta_a = \tan^{-1}\left(\frac{a_x}{a_z}\right). \tag{4.4}$$

This calculation assumes two things: that the initial orientation of the accelerometer is aligned with the gravitational field vector along the z axis, and that there are no external accelerations. The second clearly fails when the robot is moving, so filtering of this measurement is necessary. A complementary filter [53] is used. The same task could be performed by a Kalman filter, but the computational limitations of the microcontroller make the complementary filter the appropriate choice. The filter combines the accelerometer and gyroscope measurements to obtain a more accurate angle estimate using a simplified high-pass and low-pass pair. The form implemented on the robot is

$$\theta_{k+1} = \lambda_c\big(\theta_k + g_y \Delta t\big) + \big(1 - \lambda_c\big)\theta_a, \tag{4.5}$$

where λ_c is a design parameter set to 0.99, g_y is the y-axis gyroscope reading, θ_a is the accelerometer tilt angle from (4.4), and θ_k is the filter angle estimate. The complementary filter is simple to implement and well suited to an implementation where computation is limited.

The tilt angle is the only measurement that must be pre-filtered before the DMSO or the Kalman filter are used to estimate the full state. The gyroscope measurements are used directly for the tilt rate. The motor encoder outputs are processed using hardware interrupts: each time a Hall effect sensor signal changes state the interrupt fires and increments or decrements the encoder position. Velocity is obtained by differencing the position readings.

Remark 4.1 (The pre-filter is part of the measurement model). Equation (4.5) is a dynamic element placed between the sensors and the estimator, so the quantity presented to the DMSO as y_k is not a raw measurement but a filtered one with its own lag and its own error characteristics. Two consequences follow. First, the measurement noise covariance used by the Kalman filter baseline must describe the complementary filter output, not the accelerometer. Second, the comparison reported in Section 8.4, in which the complementary filter outperforms the DMSO fed raw accelerometer tilt but not a Kalman filter weighted

by that tilt's actual error, is a statement about this pre-filtering step rather than about the estimators.

4.4 DIFFICULTIES

During assembly and programming many difficulties were encountered. Initially the motors were a substantial problem and were eventually replaced. The original motors were 6 V DC units with lower torque output and an enormous deadzone nonlinearity, giving a correspondingly small range of controllable output. Simulation studies showed that the best achievable behavior with those motors was a large stable oscillation, and sensor noise made even that unachievable. It was during this period that the potentiometers were added, so that controller gains could be adjusted quickly without reprogramming the microcontroller.

The motors were replaced with larger 12 V units that do not have the same large deadzone and allow a wider output range for control. This created a problem with the plastic mounting plates, which twisted under the additional weight until stiffening rods were added to the underside. It also exposed an initial design flaw: plastic rods were used to separate the two mounting plates, and the resulting flexure allowed vibrations to contaminate the sensor measurements. Steel threaded rods were substituted and stiffened the structure satisfactorily.

The robot is limited in available computation, so computations were minimized wherever possible. Serial output proved to be one of the largest consumers of computational effort. Serial data is essential for evaluating performance, so its content had to be selected carefully and limited in order to keep the loop time acceptable. The minimum loop execution time obtained was 0.01 s.

The LQR gains had to be modified from the values computed from the measured parameters of the robot. The model is not entirely accurate to the true system: the pendulum is modeled as a point mass attached by a thin beam to the motor axis, whereas the mounting plates create more complex interactions than this. Once the gains were retuned the robot bal-

anced, for up to 21 s. The logs point to a more specific cause for the retuning (Section 5.3.6). The switch-on transients indicate that the firmware's encoder position is mirrored relative to the model. In that frame, position and velocity gains entered with the signs of the LQR design act with the opposite sign on the robot, and the position and velocity potentiometers, which the firmware maps to $[-15, 0]$, could not set the signs the design requires. Both v9 runs balanced with the base held within a few millimeters; in the two v8 runs the base drove away until the robot was switched off (test 1) or until a chatter ended the run (test 2). The modification of the gains is therefore not evidence of parameter error alone.

Remark 4.2 (These are the case for the work). Three of the observations above are the empirical case for Chapters 3 and 6 rather than incidental setbacks. The parameters are uncertain, which motivates online uncertainty estimation. The actuator is nonlinear in a way the linear model does not capture, which motivates the unmatched uncertainty formulation. The available computation constrains the achievable sample rate, which is an implementation constraint treated as a design variable in Appendix A rather than as a fixed property of the platform.

Chapter 5

THE BALANCING ROBOT MODEL

5.1 MODEL OF A DIRECT CURRENT MOTOR

The armature circuit of a brushed DC motor with terminal voltage V_a, resistance R, inductance L, and back-EMF constant k_e obeys Kirchhoff's voltage law [55],

$$L\frac{\mathrm{d}i}{\mathrm{d}t} + Ri + k_e\omega = V_a, \tag{5.1}$$

where i is the armature current and $\omega = \dot{\vartheta}$ the shaft rate relative to the stator. Rearranging,

$$\frac{\mathrm{d}i}{\mathrm{d}t} = \frac{V_a}{L} - \frac{R}{L}i - \frac{k_e}{L}\omega. \tag{5.2}$$

The electromagnetic torque developed is proportional to the current,

$$\tau_m = k_m i, \tag{5.3}$$

and the viscous friction torque is proportional to the shaft rate,

$$\tau_f = k_f\omega. \tag{5.4}$$

Applying Newton's second law and summing moments about the motor output shaft against an applied load torque τ_a and load inertia I_m,

$$I_m\dot{\omega} = \tau_m - \tau_f - \tau_a. \tag{5.5}$$

Substituting (5.3) and (5.4) into (5.5) and rearranging gives the second equation of motion,

$$\frac{\mathrm{d}\omega}{\mathrm{d}t} = \frac{k_m}{I_m}i - \frac{k_f}{I_m}\omega - \frac{\tau_a}{I_m}. \tag{5.6}$$

Equations (5.2) and (5.6) describe the fundamental behavior of a DC motor as a pair of first-order differential equations relating circuit current, motor speed, and applied torque. For a balancing robot a simplified model provides acceptable performance. Assuming the motor inductance L and the friction coefficient k_f are negligible,

$$i = \frac{V_a}{R} - \frac{k_e}{R}\omega, \tag{5.7}$$

and

$$\frac{\mathrm{d}\omega}{\mathrm{d}t} = \frac{k_m}{I_m}i - \frac{\tau_a}{I_m}. \tag{5.8}$$

Substituting (5.7) into (5.8) gives a model that does not include the circuit current,

$$\frac{\mathrm{d}\omega}{\mathrm{d}t} = -\frac{k_m k_e}{I_m R}\omega + \frac{k_m}{I_m R}V_a - \frac{\tau_a}{I_m}. \tag{5.9}$$

This elimination is desirable: the current is not directly commandable on the physical system and would add a state with no observable benefit. Neglecting the inductance amounts to assuming the current reaches steady state immediately relative to the motor speed, which is justified because the electrical time constant L/R is orders of magnitude shorter than the

mechanical one. In state-space form,

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix}\vartheta\\ \omega\end{bmatrix} = \begin{bmatrix}0 & 1\\ 0 & -\dfrac{k_m k_e}{I_m R}\end{bmatrix}\begin{bmatrix}\vartheta\\ \omega\end{bmatrix} + \begin{bmatrix}0 & 0\\ \dfrac{k_m}{I_m R} & -\dfrac{1}{I_m}\end{bmatrix}\begin{bmatrix}V_a\\ \tau_a\end{bmatrix}. \tag{5.10}$$

5.2 TWO WHEELED INVERTED PENDULUM DYNAMIC MODEL

The TWIP shares many similarities with the pendulum-on-a-cart problem. The motion is comparable but the dynamics are more involved. The analysis begins with the two wheels, the pendulum is analyzed separately, and the two sets of equations are combined by eliminating the interface reactions, yielding two dynamic equations describing the motion of the system.

5.2.1 Wheel Dynamics

The free-body diagram of the left wheel is shown in Figure 5.1. The reaction forces between the wheel and the pendulum are P_L and H_L, the torque applied to the wheel by the DC motor is C_L, and the friction force between the ground and the wheel is H_{fL}. The equations for the left and right wheels are completely analogous, so only the left wheel is analyzed. The wheels are assumed to remain in ground contact at all times with no slip.

Summing forces in the x direction,

$$M_w\ddot{x} = H_{fL} - H_L. \tag{5.11}$$

Summing moments about the center of the wheel,

$$I_w\ddot{\theta}_w = C_L - H_{fL}r, \tag{5.12}$$

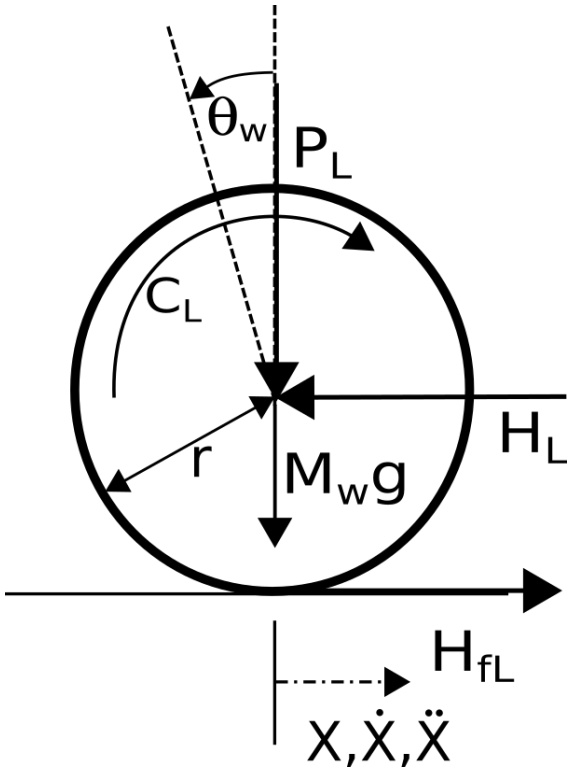


Figure 5.1: Left wheel free body diagram.

where I_w is the wheel moment of inertia and $\ddot{\theta}_w$ the wheel angular acceleration. The wheel angle θ_w and the motor torque C_L are both taken clockwise, the sense in which a wheel turns when the robot rolls in the positive x direction; equations (5.12), (5.17) and the back-EMF term of (5.13) require this. (The θ_w arrow in Figure 5.1 is drawn in the opposite sense.) The motor stator is fixed to the pendulum body, so the shaft rate ω in (5.7) is the rate of the wheel relative to the body. With the pendulum angle θ of the next section taken counterclockwise, that relative rate is $\dot{\theta}_w + \dot{\theta}$. The armature inertia is neglected along with L and k_f, so that in (5.5) the torque delivered to the wheel equals the electromagnetic torque τ_m. Both k_m and k_e are referred to the gearbox output shaft, with the losses of the gear train lumped into k_m (Section 4.2.2). Substituting (5.7), and writing V for the terminal voltage V_a, the motor torque delivered to either wheel is

$$C_L = C_R = \tau_m = -\frac{k_m k_e}{R}\left(\dot{\theta}_w + \dot{\theta}\right) + \frac{k_m}{R}V. \tag{5.13}$$

Using (5.13) in (5.12) and rearranging gives the friction reaction force,

$$H_{fL} = -\frac{k_m k_e}{Rr}\big(\dot{\theta}_w + \dot{\theta}\big) + \frac{k_m}{Rr}V - \frac{I_w}{r}\ddot{\theta}_w. \tag{5.14}$$

Substituting (5.14) into (5.11) gives the equation of motion for the left wheel,

$$M_w\ddot{x} = -\frac{k_m k_e}{Rr}\big(\dot{\theta}_w + \dot{\theta}\big) + \frac{k_m}{Rr}V - \frac{I_w}{r}\ddot{\theta}_w - H_L, \tag{5.15}$$

and an identical development for the right wheel gives

$$M_w\ddot{x} = -\frac{k_m k_e}{Rr}\big(\dot{\theta}_w + \dot{\theta}\big) + \frac{k_m}{Rr}V - \frac{I_w}{r}\ddot{\theta}_w - H_R. \tag{5.16}$$

The no-slip assumption provides the transformation between rotational and linear motion,

$$\ddot{\theta}_w = \frac{\ddot{x}}{r}, \qquad \dot{\theta}_w = \frac{\dot{x}}{r}. \tag{5.17}$$

Applying (5.17) to (5.15) and (5.16),

$$M_w\ddot{x} = -\frac{k_m k_e}{Rr}\Big(\frac{\dot{x}}{r} + \dot{\theta}\Big) + \frac{k_m}{Rr}V - \frac{I_w}{r^2}\ddot{x} - H_L, \tag{5.18}$$

$$M_w\ddot{x} = -\frac{k_m k_e}{Rr}\Big(\frac{\dot{x}}{r} + \dot{\theta}\Big) + \frac{k_m}{Rr}V - \frac{I_w}{r^2}\ddot{x} - H_R. \tag{5.19}$$

Adding (5.18) and (5.19) gives the equation of motion for the combined wheel system,

$$2\left(M_w + \frac{I_w}{r^2}\right)\ddot{x} = -\frac{2k_m k_e}{Rr}\Big(\frac{\dot{x}}{r} + \dot{\theta}\Big) + \frac{2k_m}{Rr}V - \big(H_L + H_R\big). \tag{5.20}$$

5.2.2 Pendulum Dynamics

Figure 5.2 shows the free-body diagram of the inverted pendulum. Two reference frames are used: an inertial frame attached to the base of the pendulum, and a translating frame at the center of gravity. Here l is the distance from the base to the center of gravity; α_p, ω_p, and θ are the angular acceleration, angular velocity, and angular position of the pendulum; $x, \dot{x}, \ddot{x}$ are the position, velocity, and acceleration of the pendulum base; and $x_p, \dot{x}_p, \ddot{x}_p$ those of the center of gravity.

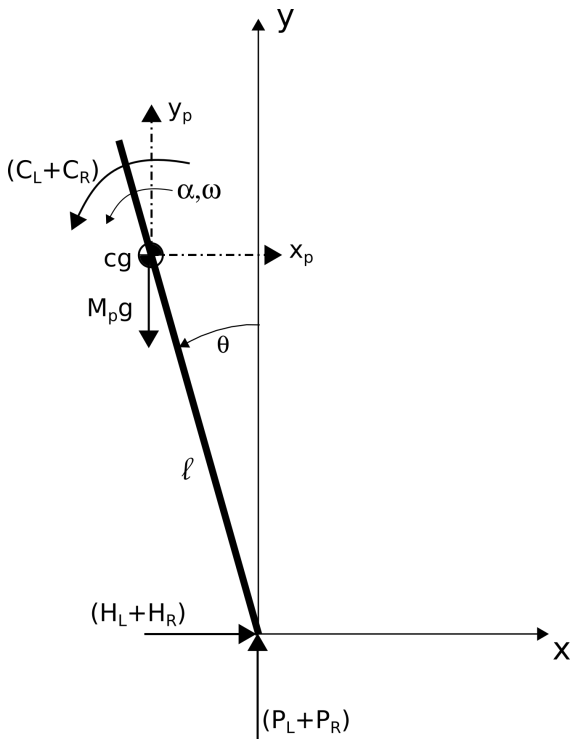


Figure 5.2: Inverted pendulum free body diagram.

Summing forces in the x_p direction,

$$\sum F_{xp} = H_L + H_R = M_p \ddot{x}_p, \tag{5.21}$$

and in the y_p direction,

$$\sum F_{yp} = P_L + P_R - M_p g = M_p \ddot{y}_p. \tag{5.22}$$

The acceleration of a point in a translating frame follows from relative motion analysis [54]. The acceleration of the pendulum center of gravity, denoted $\mathbf{a}_B$, is

$$\mathbf{a}_B = \mathbf{a}_A + \boldsymbol{\alpha}_p \times \mathbf{r}_{B/A} + \boldsymbol{\omega}_p \times \big(\boldsymbol{\omega}_p \times \mathbf{r}_{B/A}\big), \tag{5.23}$$

with

$$\mathbf{a}_B = \ddot{x}_p \hat{\imath} + \ddot{y}_p \hat{\jmath}, \tag{5.24}$$

$$\mathbf{r}_{B/A} = l \sin\theta\, \hat{\imath} - l \cos\theta\, \hat{\jmath}, \tag{5.25}$$

$$\boldsymbol{\omega}_p = \dot{\theta}\, \hat{k}, \tag{5.26}$$

$$\boldsymbol{\alpha}_p = \ddot{\theta}\, \hat{k}, \tag{5.27}$$

$$\mathbf{a}_A = \ddot{x}\, \hat{\imath}. \tag{5.28}$$

Using (5.25)–(5.28) in (5.23),

$$\mathbf{a}_B = \Big(\ddot{x} + \ddot{\theta} l \cos\theta - \dot{\theta}^2 l \sin\theta\Big)\hat{\imath} + \Big(\ddot{\theta} l \sin\theta + \dot{\theta}^2 l \cos\theta\Big)\hat{\jmath}. \tag{5.29}$$

Substituting (5.29) into (5.21) and (5.22),

$$H_L + H_R = M_p\Big(\ddot{x} + \ddot{\theta} l \cos\theta - \dot{\theta}^2 l \sin\theta\Big), \tag{5.30}$$

$$P_L + P_R = M_p g + M_p\Big(\ddot{\theta} l \sin\theta + \dot{\theta}^2 l \cos\theta\Big). \tag{5.31}$$

Summing moments about the pendulum center of gravity, with the counterclockwise direction (the sense of θ, since $\boldsymbol{\omega}_p = \dot{\theta}\, \hat{k}$) taken as positive,

$$\sum M = \big(C_L + C_R\big) - \big(H_L + H_R\big) l \cos\theta - \big(P_L + P_R\big) l \sin\theta = I_p \ddot{\theta}. \tag{5.32}$$

The motors are mounted on the pendulum, so the torque $C_L + C_R$ that turns the wheels clockwise reacts on the pendulum counterclockwise, with positive sign. Remark 5.2 gives an independent check of this sign, and Section 5.3 tests it against the robot's recorded motion.

5.2.3 Combined Equations of Motion

Combining (5.13), (5.30), (5.31), and (5.32),

$$\begin{aligned}\left(-\frac{2k_m k_e}{R}\Big(\frac{\dot{x}}{r}+\dot{\theta}\Big)+\frac{2k_m}{R}V\right)-\Big(M_p\big(\ddot{x}+\ddot{\theta}l\cos\theta-\dot{\theta}^2 l\sin\theta\big)\Big)l\cos\theta \\ -\Big(M_p g+M_p\big(\ddot{\theta}l\sin\theta+\dot{\theta}^2 l\cos\theta\big)\Big)l\sin\theta=I_p\ddot{\theta}.\end{aligned} \tag{5.33}$$

Cancelling terms, rearranging, and applying the identity $\sin^2\theta+\cos^2\theta=1$ gives the first nonlinear equation of motion,

$$\big(I_p+M_p l^2\big)\ddot{\theta}+\frac{2k_m k_e}{R}\Big(\frac{\dot{x}}{r}+\dot{\theta}\Big)-\frac{2k_m}{R}V+M_p g l\sin\theta=-M_p l\ddot{x}\cos\theta. \tag{5.34}$$

Using (5.30) to remove the body forces from (5.20) gives the second nonlinear equation of motion,

$$\frac{2k_m}{Rr}V=\left(2M_w+\frac{2I_w}{r^2}+M_p\right)\ddot{x}+\frac{2k_m k_e}{Rr}\Big(\frac{\dot{x}}{r}+\dot{\theta}\Big)+M_p l\ddot{\theta}\cos\theta-M_p l\dot{\theta}^2\sin\theta. \tag{5.35}$$

Equations (5.34) and (5.35) describe the full nonlinear motion of the two-wheeled inverted pendulum. The motor torque acts between the body and the wheels through the relative angle $x/r+\theta$: it enters the wheel equation (5.35) as T/r and the pendulum equation (5.34) as $+T$, with $T=(2k_m/R)\big(V-k_e(\dot{x}/r+\dot{\theta})\big)$ the total motor torque, which is the virtual work of T acting through that angle. At zero voltage the power delivered by the motors is $-(2k_m k_e/R)(\dot{x}/r+\dot{\theta})^2\le 0$, so the back-EMF is strictly dissipative.

5.2.4 Linearization

In the convention of (5.25), $\theta = \pi$ is the upright vertical configuration (the angle drawn in Figure 5.2 is measured from the upward vertical, so it equals ϕ below). A set of linearized equations is obtained by writing $\theta = \pi + \phi$, where ϕ is a small angle from the vertical. The small-angle assumptions are

$$\cos\theta = -1, \qquad \sin\theta = -\phi, \tag{5.36}$$

together with the assumption that the squared angular rate is small,

$$\dot{\theta}^2 = 0. \tag{5.37}$$

Applying (5.36) and (5.37) to (5.34) and (5.35) yields the linearized equations of motion,

$$\left(I_p + M_p l^2\right)\ddot{\phi} + \frac{2k_m k_e}{R}\left(\frac{\dot{x}}{r} + \dot{\phi}\right) - \frac{2k_m}{R}V - M_p g l \phi = M_p l \ddot{x}, \tag{5.38}$$

$$\frac{2k_m}{Rr}V = \left(2M_w + \frac{2I_w}{r^2} + M_p\right)\ddot{x} + \frac{2k_m k_e}{Rr}\left(\frac{\dot{x}}{r} + \dot{\phi}\right) - M_p l \ddot{\phi}. \tag{5.39}$$

Remark 5.1 (Sign of the gravity term). Rearranging (5.38) gives $\left(I_p + M_p l^2\right)\ddot{\phi} = M_p g l \phi + \cdots$, so the gravitational term enters with positive sign and is destabilizing. This is the expected signature of an upright linearization and confirms the convention: ϕ is the displacement from the *unstable* equilibrium, not from the hanging one. It is also consistent with (5.40), whose $(2,3)$ and $(4,3)$ entries are both positive. Section 5.3.2 checks both equilibria of the implemented equations.

Rearranging (5.38) and (5.39) into the state-space form $\dot{\mathbf{x}} = \mathcal{A}\mathbf{x} + \mathcal{B}V$, with state $\mathbf{x} =$

$[\,x\ \ \dot{x}\ \ \phi\ \ \dot{\phi}\,]^T$, gives

$$\mathcal{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\dfrac{2k_m k_e\big(I_p + M_p l^2 + M_p l r\big)}{Rr^2\alpha} & \dfrac{M_p^2 g l^2}{\alpha} & -\dfrac{2k_m k_e\big(I_p + M_p l^2 + M_p l r\big)}{Rr\alpha} \\ 0 & 0 & 0 & 1 \\ 0 & -\dfrac{2k_m k_e\big(r\beta + M_p l\big)}{Rr^2\alpha} & \dfrac{M_p g l \beta}{\alpha} & -\dfrac{2k_m k_e\big(r\beta + M_p l\big)}{Rr\alpha} \end{bmatrix}, \tag{5.40}$$

$$\mathcal{B} = \begin{bmatrix} 0 & \dfrac{2k_m\big(I_p + M_p l^2 + M_p l r\big)}{Rr\alpha} & 0 & \dfrac{2k_m\big(r\beta + M_p l\big)}{Rr\alpha} \end{bmatrix}^T,$$

where

$$\beta := 2M_w + \frac{2I_w}{r^2} + M_p \tag{5.41}$$

and

$$\alpha := I_p\beta + 2M_p l^2\left(M_w + \frac{I_w}{r^2}\right). \tag{5.42}$$

The tilt-rate entries $\mathcal{A}_{24} = r\mathcal{A}_{22}$ and $\mathcal{A}_{44} = r\mathcal{A}_{42}$ come from the relative rate in (5.13); they are what make the motor dissipative at zero voltage, and they are small (Table 7.1).

Remark 5.2 (Check on the input direction). The motor torque acts between the pendulum and the wheels, so it cannot change the angular momentum of the robot about the wheel contact point; only gravity, which has no moment about that point when the robot is upright, and the motion of the point itself can. At the upright equilibrium with zero rates, the acceleration produced by the voltage alone must therefore leave that angular momentum unchanged. Measured clockwise, the angular momentum about the contact point is $(r\beta + M_p l)\dot{x} - (I_p + M_p l^2 + M_p l r)\dot{\phi}$ at $\phi = 0$, and its rate of change is zero only if

$$\frac{\mathcal{B}_2}{\mathcal{B}_4} = \frac{I_p + M_p l^2 + M_p l r}{r\beta + M_p l}. \tag{5.43}$$

The input vector of (5.40) satisfies (5.43), and the same ratio relates $\mathcal{A}_{22}$ to $\mathcal{A}_{42}$ and $\mathcal{A}_{24}$ to

$\mathcal{A}_{44}$, since the back-EMF torque is internal too. Both entries of $\mathcal{B}$ are positive: a positive voltage pushes the base forward and turns the body backward. Had the reaction torque been entered in (5.32) with the opposite sign, the factors would read $I_p + M_p l^2 - M_p l r$ and $M_p l - r\beta$, and the check would fail.

For the development in Chapter 6 it is convenient to name the entries of (5.40) directly. Writing $x_1 = x$, $x_2 = \dot{x}$, $x_3 = \phi$, $x_4 = \dot{\phi}$, $u = V$, and appending the unmodeled and parametric uncertainty as d_1 and d_2,

$$\begin{aligned} \dot{x}_1 &= x_2, \\ \dot{x}_2 &= A_1 x_2 + A_2 x_3 + A_5 x_4 + B_1 u + d_1(\mathbf{x}), \\ \dot{x}_3 &= x_4, \\ \dot{x}_4 &= A_3 x_2 + A_4 x_3 + A_6 x_4 + B_2 u + d_2(\mathbf{x}), \end{aligned} \tag{5.44}$$

with the coefficients read off from (5.40): $A_1 = \mathcal{A}_{22}$, $A_2 = \mathcal{A}_{23}$, $A_5 = \mathcal{A}_{24}$, $A_3 = \mathcal{A}_{42}$, $A_4 = \mathcal{A}_{43}$, $A_6 = \mathcal{A}_{44}$, $B_1 = \mathcal{B}_2$, $B_2 = \mathcal{B}_4$. The designs of Chapter 6 carry the known coefficients as such, and the uncertainties d_1 and d_2 absorb whatever the nominal model leaves out.

Remark 5.3 (Structural consequences). Two features of (5.44) drive everything that follows. First, B_1 and B_2 are both nonzero and the input is scalar, so the system is underactuated and no choice of u cancels both d_1 and d_2: the uncertainty is unmatched. Second, the tilt drives the translational motion. With the input eliminated between the two acceleration equations, the tilt enters with the gain $a_2 = A_2 - (B_1/B_2)A_4 = -M_p g l/(r\beta + M_p l)$, which is nonzero for any $l > 0$ (Section 6.4). This is what makes x_3 usable as a virtual control for the position subsystem, and it is also the term that couples the two error subsystems in the backstepping design. It must be retained; see Section 6.5.

5.3 MODEL VALIDATION

The model of Section 5.2 is used three times in this thesis: it sets the controller gains of Chapter 6, it is the prediction model F, G inside every observer, and it is the truth model of Chapter 8. A simulation shows that the equations are implemented as written, not that they describe the robot, so this section tests the model against the robot's recorded data. The feature of the derivation that the data can decide is the sign with which the motor torque reacts on the body in (5.32), so the model is tested alongside two alternatives that carry the opposite sign. Each test is stated with the value every candidate model predicts. A candidate is called contradicted by a test when its prediction misses the measurement by several times the error of the best candidate, or, for a fitted coefficient, lies well outside the bootstrap interval of the measurement; by the second rule some measured coefficients contradict the model of this chapter as well, and they are reported.

5.3.1 Candidate Models

Three candidates span the questions the derivation leaves open: the sign of the motor reaction on the body, the motor constants, and the mass properties. They share the mass matrix and differ in how the motor couples the wheels to the body.

M1. The model of this chapter: reaction $+(C_L + C_R)$ on the body, as in (5.32); back-EMF on the wheel rate relative to the body, (5.13); parameters of Chapter 4. This is the model behind the results of Chapter 8.

M2 (reversed sign). Reaction $-(C_L + C_R)$, with the back-EMF on $\dot{x}/r - \dot{\phi}$, the relative rate that makes the reversed reaction energy-consistent; parameters of Chapter 4. It differs from M1 only in the sign of the reaction.

M3 (firmware model). The model programmed into the robot's on-board estimator for the recorded runs (Section 5.3.3): reaction $-(C_L + C_R)$, back-EMF on the absolute wheel

rate $\dot{x}/r$, $k_e = 0.003\,61\,\mathrm{V\,s/rad}$, and $I_p = 0.025\,\mathrm{kg\,m^2}$, the inertia about the wheel axis used as the inertia about the center of gravity.

Table 5.1 lists the predictions used below. Both input gains are positive in every candidate at the parameters used here, so a positive voltage pushes the base forward and turns the body backward in all three. (For M2 and M3, $\mathcal{B}_4$ changes sign below $l \approx 0.06\,\mathrm{m}$, inside the range searched in Section 5.3.6; the M2 plants there fall with either sign of the command.) The sign question therefore does not show up as a reversed response. It shows up in the magnitudes: per volt, M1 turns the body seven to ten times faster than M2 and M3, and it moves the base much less per unit of body rotation.

Table 5.1: Predictions of the three candidate models (linearized about upright)

Quantity	M1	M2	M3
$\mathcal{B}_2$ ($\mathrm{m/s^2}$ per V)	1.855	1.253	1.222
$\mathcal{B}_4$ ($\mathrm{rad/s^2}$ per V)	11.67	1.692	1.179
$\mathcal{B}_2/\mathcal{B}_4$ (m/rad)	0.159	0.741	1.037
Unstable pole (rad/s)	5.95	7.50	6.30
c_1 of (5.47) (m/rad)	0.204	0.786	1.082
c_2 of (5.47) ($\mathrm{m\,rad^{-1}\,s^{-2}}$)	−5.62	−38.7	−38.7

5.3.2 Simulation Checks

The uncontrolled nonlinear equations (5.34)–(5.35) are integrated from two initial conditions with the motor voltage held at zero. In the convention of (5.36), $\theta = \pi$ is the upright configuration and $\theta = 0$ the hanging one. Started at $\theta = 180°$ with zero rate, the system remains there (Figure 5.3). This confirms that the upright configuration is an equilibrium of the implemented equations; it is not evidence of stability, and as Remark 5.1 establishes, a perturbation of 0.1° grows to 10° in 0.88 s.

Started at $\theta = 10°$, the system oscillates about the hanging equilibrium (Figure 5.4) with a period of 0.84 s. The swing is lightly damped, falling to about 7° in 10 s, because the

back-EMF damps only the rotation of the wheels relative to the body, and in a free swing the wheels roll nearly in step with it. The base moves initially in the positive direction and oscillates between -1 mm and 17 mm.

These checks would catch a sign or transcription error in the implemented equations, but they are not validation. Table 5.2 shows that all three candidates pass them, so they cannot decide between them.

Table 5.2: Simulation checks for the three candidates (zero voltage)

	M1	M2	M3
Equilibrium at $\theta = \pi$	yes	yes	yes
$0.1°$ to $10°$ from $\theta = \pi$ (s)	0.88	0.71	0.84
Initial base acceleration from $10°$ (m/s^2)	+0.58	+0.58	+0.41
Swing period about $\theta = 0$ (s)	0.840	1.144	1.001

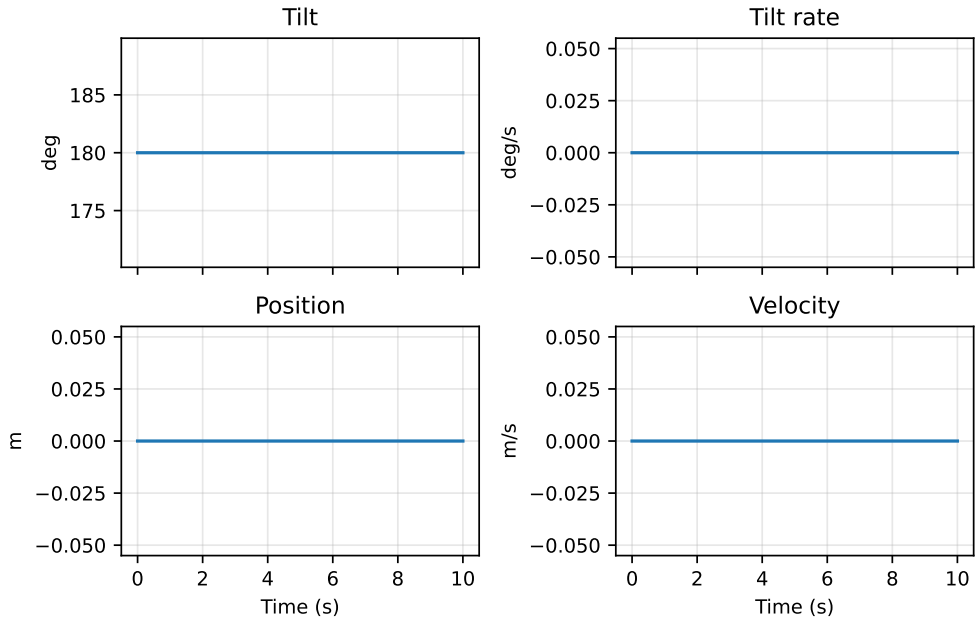


Figure 5.3: Uncontrolled response from $\theta = 180°$ (model M1).

5.3.3 Recorded Data and Sign Conventions

Four logs of the robot balancing under LQR control are analysed here:

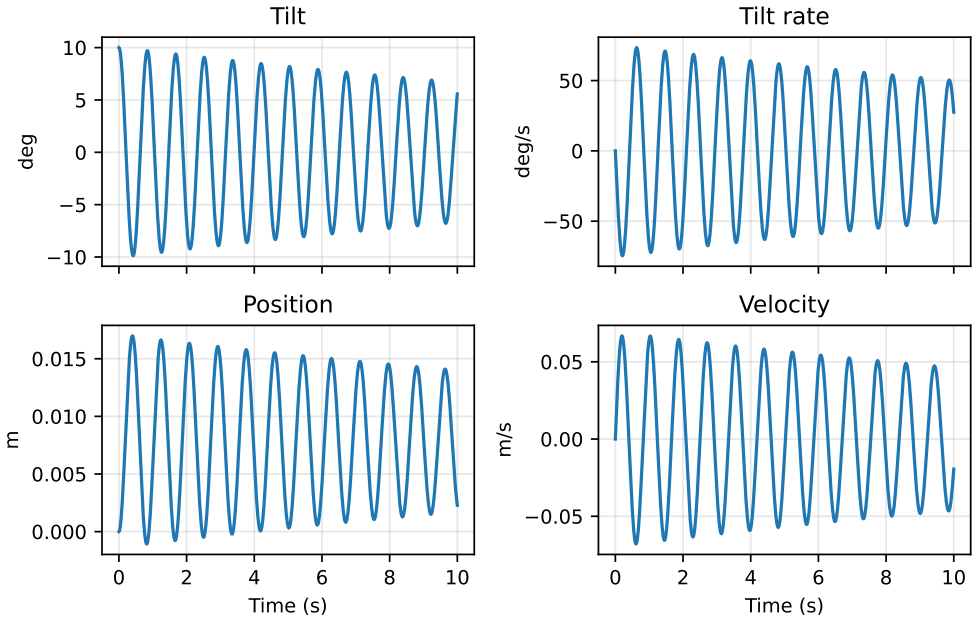


Figure 5.4: Uncontrolled response from $\theta = 10°$ (model M1).

- `implementationtest1.txt` and `implementationtest2.txt`: firmware v8, 13 ms loop, 3.0 s and 3.9 s long, with the potentiometer gains printed;
- `implementationtest4.txt` and `implementationtest5.txt`: firmware v9, 10 ms loop, 12.6 s and 21.2 s long, gains not printed.

They are called tests 1, 2, 4 and 5 below. Each row holds the raw gyroscope rate, the raw accelerometer components, the encoder position p and its backward difference, the filtered tilt, the on-board estimates, and the motor command in PWM counts (5904.5 counts per volt). Other balancing logs exist and were not analysed: 26 logs of firmware v7 in `Tests/LQR Tests` (February 2015, on the present gearmotors and encoders, most with tilt, position, accelerometer, gyroscope and command) and `newmotortest1.txt`. They are a replication set for the tests below. No free-swing, mass property or locked-rotor measurement exists; the motor sweeps in `Tests/Motor Tests` were recorded on the gearmotors that firmware v7 replaced. The analysis is the script `scripts/validate_plant.py` of the accompanying code, which writes its results to `generated/plant_validation.json`.

Three rules apply throughout. The on-board estimates are not used as measurements, because they were computed with the matrices F and G of M3; they are used only to recover the gains. The tilt is the integrated gyroscope, started from the filtered tilt, because the accelerometer and the complementary filter read tilt relative to the apparent vertical, which is displaced by about $\ddot{x}/g$ whenever the robot accelerates. And the inertial signals are advanced by 7 ms, the estimated lag of the IMU behind the encoders (the 4.8 ms group delay of its 42 Hz low-pass filter plus sampling), in the v8 runs as well. Without this correction the ranking of the candidates below is unchanged, but the fitted encoder coefficient c_1 of tests 4 and 5 falls from 0.205 and 0.203 to 0.15 and 0.14.

Sign conventions. Two relative signs must be fixed before the logs can be compared with a model. The first is that of the command relative to the tilt. The gains recovered in Section 5.3.6 feed the logged tilt back with a positive gain (+118 in v9, +27 to +73 in v8), which stabilizes the body only if the logged command and tilt have the signs of V and ϕ, because $\mathcal{B}_4 > 0$ in every candidate. The second is that of the encoder relative to the tilt. Each v9 run starts with the robot leaning (24.8° in test 4, 15.7° in test 5) and the command saturated at −10 V. Within 100 ms the gyroscope shows the body turning back toward upright at 178 °/s to 193 °/s while the logged position rises by 51 mm, whereas every candidate, driven by the logged −10 V, predicts $p = x + r\phi$ to fall by 33 to 58 mm. The same holds from 0.1 s to 0.8 s, after any contact with the support has ended: since $c_1 > 0$ in every candidate, (5.47) below requires p to fall with the tilt, and the encoder fit there gives $c_1 = +0.20$ only if the logged position is mirrored. So the logged position is mirrored: it equals $-p$. The encoder tests below use this same transient, so their positive c_1 is not an independent check of the mirror. The accelerometer is, because its sign is fixed by gravity through the tilt: in this frame it gives an axle coefficient of the same sign as the encoder (0.19 and 0.24, against 0.16 and 0.16 from the encoder). The mirror is inferred from these two v9 transients and assumed for v8. The closed loop of Section 5.3.6 does not decide it,

since the model holds the base in either frame given enough rolling resistance, and the hand check of Table 5.7 settles it.

5.3.4 Input-Free Angular Momentum Test

Multiplying (5.39) by r and adding it to (5.38) eliminates the voltage and every back-EMF term:

$$\big(r\beta + M_p l\big)\ddot{x} = \big(I_p + M_p l^2 + M_p l r\big)\ddot{\phi} - M_p g l \phi. \tag{5.45}$$

This is the angular momentum balance about the wheel contact point of Remark 5.2: the motor torque acts between body and wheels and drops out, and only gravity remains. With the reversed sign of the reaction the same elimination gives instead

$$\big(M_p l - r\beta\big)\ddot{x} = \big(I_p + M_p l^2 - M_p l r\big)\ddot{\phi} - M_p g l \phi. \tag{5.46}$$

In terms of the encoder reading $p = x + r\phi$, (5.45) becomes

$$\ddot{p} = c_1\ddot{\phi} + c_2\phi, \qquad c_1 = \frac{I_p + M_p l^2 + M_p l r}{r\beta + M_p l} + r, \qquad c_2 = -\frac{M_p g l}{r\beta + M_p l}, \tag{5.47}$$

and (5.46) gives the same form with the coefficients of M2 and M3 in Table 5.1. The axle acceleration $\ddot{x}$ satisfies the same relation with $c_1 - r$ in place of c_1, and it can be measured without the encoder by the accelerometer, corrected for gravity and for its height above the axle.

The test needs neither the voltage nor k_m, k_e or R, so the battery level, PWM scaling, deadzone and motor constants do not enter. It separates the sign question from the motor-constant question. Because it holds whatever the motor does, it can be applied to any stretch of data, but it identifies c_1 and c_2 only where the motion is large compared with the gear backlash, which sits between the encoder and the wheel. It also requires that no external

force act on the robot other than gravity and the support of the floor at the contact point: contact with a hand or stand at switch-on, and rolling resistance of the floor, break it.

Switch-on transients. Between 0.1 s and 0.8 s after switch-on the motions are the largest in the logs: the encoder travels 39 mm and the tilt spans 11° in test 4, and 96 mm and 3.9° in test 5. The first 0.1 s is left out because the robot may still be touching the hand or stand that held it at its initial lean. For each candidate, (5.47) turns the measured tilt history into a predicted encoder trajectory and a predicted axle trajectory. These are compared with the logged encoder and with the doubly integrated accelerometer, with the initial position and velocity and a constant offset fitted per window. Figure 5.5 shows the result for the v9 runs, and Table 5.3 lists all four runs. M1 has the smallest error in every run and on both sensors, by a factor of 2.5 to 10 against M2 and 3 to 16 against M3. Only tests 4 and 5 are diagnostic on the encoder, however: in tests 1 and 2 the encoder moves only 5.7 mm and 8.2 mm, and there stick-slip in the gearbox can let the candidate with the smallest predicted coupling win by default. The accelerometer is not behind the gearbox.

Fitting c_1 freely, instead of taking it from a candidate, gives $c_1 = 0.205$ (bootstrap interval 0.12 to 0.25) in test 4 and 0.203 (0.15 to 0.26) in test 5, against 0.204 for M1 and 0.79 or 1.08 for the reversed-sign models. Without the 7 ms IMU correction these become 0.15 and 0.14, so the close agreement with M1 depends on the correction, while the separation from M2 and M3 does not. The accelerometer, which the correction does not affect, gives the axle coefficient $c_1 - r = 0.189$ (0.169 to 0.210) and 0.242 (0.208 to 0.274) in tests 4 and 5 (M1: 0.159; M2: 0.741; M3: 1.037). These values are far from the reversed-sign models, but they are 20 to 50% above M1 with intervals that exclude it, and in test 5 the value lies above the whole encoder interval ($c_1 - r = 0.158$, 0.101 to 0.211). Within M1 the axle coefficient depends mainly on I_p, and the test 4 value corresponds to $I_p \approx 0.023\,\mathrm{kg\,m^2}$ at the nominal l, against the $0.0169\,\mathrm{kg\,m^2}$ of Table 4.1. The test 5 value is not converted, because the same fit gives a gravity coefficient of +43 (28 to 59), which no candidate allows.

A rolling resistance of the floor would not explain the excess: in simulation it biases the coefficient low (Section 5.3.6). In tests 1 and 2 the fits are mixed: the test 1 accelerometer gives $c_1 - r = 0.18$ (0.13 to 0.22) and the test 2 encoder $c_1 = 0.19$ (0.13 to 0.25), both consistent with M1 and far from M2 and M3, while the test 1 encoder gives $c_1 = 0.09$ (-0.01 to 0.20) and the test 2 accelerometer $c_1 - r = -0.10$ (-0.20 to 0.01), a sign no candidate allows. The test 1 accelerometer is the only accelerometer interval that contains M1, and it comes from a third event. The gravity coefficient c_2 is not identified in the v9 encoder fits, and where a free fit does identify it (the test 5 accelerometer, the test 1 accelerometer, the test 2 encoder) it is positive, the wrong sign for every candidate.

Table 5.3: Input-free test on the switch-on transients (0.1 s to 0.8 s): RMS error of each candidate's predicted trajectory, mm

	Encoder			Accelerometer		
Run	M1	M2	M3	M1	M2	M3
Test 4	1.8	12.3	18.9	1.1	11.1	17.7
Test 5	1.3	5.8	8.5	1.4	5.4	8.0
Test 1	0.4	2.4	2.9	0.3	2.3	2.8
Test 2	0.5	1.7	2.1	0.8	2.0	2.5

Steady balancing. After the first second the robot is in a stick-slip limit cycle: the encoder stays frozen for several tenths of a second while the body sways, then jumps by a few counts. During those intervals p does not follow the wheel, and (5.47) fails for every candidate (R^2 of 0.05 to 0.11 for the filtered encoder fit). The accelerometer, which sees the axle directly, gives a $\ddot{\phi}$ coefficient of 0.173 (0.158 to 0.188) and 0.158 (0.145 to 0.172) in tests 4 and 5 at 8 Hz (predicted $c_1 - r$ less the IMU height: 0.139 for M1, 0.721 for M2, 1.017 for M3), closest to M1 but excluding it. Its ϕ coefficient, 14.1 in both runs (11.6 to 19.6 and 11.6 to 17.8), contradicts all three, which predict $c_2 + g = 4.2$ for M1 and -28.9 for M2 and M3. Gear stiction cannot cause this, because it acts between body and wheels and drops out of (5.45) like the motor torque. An external force is needed, or an error in

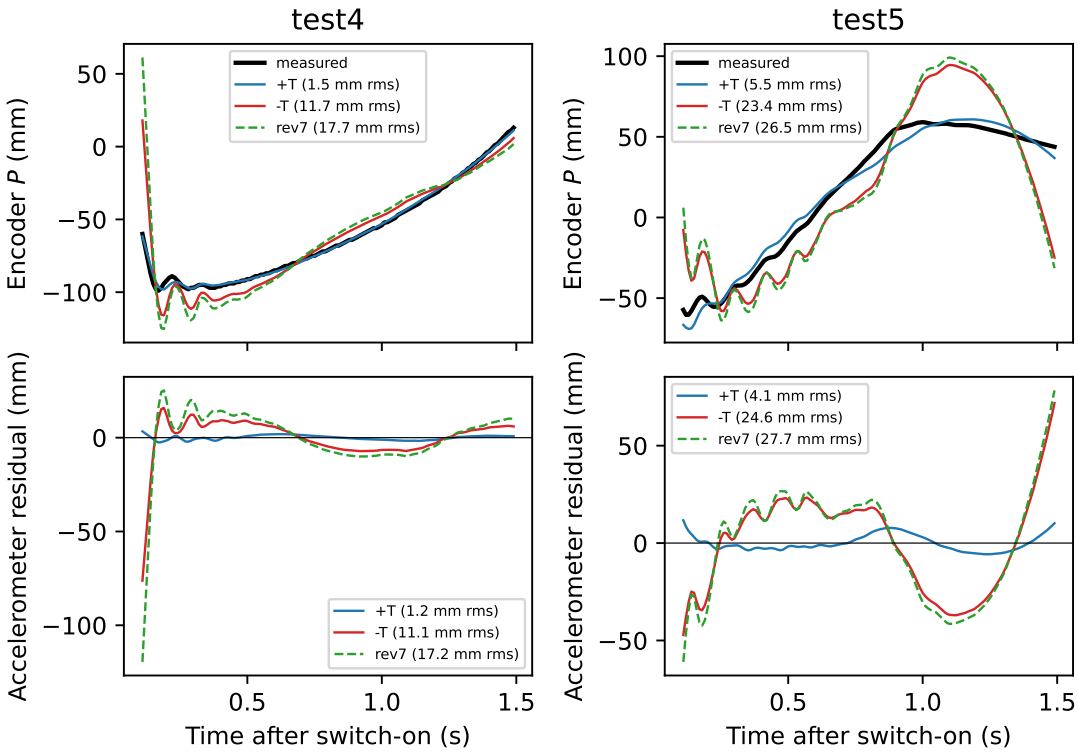


Figure 5.5: Input-free test on the switch-on transients of tests 4 and 5, plotted over 0.1 s to 1.5 s (Table 5.3 uses 0.1 s to 0.8 s; over the longer window the errors of M1 stay near 1.5 and 1.2 mm in test 4 but grow to 5.5 and 4.1 mm in test 5). Top: the logged encoder position and each candidate's prediction from the measured tilt history. Bottom: residual of the doubly integrated accelerometer against each candidate. In the legend, $+T$ is M1, $-T$ is M2 and rev7 is M3.

the integrated tilt or in l. Rolling resistance of the floor is a candidate, but it has not been shown to fit: in the simulated v9 loop, the 6 to 10% of the weight that holds the base in the mirrored frame (Section 5.3.6) gives a ϕ coefficient of 5 to 9 and lowers the $\ddot{\phi}$ coefficient to 0.05 to 0.08, against 14.1 and 0.16 to 0.17 measured. With the encoder not mirrored, 2 to 6% gives a ϕ coefficient of 9 to 13, inside the measured intervals at 6%, but a $\ddot{\phi}$ coefficient of only 0.05 to 0.09. The frequency response from the gyroscope to the encoder rate does not discriminate either, because a simulated M2 plant with backlash reproduces the low values measured.

Validation of the method. Every estimator above was first run on eight simulated closed-loop runs: M1 and M2 plants, clean and with gearbox friction, 2° of backlash and a rolling resistance of 2% of the weight, two noise seeds each, passed through the robot's sensor models and loop. The M2 runs use the gyroscope alone ($\lambda_c = 1$), since with the robot's filter they fall, and three M2 runs with friction that fell were replaced by the next seed. On 0.1 s to 0.8 s the switch-on test ranked the true plant first in all 16 run and sensor cases, and the free fit recovered c_1 between 0.200 and 0.204 for M1 and between 0.70 and 0.79 for M2. Figure 5.6 collects the simulated and measured coefficients. The check has limits. The simulated plants obey the input-free relation by construction apart from the rolling resistance, they have no IMU delay, and they use the same IMU height as the analysis. The M2 runs move 300 to 760 mm in the window, against 39 to 96 mm on the robot. Over the longer window 0.1 s to 1.5 s the free fit returned $c_1 = -0.36$ and 0.27 for the M2 plants with friction, far from their true 0.79, although the ranking was still right, and on 0.1 s to 0.8 s the accelerometer fit is 3 to 12% low in seven of the eight simulated runs (and unbiased in the eighth). The bootstrap intervals quoted here are therefore not calibrated 95% intervals. With a rolling resistance of 6% or 10% of the weight (simulated by the script of Section 5.3.6), M1 plants are still ranked first, with the accelerometer coefficient 6 to 8% low at 6% and 25 to 52% low at 10%; M2 plants with that much resistance do not balance at all. The stick-slip of the robot is stronger

than the simulated backlash and friction produce, which is why the steady-state encoder fit works in simulation and not on the logs.

5.3.5 Response to the Voltage

The switch-on transient also tests the input gain, because for at least the first 60 ms the command is saturated at a known −10 V and the motion is large. It is a consistency check rather than a clean test: the robot may still have been touching the hand or stand that held it, and the models start from rest at the logged lean. Each candidate is simulated open loop with the logged voltages (Figure 5.7). The cleanest measured quantity is the body rate from the gyroscope, because the on-board filtered tilt carries part of the accelerometer's $\ddot{x}/g$ error during these accelerations: at 50 ms the filtered tilt has fallen by 5.5° and 5.4°, while the gyroscope rate, −98 °/s and −110 °/s, implies about 2.5°.

At 50 ms the body is therefore already turning back toward upright. M1 predicts −139 °/s and −168 °/s. M2 predicts 44 °/s and 13 °/s, and M3 28 °/s and 5 °/s: with their tilt input gain, 10 V does not overcome gravity at these angles, and the body is still falling further into its lean. With the command reversed they fall faster still. At 100 ms the measured rate is −193 °/s and −178 °/s, against −206 °/s and −130 °/s for M1, while M2 and M3 are still turning the wrong way. The direction excludes the reversed-sign models only for k_m near its data-sheet value, however. At nominal I_p and l, M3 turns the body back within 50 ms from 2.5 times that k_m in test 4 but from 1.2 times in test 5, inside the 35% spread of the data sheets, and M2 from 7.5 and 1.6 times. Reaching the measured rate needs at least 5.4 times for M3 and is beyond M2 up to 12 times.

The size of the response fixes the tilt input gain of M1 only loosely. Scaling k_m to fit the filtered tilt over the first 100 ms needs a factor of 1.26 and 1.08, but fitting the gyroscope rate at 50 ms needs 0.73 and 0.64. The switch-on response therefore bounds the tilt input gain of M1 only to within about 40%, in either direction, and contact with the support would slow

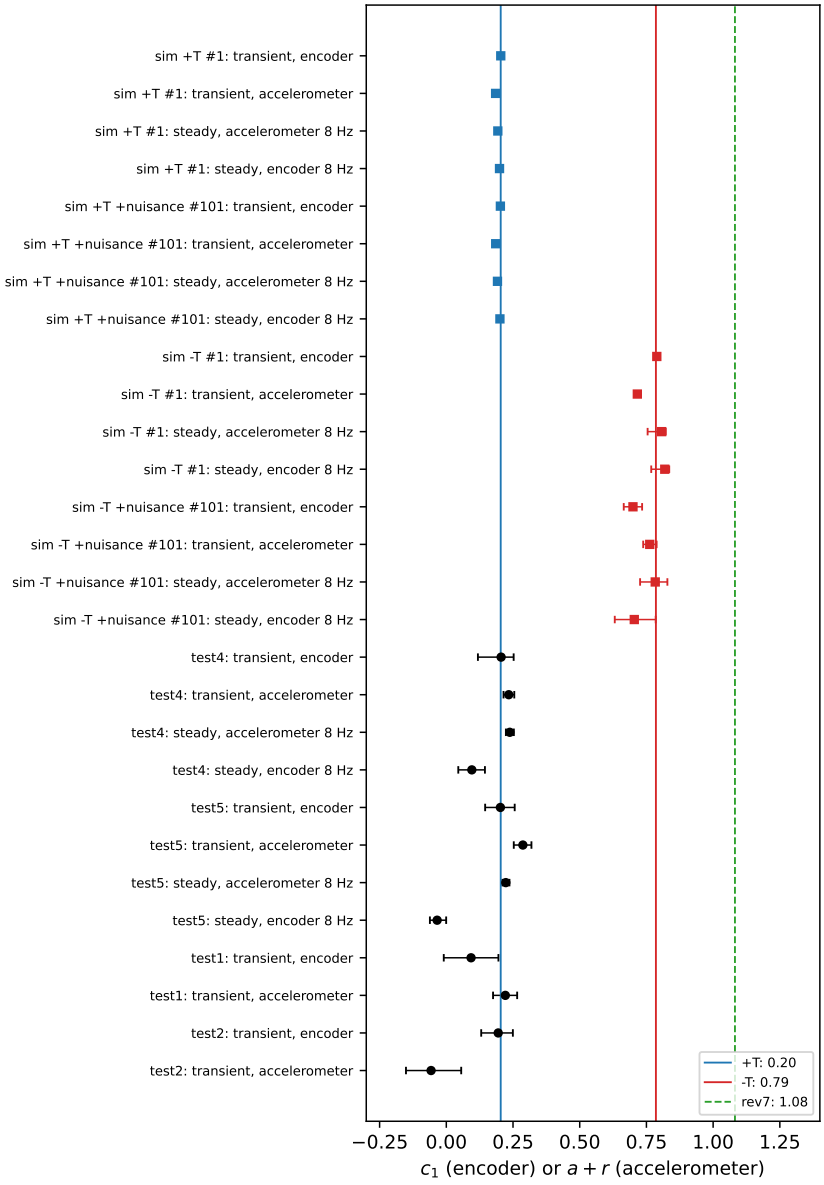


Figure 5.6: The coefficient c_1 with bootstrap intervals, from the encoder (transient and steady fits) and from the accelerometer (the transient fit a shifted by r, and the steady $\ddot{\phi}$ coefficient shifted by r plus the IMU height): one seed of each simulated case (top) and the recorded runs (bottom). Vertical lines: M1 ($+T$), M2 ($-T$) and M3 (rev7). Eight of the twelve measured intervals exclude M1; none includes M2 or M3.

the measured rotation further. With I_p, l and k_m freed ($0.008\,\mathrm{kg\,m^2}$ to $0.030\,\mathrm{kg\,m^2}$, 0.05 m to 0.15 m, and 0.5 to 5 times) and the command held at $-10\,\mathrm{V}$, M2 can match the rotation over the first 100 ms, whether the target is the filtered tilt, the integrated gyroscope or the gyroscope rate, with k_m from 1.25 times its data-sheet value in test 5 (two to three times in test 4) and l of at least 0.08 m. It then predicts the base to move 70 to 135 mm, against 51 mm measured, while M1 matches both. That discrimination is the input-free relation (5.47) applied to the first 100 ms, so it is not independent of Section 5.3.4. At nominal parameters the encoder does not discriminate on its own, since the encoder prediction of M3 (-47 mm and -48 mm) is closer than that of M1 (-58 mm) to the measured -51 mm. This test shares its two events with Section 5.3.4 and with the frame of Section 5.3.3.

This is how the reaction sign shows in the voltages: not as a reversed response to the voltage, but, at the data-sheet motor constant, as a tilt authority large enough to turn the body back against gravity within the first 50 ms, which the robot did and which the reversed-sign models cannot.

5.3.6 Gains Used on the Robot

The v9 firmware computes the command as a linear function of the logged on-board estimates, so the gains are recovered by least squares. In the firmware's frame both v9 runs used

$$K = \begin{bmatrix} -5.49 & -5.86 & 118.4 & 0.90 \end{bmatrix},$$

with an RMS residual of at most 2.4 mV ($R^2 > 0.99999$). The same fit recovers the v8 gains, $K = [-0.22,\ -0.38,\ K_3,\ 2.61]$ with $K_3 = 26.9$ in test 1 and 72.5 in test 2. If the encoder is mirrored, as Section 5.3.3 infers for v9 and assumes for v8, the position and velocity gains change sign, and all four runs fed back position and velocity with the opposite sign to the LQR gains designed for this plant, $K = [-0.30,\ -13.4,\ 22.1,\ 3.12]$ on M1 and $[-0.30,\ -0.90,\ 74.9,\ 11.9]$ on M3. They contribute little to the tilt loop: at the few

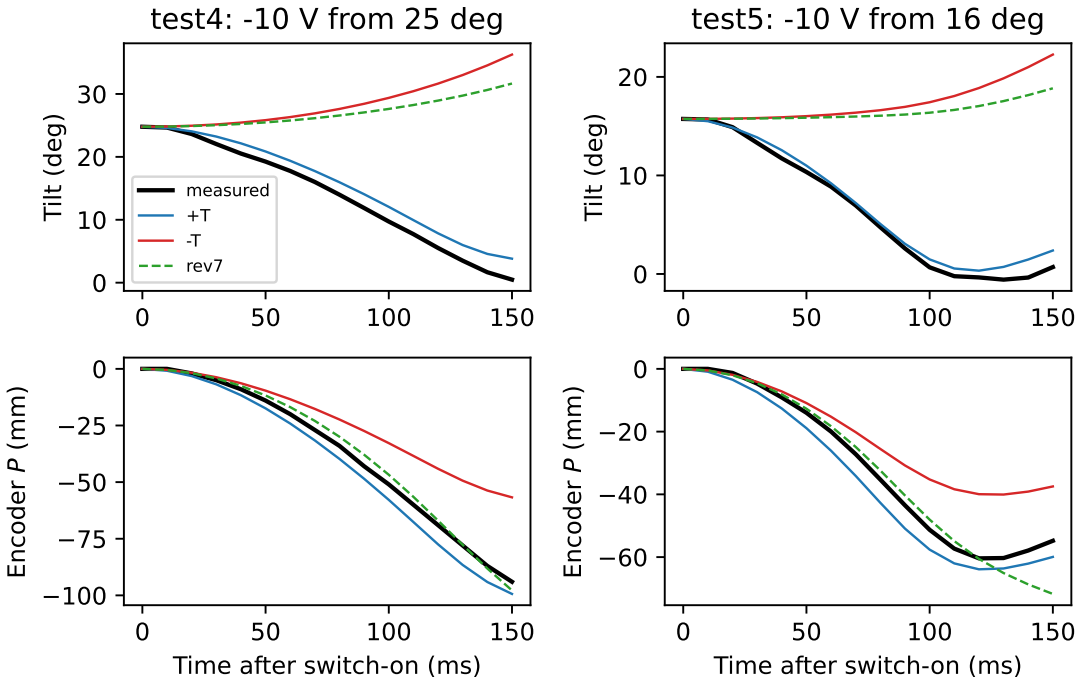


Figure 5.7: The first 150 ms after switch-on at −10 V: the on-board filtered tilt and the encoder position, and each candidate simulated open loop from rest at the logged initial tilt with the logged voltages. The filtered tilt overstates the fall during these accelerations; the gyroscope rates quoted in the text are the cleaner comparison. In the legend, $+T$ is M1, $-T$ is M2 and rev7 is M3.

millimeters the base moves near upright they contribute tens of millivolts, against about 2 V per degree of tilt in v9 (0.5 and 1.3 V in v8). They set the slow base mode, however: with the v9 gains M1 has a mode at $0.9\,\mathrm{s}^{-1}$ in the mirrored frame and $0.04\,\mathrm{s}^{-1}$ with the encoder not mirrored (without the estimator, Table 5.4).

Table 5.4 gives the least stable mode of each loop on each candidate. Test 1 drove away for 3 s, growing at $1.49\,\mathrm{s}^{-1}$, until the robot was switched off 74 mm later. M1 predicts $1.48\,\mathrm{s}^{-1}$, M2 $5.48\,\mathrm{s}^{-1}$ and M3 $2.72\,\mathrm{s}^{-1}$. This rate is set mainly by the weak tilt gain, not by the frame. Without the estimator, the rate of M1 is $1.44\,\mathrm{s}^{-1}$ with the position and velocity gains removed and $1.35\,\mathrm{s}^{-1}$ with the encoder not mirrored; that of M2 stays above $5.4\,\mathrm{s}^{-1}$ in every case, while that of M3 falls to $1.48\,\mathrm{s}^{-1}$ with the encoder not mirrored. So test 1 contradicts M2 but separates M1 from M3 only in the mirrored frame. The rate also depends on k_e: with the values of the motor sweeps scaled to the present gear ratio (0.44 to 0.50 V s/rad, Section 5.3.7) M1 predicts 1.9 to $2.0\,\mathrm{s}^{-1}$. Test 2 drifted at $1.6\,\mathrm{s}^{-1}$, between M1 and M3 (0.65 and 0.75) and M2 (3.89), so M1 and M2 miss it by similar factors; with the encoder not mirrored the test 2 loop on M3 is stable. The run ended in a chatter at the 38 Hz Nyquist frequency of its loop, at ± 10 V, which no linear model predicts.

With the v9 gains every candidate is linearly unstable at the nominal input gain, and in the mirrored frame no input-gain multiplier between 0.1 and 10 stabilizes any of them. With the encoder not mirrored, 1.3 to 5 times stabilizes M1 and 1.3 to 1.5 times M2 (with the estimator), although the simulated M2 loop still falls. Yet the robot balanced for 12.6 s and 21.2 s with the base held within a few millimeters. The v9 loop was therefore simulated with the logged gains for 21 s, using the sensors, filter, estimator, saturation and PWM truncation of the firmware (the simulator of the method validation), on each candidate over 160 parameter sets (I_p from 0.008 to $0.030\,\mathrm{kg\,m^2}$, l from 0.05 to 0.15 m, k_m from 0.5 to 4 times and k_e from 1 to 1.7 times nominal), clean and with gearbox friction, backlash and rolling resistance, with the encoder mirrored, not mirrored, and with the tilt gains alone

(Table 5.5). With the robot's filter no M2 plant stayed up longer than 5.3 s. This does not rest on the complementary filter: with the gyroscope alone ($\lambda_c = 1$), the nominal M2 loop falls within 0.7 to 5.5 s in every frame, clean or with friction and backlash. M3 stayed up only with the encoder not mirrored, and then drove at least 1.9 m away. With gearbox friction, backlash and 2% rolling resistance, M1 balanced with the base held in 61 of the 160 sets with the encoder not mirrored and in 115 with the tilt gains alone, but in only 2 in the mirrored frame, where its 0.9 s^{-1} mode drives the base off. Clean, it held the base in none of the 160 sets in any frame. M1 therefore holds its base only if the floor resists rolling, in either frame. For the nominal M1 in the mirrored frame, a resistance of 4% of the weight keeps it up with the base wandering 23 mm to 28 mm over the last 10 s, and 6% holds the base within 8 mm; with the encoder not mirrored, 2% holds it within 6 to 10 mm. M2 and M3 fall within 3 s at every level from 0 to 15% in every frame. The frame thus sets how much floor resistance the v9 runs require, not whether they require it. The floor the robot ran on was not recorded. The simulation, from `scripts/validate_closed_loop.py`, writes its results to `generated/plant_closedloop.json`.

This is how the reaction sign shows in the controller results. With the gains the robot actually ran, no reversed-sign plant balances with its base held over the parameter range tried: M2 falls within 5.3 s in every set and frame, and M3 stays up only with the encoder not mirrored, where it drives at least 1.9 m away, while the robot balanced for 21 s with its base held within a few millimeters. Only M1 balances with its base held, and only if the floor resists rolling, by about 2% of the weight with the encoder not mirrored and about 6% in the frame inferred from the switch-on transients. In the simulated loop that resistance does not reproduce the steady-balancing coefficients of Section 5.3.4, so the two do not yet support each other. The closed loop therefore confirms the sign but does not decide the frame. The floor resistance and the encoder frame are what the logs leave open about how the v9 runs held their base; two bench checks settle them (Section 5.3.8).

A last consistency check concerns the tilt measurement. In closed loop, the base acceleration needed to balance an M2 plant makes the accelerometer read about -2.95ϕ at low frequency, against $+0.43\phi$ for M1. With the robot's complementary filter ($\lambda_c = 0.99$) in the loop, simulated clean M2 plants fell within 1.5 s to 2.2 s under each of three LQR weightings, while M1 balanced under all of them. M3 and plants with friction and backlash were not simulated, so this check weighs only weakly against M2.

Table 5.4: Gains recovered from the logs and the least stable closed-loop mode (s^{-1}, positive is unstable)

Run (gains, firmware frame)	M1	M2	M3	Observed
Test 1 $(-0.22, -0.38, 26.9, 2.61)$	1.48	5.48	2.72	base ran away at 1.49
Test 2 $(-0.22, -0.38, 72.5, 2.61)$	0.65	3.89	0.75	drift at 1.6, then 38 Hz chatter
Tests 4, 5 $(-5.49, -5.86, 118.4, 0.90)$	0.9	3.95	2.6	balanced 12.6 s, 21.2 s
encoder not mirrored, no estimator	0.04	0.49	4.24	
tilt gains only, no estimator	0.37	2.87	0.04	

Rows without a note include the on-board estimator (the v8 filter or the v9 DMSO).

Table 5.5: The v9 loop simulated with the logged gains for 21 s: parameter sets, of 160, in which the robot stays up with the base held (range under 10 mm over the last 10 s), with gearbox friction, backlash and a rolling resistance of 2% of the weight

Encoder frame	M1	M2	M3
Mirrored (Section 5.3.3)	2	0	0
Not mirrored	61	0	0
Tilt gains only	115	0	0
Any frame, clean (no friction, backlash or floor)	0	0	0
Longest up, any set or frame, clean or not	21 s	5.3 s	21 s, drives off
Nominal, mirrored, floor 4, 6, 10%: base range	28, 8, 4 mm	falls	falls

5.3.7 Input Regression, Prediction and Motor Data

Regressing $\ddot{\phi}$ on the states and the voltage over the switch-on transients does not settle $\mathcal{B}_4$. With the command acting in the same loop, tests 1, 2, 4 and 5 give 8.2, 1.2, 2.5 and 9.5 per volt; with the command acting in the next loop they give -0.3, 2.8, 4.2 and 6.7. With the

same-loop timing two runs point toward M1 and two toward M2; with the next-loop timing three are nearer M2 and the fourth lies midway. Tests 1 and 2 with the command in the same loop also give a negative ratio $\mathcal{B}_p/\mathcal{B}_4$ (-0.09 and -0.42), which no candidate allows. The bootstrap has only about three blocks per window, and several estimates fall outside their own intervals, so no interval is quoted. k_e is not identifiable from the back-EMF term. The regression is confined to the transients because elsewhere the command is nearly a linear function of the states.

As a predictor of the base after switch-on, M1 is best in the v9 runs: over 100 ms and 200 ms from the logged states and voltages in test 4 its RMS encoder error is 2.4 mm and 5.3 mm, against 6.3 and 16.9 for M2, 9.2 and 27.4 for M3, and 9.9 and 18.5 for holding the state. In the v8 runs, with a few millimeters of motion, M2 is better (0.54 mm and 1.14 mm against 0.80 and 1.83 in test 1; 1.50 and 3.95 against 2.11 and 5.05 in test 2), and in steady balancing M2 predicts the base better than M1 in both v9 runs. These comparisons do not discriminate, however: there the base barely moves, M2 predicts less base motion per volt, and holding the state or extrapolating at constant velocity does about as well or better (in test 2, holding the state gives 1.15 mm and 1.44 mm). For the tilt, holding the state beats every candidate. In steady balancing M2 and M3 predict the tilt about three times better than M1 in test 5 (0.22° and 0.20° against 0.61° at 100 ms); in test 4 M3 is slightly better than M1 and M2 slightly worse (0.61, 0.65 and 0.65°). This comparison does not discriminate either. In the simulated v9 loop with gearbox friction, backlash and floor resistance, a plant that is M1 is also predicted better by M2 and M3 (0.25 to 0.30° against 0.42 to 0.50° at 100 ms), in either frame, because small commands barely move the wheels through the backlash and stiction and the larger input gain of M1 overstates their effect.

The motor sweeps of `Tests/Motor Tests` (test 9) give $k_e = 0.49\,\mathrm{V\,s/rad}$ to $0.57\,\mathrm{V\,s/rad}$, assuming 12 V at full PWM, with fitted deadzones of 1.9 V to 2.4 V and, for one motor, $R = 2.4\,\Omega$ (the fitted R of the other motor is negative). They were recorded on the earlier

34:1 gearmotors, and no torque was logged, so k_m cannot be found from them. Scaled by the gear ratio they still exceed the k_e of M1, 0.307 V s/rad, by 40 to 65%. Since the motors differ, they confirm only the order of magnitude of k_e in M1 and M2; the value 0.003 61 V s/rad of M3 is two orders too small.

5.3.8 Conclusions and Remaining Measurements

Does the sign of the motor reaction show in the voltages and in the controller results? Not as a reversed response: in all three candidates a positive voltage pushes the base forward and turns the body backward. It shows in the size of the response, and the logs measure it in three places (Table 5.6). First, at switch-on, with the command saturated at −10 V, the gyroscope reads −98 °/s and −110 °/s after 50 ms: the body is already turning back toward upright. M1 predicts −139 °/s and −168 °/s, while M2 and M3 have the body still falling further into its lean (44 °/s and 13 °/s, 28 °/s and 5 °/s); in both runs they turn it back only with k_m at least 2.5 times its data-sheet value. Second, per unit of body rotation, the base moved about three to five times less than M2 and M3 predict, on the encoder and on the accelerometer, in a relation that involves neither the voltage nor the motor constants; on the two v9 switch-on transients M1 predicts the motion 4 to 16 times better than M2 and M3. The encoder agrees with M1 in the frame inferred from these same transients, while the accelerometer reads 20 to 50% above M1. Third, with the test 1 gains the base drove away at 1.49 s^{-1}; M1 predicts 1.48 (1.9 to 2.0 with the k_e of the motor sweeps) and M2 more than 5.4 in every encoder frame, while M3 predicts 2.72, or 1.48 with the encoder not mirrored, so this run separates M1 from M3 only in the inferred frame. The 1.6 s^{-1} drift of test 2 is missed by M1 and M2 by similar factors. The first two come from the same two switch-on events, the third from a third run. With the v9 gains no linear model is stable in the inferred frame, so those runs speak only through the simulated loop, where no reversed-sign plant balances with its base held over the parameter range tried: with the robot's filter M2 falls within

5.3 s, and M3 stays up only with the encoder not mirrored, then driving at least 1.9 m away, while the robot balanced for 21 s with its base held within a few millimeters. This confirms the reaction sign of (5.32) by measurement, independently of the momentum argument of Remark 5.2. The 26 unanalysed v7 logs are the natural further replication.

The tests do not validate M1 quantitatively:

- the axle coefficient measured by the accelerometer exceeds that of M1 by 20 to 50%, with intervals that exclude it; in test 4 this corresponds to an I_p of about $0.023\,\mathrm{kg\,m^2}$, but the test 5 fit also has a gravity coefficient of the wrong sign, so the excess may come from an unmodelled force rather than from I_p;
- the steady-balancing coefficients contradict all three models, and the floor resistance that the closed loop needs does not reproduce them;
- in the simulated loop M1 holds the base of the v9 runs (range under 10 mm) only if the floor resisted rolling, by about 6% of the weight in the encoder frame inferred from the switch-on transients and about 2% with the encoder not mirrored; the resistance was not measured, so the closed loop does not decide the frame;
- the switch-on response bounds the tilt input gain of M1 only to within about 40%: the gyroscope rate at 50 ms asks for 0.64 to 0.73 times the data-sheet k_m, the filtered tilt for 1.08 to 1.26 times;
- in tests 1 and 2 only two of the four switch-on fits agree with M1 on c_1, and both, like the test 5 accelerometer fit, give a gravity coefficient of the wrong sign;
- M2 and M3 predict the tilt near upright better than M1 in test 5, and M2 predicts the base better in tests 1 and 2 and in steady balancing, but the simulator shows that M2 and M3 also predict the tilt of an M1 plant with backlash and friction better than M1 does, and that near upright holding the state predicts as well, so these comparisons do not discriminate;

- the drift of test 2 is missed by M1 and M2 by similar factors.

The same data show the limits of all three linear models. Near upright the robot is dominated by gear backlash and stick-slip, by an external force such as friction with the floor, and possibly by a motor deadzone (about 2 V on the earlier gearmotors, not measured on the present ones), none of which the models contain. These effects are what the uncertainty terms d_1 and d_2 of (5.44) must absorb on the real robot, and they bound how well any model-based estimator can predict the tilt over 100 ms or more.

The parameters themselves are less certain than the sign. I_p and l come from a lumped-mass idealization, and k_m from data sheet values that differ by 35%. The input-free test does not depend on them much: over $I_p \in [0.010, 0.025]\,\mathrm{kg\,m^2}$ and $l \in [0.065, 0.085]\,\mathrm{m}$, the c_1 of M1 stays between 0.16 and 0.25, while that of M2 stays above 0.40. The c_1 of M2 falls with larger l, to 0.29 at $I_p = 0.010\,\mathrm{kg\,m^2}$ and $l = 0.12\,\mathrm{m}$, outside the v9 encoder intervals but with an axle coefficient, 0.25, inside the test 5 accelerometer interval, and to 0.24, inside the v9 encoder intervals, only at $I_p = 0.005\,\mathrm{kg\,m^2}$, less than a third of the lumped-mass value. The switch-on response, tilt and encoder together, and the closed loop exclude M2 over the whole range tried; the direction of the body rate at 50 ms alone excludes the reversed-sign models, at nominal I_p and l, only for k_m below 2.5 times its data-sheet value in test 4 and 1.2 times in test 5. A few bench measurements would complete the validation (Table 5.7). The swing period of the body with the wheels clamped gives I_p at a measured l. Free-run and locked-rotor tests give k_e, k_m, R and the reflected armature inertia. Blocking the wheels and applying 1 V is a direct, model-free check of the reaction sign, since the body must pitch opposite to the way the wheels would roll. Rolling the robot by hand with the motors off fixes the encoder frame, and the force needed to push it slowly over the floor it balanced on gives the rolling resistance; together they decide how the v9 runs held their base.

Table 5.6: Validation summary. The switch-on rows share the same two events.

Test	M1	M2	M3
Simulation checks (Table 5.2)	pass	pass	pass
Input-free, switch-on, error (tests 4, 5)	best	contradicted	contradicted
Input-free, switch-on, fitted coefficient	encoder agrees; accelerometer 20–50% high	contradicted	contradicted
Input-free, switch-on, tests 1, 2	two of four fits agree on c_1, both with c_2 of the wrong sign	no fit agrees	no fit agrees
Input-free, steady balancing	contradicted	contradicted	contradicted
Body rate 50 ms after switch-on at -10 V	turning back, as measured; gain within about 40%	still falling at nominal k_m (in test 5 up to 1.6×)	still falling at nominal k_m (in test 5 up to 1.2×)
Drive-away rate with the test 1 gains	consistent (1.9–2.0 with the swept k_e)	contradicted	depends on frame
Drift rate with the test 2 gains	2.5× too slow	2.4× too fast	2.2× too slow; stable if not mirrored
Balancing 21 s with the v9 gains (simulated loop)	consistent if the floor resists rolling (2% not mirrored, 6% mirrored)	contradicted (falls)	contradicted (falls or drives off)
Complementary filter in the loop	consistent	weakly against	not tested
Input regression, $\mathcal{B}_4$	no decision	no decision	no decision
Prediction, base, transients of tests 4, 5	best	worse	worse
Prediction, base, tests 1, 2 and steady	no decision	no decision	no decision
Prediction, tilt near upright	no decision	no decision	no decision
Motor sweeps (earlier motors), k_e	order of magnitude	order of magnitude	contradicted

Table 5.7: Bench measurements that would complete the validation, with each candidate's prediction. The free-run speed assumes the data-sheet no-load current of 0.3 A.

Test	M1	M2	M3
Wheels blocked, 1 V: body pitches	backward	forward	forward
$\ddot{\phi}$ per volt (rad/s^2)	+3.7	−3.7	−2.8
Back-driven motor at 1 rev/s, open-circuit voltage (V)	1.93	1.93	0.023
Wheels clamped, body swinging, leads open: period (s)	0.967	0.967	1.112
Same, leads shorted: added decay rate (s^{-1})	0.57	0.57	0
Free run at 6 V (rad/s)	17	17	1450
Motors off, leaned so the tilt is positive and rolled toward the lean: logged position	rises if the encoder is mirrored, falls if not		

Chapter 6

CONTROL DESIGN

The TWIP is unstable and must be actively controlled. Linear control is effective near the upright equilibrium when the parameters are accurately known, so a linear quadratic regulator is used as the nominal control. An additive signal is then formulated to compensate for the unmatched uncertainties appearing in the two acceleration equations of (5.44).

6.1 LQR CONTROL DESIGN

Because the controller is implemented digitally, the discrete-time infinite horizon formulation is used. The performance index

$$J = \sum_{k=0}^{\infty} \left(x_k^T Q_{\mathrm{L}} x_k + u_k^T R_{\mathrm{L}} u_k \right) \tag{6.1}$$

is minimized by solving the discrete algebraic Riccati equation [58]

$$P = Q_{\mathrm{L}} + F^T \Big(P - PG \big(R_{\mathrm{L}} + G^T PG \big)^{-1} G^T P \Big) F, \tag{6.2}$$

with F and G from the discretized model of Section 7.1 and $Q_{\mathrm{L}} \succeq 0$, $R_{\mathrm{L}} \succ 0$ the weighting matrices. The control law is

$$u_{\mathrm{nom},k} = -K_{\mathrm{LQR}}\big(x_k - x_{\mathrm{des},k}\big), \qquad K_{\mathrm{LQR}} = \big(R_{\mathrm{L}} + G^T P G\big)^{-1} G^T P F. \tag{6.3}$$

6.2 UNMATCHED UNCERTAINTY AND THE VIRTUAL CONTROL

When the uncertainty enters the same channel as the control, the system is said to have matched uncertainty, and an estimate of that uncertainty can be used directly to cancel it. The TWIP does not have this structure. In (5.44) the scalar input u enters both $\dot{x}_2$ and $\dot{x}_4$ through B_1 and B_2, while two independent uncertainties d_1 and d_2 appear in the same two equations. No choice of u cancels both.

The device used here, following Huang [56] and related in spirit to the partial feedback linearization of Pathak et al. [12], is to treat the tilt state x_3 as a virtual control for the position subsystem, and then to design the actual control u so that x_3 tracks the value the position subsystem requires. The construction below is a four-step backstepping procedure with command filtering, carried out in coordinates built on the flat output of the linearized robot (Section 6.4).

6.2.1 Three requirements on the construction

Three requirements shape the design.

(R1) *The coupling term must be retained.* Substituting the desired value of x_3 into the position-subsystem dynamics is valid only if x_3 equals that value. It does not; the difference is precisely the error coordinate z_3 introduced at the next step. Carrying

$x_3 = z_3 + x_3^c$ through leaves a term proportional to z_3 in $\dot{z}_2$ that must be cancelled explicitly. Dropping it removes a term of the same order as the ones being cancelled.

(R2) *Every error coordinate needs damping.* If z_1 and z_3 appear in the Lyapunov function but not in its derivative, uniform ultimate boundedness cannot be concluded for them: the standard argument requires the derivative to be negative outside a compact set in the full error space, and a coordinate absent from the derivative fails this in its own direction. Concretely, $\dot{z}_1 = z_2$ is a pure integrator, so a merely bounded z_2 permits z_1 to ramp. Damping is assigned to all four coordinates below.

(R3) *Each network's target must be a function of measurable signals alone.* If the function F_1 that network one is required to approximate contained the control u, and u contained the output of network two, and the target F_2 of network two contained derivatives of the output of network one, then neither target would be a fixed continuous function of its network input and the universal approximation hypothesis would not apply. Command filtering removes this dependence: the filter outputs are states, so the required signals are available at the current instant without differentiating any adaptive law. The flat-output coordinates remove the control from the position subsystem altogether, so the first target does not contain u in the first place.

6.3 COMMAND FILTERS

For each virtual control α_i, $i = 1, 2, 3$, define a second-order command filter with bandwidth $\varpi_i > 0$ and damping $\varsigma_i \in (0, 1]$:

$$\dot{q}_{i,1} = q_{i,2}, \qquad \dot{q}_{i,2} = -2\varsigma_i \varpi_i q_{i,2} - \varpi_i^2 \big(q_{i,1} - \alpha_i\big), \tag{6.4}$$

with filtered command $x_{i+1}^c := q_{i,1}$ and its derivative $\dot{x}_{i+1}^c := q_{i,2}$, both available as filter states. Magnitude, rate, and bandwidth limits may be imposed on (6.4) to respect actuator constraints. Define the filter errors

$$\chi_i := x_{i+1}^c - \alpha_i. \tag{6.5}$$

Assumption 6.1 (Filter error)**.** On the compact operating set, the virtual controls α_i and their first derivatives are bounded, and for each i there exists $\bar{\chi}_i > 0$ with $|\chi_i| \le \bar{\chi}_i$ and $\bar{\chi}_i = O(1/\varpi_i)$.

Assumption 6.1 is the standard command-filtered backstepping result of Farrell et al. [32] and Swaroop et al. [33]: the filter error can be made arbitrarily small by raising the bandwidth, subject to the usual separation from the unmodeled dynamics and the sample rate. Section 6.8 discusses what the construction below does and does not guarantee about it.

6.4 FLAT-OUTPUT COORDINATES

Let $b := B_1/B_2$ and define

$$y := x_1 - b\,x_3, \qquad \eta := \dot{y} = x_2 - b\,x_4. \tag{6.6}$$

From (5.44),

$$\dot{\eta} = (A_1 - bA_3)\,x_2 + (A_2 - bA_4)\,x_3 + (A_5 - bA_6)\,x_4 + D_1(\mathbf{x}), \qquad D_1 := d_1 - b\,d_2, \tag{6.7}$$

which contains no control. For the TWIP the coefficients of x_2 and x_4 vanish identically, because the back-EMF acts through the same generalized-force direction as the voltage

(Remark 5.2) and cancels together with the input. Physically, $(r\beta + M_p l)\eta$ is the angular momentum of the robot about the wheel contact point, to first order. The motor torque, being internal, cannot change it; only gravity does, and $(r\beta + M_p l)a_2 x_3 = -M_p g l x_3$ is the gravity moment about that point. Hence

$$\dot{\eta} = a_2 x_3 + D_1(\mathbf{x}), \qquad a_2 := A_2 - bA_4. \tag{6.8}$$

The output y has relative degree four and no zero dynamics; it is the flat output of the linearization. Because b is a ratio of input gains, the motor constants cancel from both b and a_2, which depend only on the masses, geometry, and inertia:

$$b = \frac{I_p + M_p l^2 + M_p l r}{r\beta + M_p l}, \qquad a_2 = -\frac{M_p g l}{r\beta + M_p l}. \tag{6.9}$$

For the parameters of Chapter 4, $b = 0.159\,\mathrm{m}$ and $a_2 = -5.62\ \mathrm{m\,s^{-2}}$. The design requires $a_2 \neq 0$, which holds for any $l > 0$.

Note the sign. With the tilt held, $\dot{x}_4 = 0$ requires $B_2 u = -A_4 x_3$, and the translational acceleration is then $a_2 x_3$, not $A_2 x_3$. The quasi-static effect of tilt on acceleration is opposite in sign to A_2 and 1.6 times larger in magnitude ($a_2 = -5.62$ against $A_2 = 3.43$). The virtual-control gain of the position subsystem is therefore a_2.

Remark 6.2 (Choice of the tracked output). The position x_1 is not used as the tracked output. The transfer function from u to x_1 has a right-half-plane zero at $\sqrt{M_p g l/(I_p + M_p l^2 + M_p l r)}$, which is $5.95\,\mathrm{rad/s}$ at the parameters of Table 7.1. A backstepping recursion on x_1 drives every error coordinate to zero, so it makes x_1 follow its reference by inverting that transfer function, and the inversion places a closed-loop pole at the zero. A recursion on x_1 also puts the control into the second step, since u enters $\dot{x}_2$ through B_1, so the virtual control α_2 would contain $B_1 u$ and, through u, the filter states, and the premise of Assumption 6.1 that α_i and $\dot{\alpha}_i$ are bounded would not follow from the construction. Section 8.3.1 shows the

resulting unstable eigenvalue numerically. The flat output removes both difficulties: it has no zero dynamics, and the control enters only at the last step.

6.5 COMMAND-FILTERED NEURAL BACKSTEPPING

Let $x_{1d}(t)$ be the position reference, assumed twice continuously differentiable with bounded derivatives, and let y_d be the corresponding reference for the flat output. For a constant velocity command $y_d = x_{1d}$: the nominal model holds a constant velocity at zero tilt, because $\dot{\eta} = a_2 x_3$ must then vanish, and the voltage $k_e \dot{x}_{1d}/r$ only balances the back-EMF. Write $u = u_{\text{nom}} + u_e$ as in Section 6.1. The constant $w > 0$ that appears at Step 3 weights the attitude coordinates in the Lyapunov function of Section 6.7.

Step 1. Define $z_1 := y - y_d$ and the virtual control

$$\alpha_1 := \dot{y}_d - c_1 z_1, \tag{6.10}$$

filtered to η^c. With $z_2 := \eta - \eta^c$,

$$\dot{z}_1 = \eta - \dot{y}_d = z_2 + \chi_1 - c_1 z_1. \tag{6.11}$$

Step 2. With $x_3 = z_3 + x_3^c = z_3 + \chi_2 + \alpha_2$, from (6.8),

$$\dot{z}_2 = a_2 \alpha_2 + a_2\big(z_3 + \chi_2\big) + D_1 - \dot{\eta}^c. \tag{6.12}$$

Define the first approximation target

$$F_1 := D_1(\mathbf{x}) - \dot{\eta}^c, \tag{6.13}$$

which depends only on the measured state and on the filter state $\dot{\eta}^c$, and in particular *not* on the control. Choose

$$\alpha_2 := a_2^{-1}\Big[-\hat{F}_1 - c_2 z_2 - z_1\Big]. \tag{6.14}$$

Substituting into (6.12),

$$\dot{z}_2 = \big(F_1 - \hat{F}_1\big) + a_2 z_3 + a_2\chi_2 - c_2 z_2 - z_1. \tag{6.15}$$

The term $a_2 z_3$ is retained; it is cancelled at Step 3.

Step 3. With $z_3 := x_3 - x_3^c$ and $x_4 = z_4 + x_4^c = z_4 + \chi_3 + \alpha_3$, choose

$$\alpha_3 := \dot{x}_3^c - c_3 z_3 - \frac{a_2}{w}\, z_2, \tag{6.16}$$

filtered to x_4^c, giving

$$\dot{z}_3 = z_4 + \chi_3 - c_3 z_3 - \frac{a_2}{w}\, z_2. \tag{6.17}$$

The term $-(a_2/w)z_2$ in (6.16) is what cancels $+a_2 z_3$ in (6.15) when the two are combined in the weighted Lyapunov derivative.

Step 4. With $z_4 := x_4 - x_4^c$, define the second approximation target

$$F_2 := A_3 x_2 + A_4 x_3 + A_6 x_4 + d_2(\mathbf{x}) - \dot{x}_4^c, \tag{6.18}$$

again free of the control, and choose the extra control

$$u_e := B_2^{-1}\Big[-\hat{F}_2 - B_2 u_{\text{nom}} - c_4 z_4 - z_3\Big], \tag{6.19}$$

so that

$$\dot{z}_4 = \big(F_2 - \hat{F}_2\big) - c_4 z_4 - z_3. \tag{6.20}$$

Substituting (6.19) gives $B_2 u = -\hat{F}_2 - c_4 z_4 - z_3$: the nominal control cancels, and the LQR gain does not affect the applied voltage. The control enters only at this step, so no virtual control contains u.

Approximations. Each target splits into a part known from the nominal model and the uncertainty. The estimates are formed as

$$\hat{F}_1 := -\dot{\eta}^c + \hat{D}_1, \qquad \hat{F}_2 := A_3 x_2 + A_4 x_3 + A_6 x_4 - \dot{x}_4^c + \hat{d}_2, \tag{6.21}$$

where $\hat{D}_1$ and $\hat{d}_2$ are the outputs of the two networks of Section 6.6, so that each network has to learn only the uncertainty. The approximation errors $F_i - \hat{F}_i$ are then the network errors $D_1 - \hat{D}_1$ and $d_2 - \hat{d}_2$.

Remark 6.3 (Execution order). At each sample the following order is used: (1) read the state and the filter states $q_{i,\cdot}$; (2) form $z_1, \ldots, z_4$ from the state and the filter states; (3) evaluate $\hat{F}_1, \hat{F}_2$; (4) compute u_{nom} from (6.3) and u_e from (6.19), hence u; (5) compute $\alpha_1, \alpha_2, \alpha_3$; (6) propagate (6.4) and the weight updates one step. Every quantity in steps (2) to (5) is a function of the current state, the filter states and the current weights, so there is no algebraic loop. Without command filtering, x_3^c would be replaced by α_2 and η^c by α_1, and the derivatives $\dot{\eta}^c$ and $\dot{x}_3^c$ in (6.13) and (6.16) would have to be obtained by differentiating the virtual controls, and hence the weight update laws.

6.6 NEURAL NETWORK APPROXIMATION

A single-layer network was found inadequate for the uncertainties, so two-layer networks are used. For $i = 1, 2$,

$$F_i = \bar{F}_i + W_{i2}^T \sigma_{i2}\big(W_{i1}^T \sigma_{i1}(V_i^T P_i)\big) + \varepsilon_i, \qquad |\varepsilon_i| \le \varepsilon_{im}, \tag{6.22}$$

with $\bar{F}_i$ the known part of the target in (6.21), V_i a fixed randomly selected input layer, P_i the network input vector, and σ_{i1}, σ_{i2} bounded activation functions. The estimates are

$$\hat{F}_i = \bar{F}_i + \hat{W}_{i2}^T \hat{\sigma}_{i2}\big(\hat{W}_{i1}^T \hat{\sigma}_{i1}(V_i^T P_i)\big). \tag{6.23}$$

Network inputs are taken as $P_1 = \begin{bmatrix} x_2, & x_3, & x_4, & \dot{\eta}^c, & 1 \end{bmatrix}^T$ and $P_2 = \begin{bmatrix} x_2, & x_3, & x_4, & \dot{x}_4^c, & 1 \end{bmatrix}^T$; by construction neither contains a control signal or an adaptive parameter. Both targets depend on the velocity, the tilt and the tilt rate, so each network is given all three. The dependence of D_1 on x_4 is through the nonlinear terms and, under parameter error, through the back-EMF, which contributes a term r times its x_2 term.

Writing $\tilde{W}_{ij} := W_{ij} - \hat{W}_{ij}$ and expanding the hidden-layer activation about $\hat{W}_{i1}^T \hat{\sigma}_{i1}$ in the standard manner [45],

$$F_i - \hat{F}_i = \tilde{W}_{i2}^T \Big(\hat{\sigma}_{i2} - \hat{\sigma}'_{i2} \hat{W}_{i1}^T \hat{\sigma}_{i1}\Big) + \hat{W}_{i2}^T \hat{\sigma}'_{i2}\, \tilde{W}_{i1}^T \hat{\sigma}_{i1} + \delta_i, \tag{6.24}$$

where $\hat{\sigma}'_{i2}$ is the Jacobian of σ_{i2} evaluated at the estimate and δ_i collects ε_i with the higher-order Taylor residual. That residual contains the weight errors, so δ_i is not bounded by a constant. Because the activations and their derivatives are bounded and the ideal weights satisfy $\|W_{ij}\|_F \le W_{ijm}$, it satisfies, on the compact operating set,

$$|\delta_i| \le \delta_{i0} + \delta_{i1} \left\|\tilde{W}_i\right\|_F, \qquad \left\|\tilde{W}_i\right\|_F^2 := \left\|\tilde{W}_{i1}\right\|_F^2 + \left\|\tilde{W}_{i2}\right\|_F^2, \tag{6.25}$$

with constants $\delta_{i0}, \delta_{i1} \ge 0$ [45]. Equation (6.24) is the form that makes both layers' weight errors appear explicitly and linearly, which is what allows the first-layer update law to be matched term by term in the stability analysis below.

Let $\varrho_1 := z_2$ and $\varrho_2 := w\, z_4$ denote the error coordinate driving network i, with w the

attitude weight of Section 6.7. The weight update laws, each with σ-modification, are

$$\dot{\hat{W}}_{i2} = \gamma_{i2}\Big[\big(\hat{\sigma}_{i2} - \hat{\sigma}'_{i2}\hat{W}_{i1}^T\hat{\sigma}_{i1}\big)\varrho_i^T - \kappa_{i2}\hat{W}_{i2}\Big], \tag{6.26}$$

$$\dot{\hat{W}}_{i1} = \gamma_{i1}\Big[\hat{\sigma}_{i1}\,\varrho_i^T\,\hat{W}_{i2}^T\hat{\sigma}'_{i2} - \kappa_{i1}\hat{W}_{i1}\Big], \tag{6.27}$$

with $\gamma_{ij} > 0$ the learning rates and $\kappa_{ij} > 0$ the leakage gains of the σ-modification [47, 48]. Both laws carry the same sign convention. Because ϱ_2 carries the factor w, the learning rates of network two are divided by w, $\gamma_{2j} = \bar{\gamma}_{2j}/w$, so that its adaptation speed does not depend on the weighting; $\bar{\gamma}_{2j}$ is the rate quoted in Chapter 8.

6.7 STABILITY ANALYSIS

Theorem 6.4 (Closed-loop boundedness)**.** *Consider* (5.44) *with the control* (6.3), (6.19), *the virtual controls* (6.10), (6.14), (6.16), *the command filters* (6.4), *and the update laws* (6.26)–(6.27). *Let Assumption 6.1 hold, let the ideal weights satisfy* $\|W_{ij}\|_F \le W_{ijm}$, *and let the damping and leakage gains satisfy*

$$c_1 > \tfrac{1}{2}, \qquad c_2 > 1, \qquad c_3 > \tfrac{1}{2}, \qquad c_4 > \tfrac{1}{2}, \qquad \kappa_{1j} > 2\delta_{11}^2, \qquad \kappa_{2j} > 2w\,\delta_{21}^2, \tag{6.28}$$

the first four after the cross terms are absorbed as below, the last two so that the leakage dominates the weight-error part of (6.25). *Then all tracking errors* $z_1, \dots, z_4$ *and all weight estimation errors* $\tilde{W}_{ij}$ *are uniformly ultimately bounded.*

Proof. Take

$$\mathcal{V} = \frac{1}{2}\big(z_1^2 + z_2^2\big) + \frac{w}{2}\big(z_3^2 + z_4^2\big) + \frac{1}{2}\sum_{j=1}^{2}\frac{1}{\gamma_{1j}}\operatorname{Tr}\big(\tilde{W}_{1j}^T\tilde{W}_{1j}\big) + \frac{1}{2}\sum_{j=1}^{2}\frac{w}{\bar{\gamma}_{2j}}\operatorname{Tr}\big(\tilde{W}_{2j}^T\tilde{W}_{2j}\big), \tag{6.29}$$

so that the weight on each network term is the reciprocal of the learning rate applied to it.

Differentiating and substituting (6.11), (6.15), (6.17), (6.20),

$$\begin{aligned}\dot{\mathcal{V}} = &- c_1 z_1^2 - c_2 z_2^2 - w c_3 z_3^2 - w c_4 z_4^2 \\ &+ \underbrace{z_1 z_2 - z_2 z_1}_{=0} + \underbrace{a_2 z_2 z_3 - w\, z_3 \tfrac{a_2}{w} z_2}_{=0} + \underbrace{w\, z_3 z_4 - w\, z_4 z_3}_{=0} \\ &+ z_1 \chi_1 + a_2 z_2 \chi_2 + w\, z_3 \chi_3 + \varrho_1\big(F_1 - \hat{F}_1\big) + \varrho_2\big(F_2 - \hat{F}_2\big) - \sum_{i,j} \frac{1}{\gamma_{ij}} \operatorname{Tr}\big(\tilde{W}_{ij}^T \dot{\hat{W}}_{ij}\big),\end{aligned}$$

with $\varrho_1 = z_2$, $\varrho_2 = w z_4$ and $1/\gamma_{2j} = w/\bar{\gamma}_{2j}$. The three bracketed groups vanish identically: the first because α_1 contributes $-z_1$ to $\dot{z}_2$ via (6.14), the second because of the $-(a_2/w)z_2$ term in (6.16), and the third because of the $-z_3$ term in (6.19).

Substituting (6.24) for $F_i - \hat{F}_i$, and writing $\Lambda_i := \hat{\sigma}_{i2} - \hat{\sigma}'_{i2} \hat{W}_{i1}^T \hat{\sigma}_{i1}$ for the effective second-layer regressor, the approximation terms become

$$\varrho_i^T \big(F_i - \hat{F}_i\big) = \varrho_i^T \tilde{W}_{i2}^T \Lambda_i + \varrho_i^T \hat{W}_{i2}^T \hat{\sigma}'_{i2} \tilde{W}_{i1}^T \hat{\sigma}_{i1} + \varrho_i^T \delta_i. \tag{6.30}$$

Each of the first two is a scalar and may be written as a trace,

$$\varrho_i^T \tilde{W}_{i2}^T \Lambda_i = \operatorname{Tr}\big(\tilde{W}_{i2}^T \Lambda_i \varrho_i^T\big), \tag{6.31}$$

$$\varrho_i^T \hat{W}_{i2}^T \hat{\sigma}'_{i2} \tilde{W}_{i1}^T \hat{\sigma}_{i1} = \operatorname{Tr}\big(\tilde{W}_{i1}^T \hat{\sigma}_{i1} \varrho_i^T \hat{W}_{i2}^T \hat{\sigma}'_{i2}\big). \tag{6.32}$$

The weight error contributions to $\dot{\mathcal{V}}$ follow from $\dot{\tilde{W}}_{ij} = -\dot{\hat{W}}_{ij}$ and (6.26)–(6.27),

$$\frac{1}{\gamma_{i2}} \operatorname{Tr}\big(\tilde{W}_{i2}^T \dot{\tilde{W}}_{i2}\big) = -\operatorname{Tr}\big(\tilde{W}_{i2}^T \Lambda_i \varrho_i^T\big) + \kappa_{i2} \operatorname{Tr}\big(\tilde{W}_{i2}^T \hat{W}_{i2}\big), \tag{6.33}$$

$$\frac{1}{\gamma_{i1}} \operatorname{Tr}\big(\tilde{W}_{i1}^T \dot{\tilde{W}}_{i1}\big) = -\operatorname{Tr}\big(\tilde{W}_{i1}^T \hat{\sigma}_{i1} \varrho_i^T \hat{W}_{i2}^T \hat{\sigma}'_{i2}\big) + \kappa_{i1} \operatorname{Tr}\big(\tilde{W}_{i1}^T \hat{W}_{i1}\big). \tag{6.34}$$

Comparing (6.31) with (6.33) and (6.32) with (6.34), every explicit term containing both a weight error and an error coordinate appears exactly twice with opposite sign and cancels.

(The residual δ_i still depends on the weight errors through (6.25); it is handled with the leakage below.) This pairwise cancellation is the purpose of the specific regressors chosen in (6.26)–(6.27), and it is available because (6.24) keeps both layers' weight errors explicit and linear.

What survives is the leakage,

$$\sum_{i,j} \kappa_{ij} \operatorname{Tr}\big(\tilde{W}_{ij}^T \hat{W}_{ij}\big) = \sum_{i,j} \kappa_{ij} \operatorname{Tr}\big(\tilde{W}_{ij}^T (W_{ij} - \tilde{W}_{ij})\big) \le \sum_{i,j} \Big[-\frac{\kappa_{ij}}{2} \left\| \tilde{W}_{ij} \right\|_F^2 + \frac{\kappa_{ij}}{2} W_{ijm}^2 \Big],$$

together with the filter-error and residual cross terms. Hence

$$\begin{aligned} \dot{\mathcal{V}} \le & -c_1 z_1^2 - c_2 z_2^2 - w c_3 z_3^2 - w c_4 z_4^2 - \sum_{i,j} \frac{\kappa_{ij}}{2} \left\| \tilde{W}_{ij} \right\|_F^2 \\ & + z_1 \chi_1 + a_2 z_2 \chi_2 + w\, z_3 \chi_3 + z_2 \delta_1 + w\, z_4 \delta_2 + \sum_{i,j} \frac{\kappa_{ij}}{2} W_{ijm}^2. \end{aligned}$$

Applying $ab \le \frac{1}{2}a^2 + \frac{1}{2}b^2$ to each filter cross term, and for the residuals $z_2\delta_1 \le \frac{1}{2}z_2^2 + \frac{1}{2}\big(\delta_{10} + \delta_{11} \left\| \tilde{W}_1 \right\|_F\big)^2 \le \frac{1}{2}z_2^2 + \delta_{10}^2 + \delta_{11}^2 \left\| \tilde{W}_1 \right\|_F^2$ and $w\, z_4 \delta_2 \le \frac{w}{2} z_4^2 + w\, \delta_{20}^2 + w\, \delta_{21}^2 \left\| \tilde{W}_2 \right\|_F^2$,

$$\begin{aligned} \dot{\mathcal{V}} \le & -\Big(c_1 - \tfrac{1}{2}\Big) z_1^2 - \Big(c_2 - 1\Big) z_2^2 - w \Big(c_3 - \tfrac{1}{2}\Big) z_3^2 - w \Big(c_4 - \tfrac{1}{2}\Big) z_4^2 \\ & - \sum_j \Big(\frac{\kappa_{1j}}{2} - \delta_{11}^2 \Big) \left\| \tilde{W}_{1j} \right\|_F^2 - \sum_j \Big(\frac{\kappa_{2j}}{2} - w\, \delta_{21}^2 \Big) \left\| \tilde{W}_{2j} \right\|_F^2 + \mathcal{C}, \end{aligned} \tag{6.35}$$

where

$$\mathcal{C} := \tfrac{1}{2}\bar{\chi}_1^2 + \tfrac{1}{2}a_2^2 \bar{\chi}_2^2 + \tfrac{w}{2}\bar{\chi}_3^2 + \delta_{10}^2 + w\, \delta_{20}^2 + \sum_{i,j} \frac{\kappa_{ij}}{2} W_{ijm}^2.$$

Under (6.28) every coefficient in (6.35) is negative and $\dot{\mathcal{V}} < 0$ outside a compact set in $(z_1, z_2, z_3, z_4, \tilde{W}_{11}, \tilde{W}_{12}, \tilde{W}_{21}, \tilde{W}_{22})$. Uniform ultimate boundedness follows by the standard Lyapunov argument, with ultimate bound proportional to $\sqrt{\mathcal{C}}$. □

Remark 6.5 (What the bound depends on). The residual $\mathcal{C}$ has three distinct contributions,

and they can be reduced independently. The filter terms $\bar{\chi}_i^2$ shrink as the filter bandwidths ϖ_i are raised, limited by the sample rate, by the unmodeled structural modes of the chassis, and by the measurement noise (Section 6.8). The network residuals δ_{i0}^2 shrink with the richness of the basis, down to the part of each target that the network inputs do not determine, the input-gain terms of Section 6.8. The leakage terms $\kappa_{ij} W_{ijm}^2$ shrink with κ_{ij}, at the cost of weaker protection against weight drift. This decomposition gives the tuning order recommended in Appendix A.

Remark 6.6 (The attitude weight). The skew coupling between z_2 and z_3 oscillates at $|a_2| / \sqrt{w}$. With $w = 1$ that is about 5.6 rad/s, inside the command-filter bandwidths, and the linearized closed loop (networks off) is stable at 672 of the 720 gain and bandwidth combinations examined in Section 8.3.1, including the design gains; in case C2 the nonlinear loop at the design gains balances with 9.3 mm RMS position error. With $w = a_2^2$ the coupling term in (6.16) becomes z_2/a_2, the coupling frequency drops to about 1 rad/s, and all 720 combinations are stable. That value is used for the results of Chapter 8 for its wider stability margin over the gain grid, at some cost in tracking (16 mm RMS in C2).

6.8 WHAT THE CONSTRUCTION GUARANTEES ABOUT THE ASSUMPTIONS

The filter-error assumption, Assumption 6.1, is consistent with the construction. The virtual controls depend on the state, the filter states, and the network outputs, but not on the control, and the tracked output has no zero dynamics. So bounded tracking errors imply bounded virtual controls and bounded derivatives. This is consistency, not a proof: Theorem 6.4 is conditional on Assumption 6.1, and a result that does not assume it would use the compensated tracking errors of Farrell et al. [32] or the invariant-set argument of Swaroop et al. [33]. Nor does it follow that the filter errors shrink without limit as the bandwidth

is raised. The $O(1/\varpi_i)$ scaling of Assumption 6.1 holds for smooth virtual controls; with measurement noise the virtual controls are themselves noisy, and a wider filter passes more of that noise. Figure 8.5 shows both effects: χ_1 and χ_2 stay near 10^{-3}, while χ_3, the error of the fastest filter, grows from 0.005 to 0.18 over the bandwidths tested. The bandwidth therefore trades filter lag against noise transmission, and $\bar{\chi}_i$ in the bound has a floor set by the sensors.

The analysis is in continuous time. In Chapter 8 the controller runs at the 10 ms loop period of the robot, with the command filters discretized exactly for a held input and the weight updates integrated by forward Euler (Chapter 7). Theorem 6.4 describes that sampled loop only to the extent that the sample period is short compared with the time constants of the loop. The fastest command filter, at 60 rad/s, turns through 0.6 rad per sample, so the stability of the implemented loop rests on the simulations of Chapter 8 rather than on the theorem.

One modeling limitation remains. A parameter error changes the input gains, so the residuals contain terms proportional to the control: $(B_2^{\text{true}} - B_2)u$ in $F_2 - \hat{F}_2$ and, when the error changes b, as an error in the masses does, $(B_1^{\text{true}} - b\,B_2^{\text{true}})u$ in $F_1 - \hat{F}_1$ (in case C2, -0.14 against $B_1^{\text{true}} = 1.65\,\text{m}\,\text{s}^{-2}\,\text{V}^{-1}$). An error in k_m or R alone scales B_1 and B_2 together and leaves the second term zero. These residuals depend on the control, and neither network is given the control as an input. The approximation hypothesis therefore does not cover input-gain error; robustness to it comes from the gains. It does not depend on a narrow choice of them: in the closed-loop sweep of Section 8.3, 63 of 64 gain sets balanced in both C2 and C4.

Chapter 7

TWO WHEELED INVERTED PENDULUM SIMULATION

7.1 DISCRETIZATION

The linear model (5.40) is discretized at sample period Δt using the matrix exponential,

$$F = e^{\mathcal{A}\Delta t}, \qquad G = \left(\int_0^{\Delta t} e^{\mathcal{A}\tau} \, \mathrm{d}\tau \right) \mathcal{B}, \tag{7.1}$$

which is exact for a zero-order hold input. The plant used for truth propagation is the nonlinear model (5.34)–(5.35), integrated at a step at least an order of magnitude smaller than Δt, so that the estimator and controller see a discretized measurement of a continuous truth rather than a discretized truth.

Remark 7.1 (The sample period enters the stability conditions). The quantity $\sigma_{\max}(A) = \sigma_{\max}(F - K_{\mathrm{MSO}})$ depends on Δt through (7.1), so the admissible range of Γ in Corollary 3.9 is a function of the sample rate. Reducing Δt moves F toward the identity and requires a correspondingly larger K_{MSO} for the same $\sigma_{\max}(A)$. The design must be rechecked whenever the loop rate changes rather than assumed to carry over.

7.2 SIMULATION MODEL

The truth model used for the results of Chapter 8 is the nonlinear model (5.34)–(5.35) of Section 5.2 with the parameters of Tables 4.1 and 4.2: I_p is the inertia about the center of gravity, as the equations require, and k_e is the value obtained from the free-run data in Section 4.2.2. Table 7.1 lists the entries of the linearization (5.40). The back-EMF damping is strong, the tilt input gain $\mathcal{B}_4$ is about six times the translational gain $\mathcal{B}_2$, and the tilt-rate entries $\mathcal{A}_{24}$ and $\mathcal{A}_{44}$ from the relative back-EMF are small.

The right-half-plane zero of $x_1(s)/u(s)$, which decides the choice of tracked output in Chapter 6, lies at $\sqrt{M_p g l/(I_p + M_p l^2 + M_p l r)}$, independent of the motor constants: setting $x \equiv 0$ in (5.38)–(5.39) eliminates the back-EMF terms and leaves $(I_p + M_p l^2 + M_p l r)\ddot{\phi} = M_p g l \phi$. It nearly coincides with the unstable pole, for a physical reason. With the base still and zero voltage, the back-EMF torque is $-(2k_m k_e/R)\dot{\phi}$, while holding the base still requires $-M_p l r \ddot{\phi}$. For a mode growing at rate p the two agree when $2k_m k_e/R = M_p l r p$, and the constants of Table 4.2 give 0.0283 and 0.0288 N m s at $p = 5.95$ rad/s. The body therefore falls while the base moves by less than 1% of $r\phi$: the unstable mode barely appears in x_1 and must be stabilized through the tilt.

Table 7.1: Linearized model entries

Quantity	Value
$\mathcal{A}_{22},\ \mathcal{A}_{23},\ \mathcal{A}_{24}$ (s^{-1}, $\mathrm{m\,s}^{-2}$, $\mathrm{m\,s}^{-1}$)	$-12.65,\ 3.43,\ -0.569$
$\mathcal{A}_{42},\ \mathcal{A}_{43},\ \mathcal{A}_{44}$ ($\mathrm{rad\,m}^{-1}\,\mathrm{s}^{-1}$, s^{-2}, s^{-1})	$-79.6,\ 57.0,\ -3.58$
$\mathcal{B}_2,\ \mathcal{B}_4$ ($\mathrm{m\,s}^{-2}\,\mathrm{V}^{-1}$, $\mathrm{rad\,s}^{-2}\,\mathrm{V}^{-1}$)	$1.86,\ 11.67$
Unstable pole (rad/s)	5.953
RHP zero of x_1/u (rad/s)	5.946

Sensors. The accelerometer is modeled as the specific force at the IMU location, 2 cm above the wheel axis on the body, expressed in body axes. The tilt presented to the estimators is therefore computed from a signal that includes the robot's own acceleration, as on the

hardware, rather than being the true tilt plus white noise. The encoders measure the rotation of the wheel relative to the body, so the encoder position is $x + r\phi$; it is corrected by subtracting $r\hat{\phi}$ before use. Velocity is the difference of successive quantized positions divided by Δt.

7.3 CONTROL

The nominal control is the LQR law of Section 6.1, computed from the nominal parameters of Table 4.1 and Table 4.2. The extra control of Section 6.5 is added when the neural compensation is active. The command filters (6.4) are discretized exactly for a held input at the loop period, and the weight update laws are integrated with forward Euler at the loop rate.

7.4 UNCERTAINTY

Uncertainty is introduced by propagating the truth model with perturbed physical parameters while the observer and controller retain the nominal values. Because the physical parameters enter only through the coefficients of the two acceleration equations, this produces uncertainty in those two equations and nowhere else, which is why the selection matrix is

$$B_{\mathrm{MSO}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}^{T} . \tag{7.2}$$

The uncertainty vector has dimension four in the full state, but only the second and fourth entries are assigned, so $p = 2$ and $B_{\mathrm{MSO}} \in \mathbb{R}^{4\times 2}$ satisfies Assumption 3.2. Unmodeled dynamics are introduced separately by retaining in the truth model the nonlinear terms that the linearization (5.36)–(5.37) discards.

7.5 SIMULATED NOISE

Sensor noise is not white in general, but both the gyroscope and the accelerometer can be modeled with three principal sources of error [59, 60]. Both sensors are modeled as

$$m = m_r + c_r + b_r + w_r, \tag{7.3}$$

where m is the sensor reading, m_r the true measurement, c_r a constant bias that changes on each initialization of the sensor, b_r the walking bias, and w_r a white noise process with variance σ_r^2.

The constant bias is easily remedied and is removed on initialization of the sensor in digital logic, as described in Section 4.3.1. The other two remain and must be considered in simulation. White noise is the dominant source in low cost sensors and is characterized by finding the variance of a long run of data taken with the sensor stationary. The walking bias is more involved but can be modeled as a first-order Markov process,

$$\dot{b}_r = -\frac{1}{\tau_r} b_r + w_{b_r}, \tag{7.4}$$

where τ_r is the time constant of the Markov process and w_{b_r} is white noise with intensity $2\sigma_{b_r}^2/\tau_r$, so that $\sigma_{b_r}^2$ is the steady-state variance of b_r. The walking bias captures the tendency of the sensor bias to drift over time. Its parameters are obtained by performing an autocorrelation analysis of a long run of filtered data; further detail is given by Flenniken [60].

Allan variance plots are the standard method of characterizing the accuracy of gyroscopes and accelerometers. Figure 7.1 shows an Allan variance plot of measured gyroscope data against data simulated from (7.3)–(7.4). The agreement is not perfect and other error sources are clearly present, but the result is far more representative than simply adding white noise to the measurements.

The resulting parameters, measured from the sensors used in the TWIP implementation, are given in Table 7.2. Encoder measurements carry quantization error set by the counts per revolution and the gear ratio; the velocity quantization is the position quantization divided by Δt, because velocity is obtained by differencing successive position readings.

Table 7.2: Noise simulation parameters

Parameter	Tilt angle	Gyroscope
σ_r^2	4.0474×10^{-6} rad^2	0.0012 (°/s)^2
σ_{br}^2	1.8929×10^{-7} rad^2	1.0986×10^{-4} (°/s)^2
τ_r	2160 s	1610 s
Encoder position quantization		1.47×10^{-4} m
Encoder velocity quantization		$1.47 \times 10^{-4}/\Delta t$ m/s

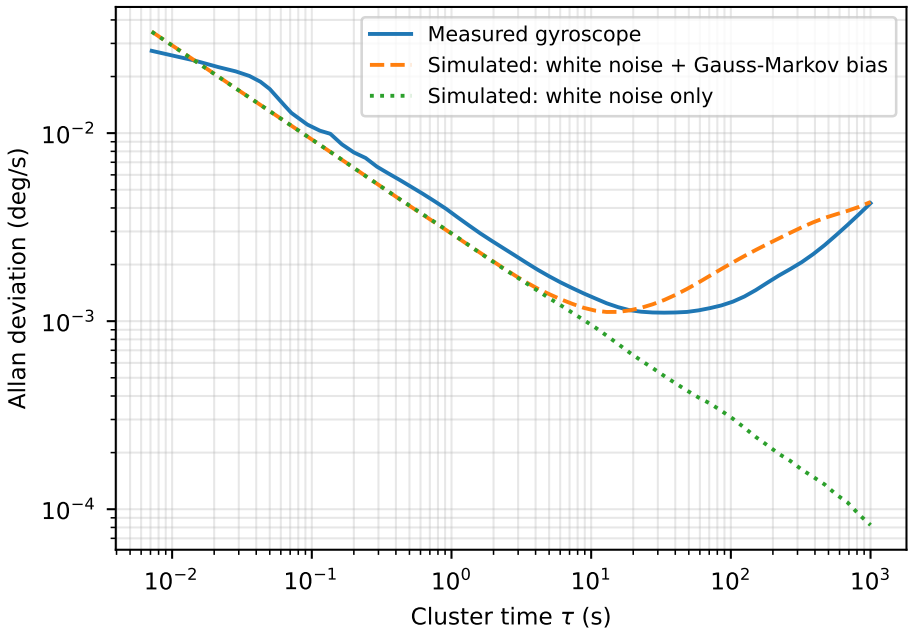


Figure 7.1: Gyroscope Allan deviation: measured (1.95 million samples at 140 Hz, stationary) against data simulated from (7.3)–(7.4) with the parameters of Table 7.2, and against white noise alone.

Remark 7.2 (The sample rate trade). The encoder velocity quantization interacts with the sample rate in the opposite direction from everything else. Halving Δt improves the fidelity

of (7.1) but doubles the velocity quantization noise. There is therefore an interior optimum in Δt for a fixed encoder resolution, and raising the loop rate without raising the encoder resolution eventually degrades the velocity estimate. A procedure for locating this optimum from logged data is given in Appendix A.

7.6 NONLINEARITIES

The Pololu DC motors exhibit several obvious nonlinearities that can be accounted for in simulation. Friction is an important nonlinearity to consider, but owing to the difficulty of evaluating the forces and obtaining an accurate model it was left out of the simulation; its effect is absorbed into d_1 and d_2 and forms part of what the networks estimate.

7.6.1 Saturation

The commanded voltage is limited by the supply rail,

$$\mathrm{sat}(u) = \begin{cases} u_{\max}, & u > u_{\max}, \\ u, & |u| \le u_{\max}, \\ -u_{\max}, & u < -u_{\max}. \end{cases} \tag{7.5}$$

Saturation is applied to the control signal output in all simulation cases.

7.6.2 Deadzone

Below a threshold voltage the motor does not turn. The standard deadzone model is continuous at the breakaway points, but the physical motors exhibit a jump discontinuity: the output steps to a finite value at breakaway rather than rising continuously from zero. The

model used is

$$\mathcal{D}(u) = \begin{cases} m_d\big(u - \delta_+\big) + j_+, & u > \delta_+, \\ 0, & -\delta_- \le u \le \delta_+, \\ m_d\big(u + \delta_-\big) - j_-, & u < -\delta_-, \end{cases} \tag{7.6}$$

with $\delta_\pm$ the breakaway voltages, m_d the slope outside the deadband, and $j_\pm$ the jump magnitudes.

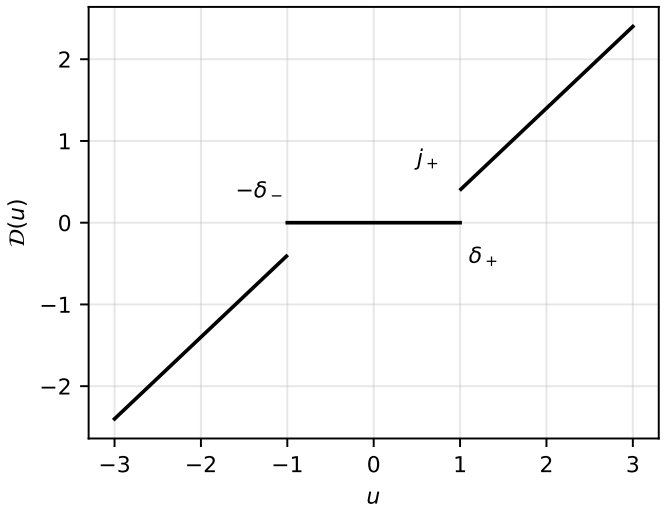


Figure 7.2: Deadzone nonlinearity with jump discontinuities.

7.6.3 Backlash

Gear backlash is modeled as hysteresis [31, 19] in which the output holds while the input reverses through a deadband of half-width Δ_b,

$$\dot{y} = \begin{cases} m_b\dot{u}, & \dot{u} > 0 \text{ and } y = m_b\big(u - \Delta_b\big), \\ m_b\dot{u}, & \dot{u} < 0 \text{ and } y = m_b\big(u + \Delta_b\big), \\ 0, & \text{otherwise.} \end{cases} \tag{7.7}$$

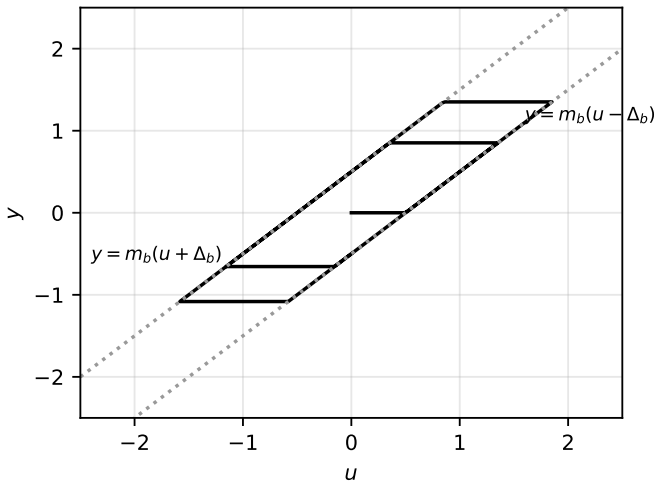


Figure 7.3: Backlash nonlinearity: response to a decaying sinusoidal input with $m_b = 1$, $\Delta_b = 0.5$.

Remark 7.3 (Backlash is outside the approximation hypothesis). The backlash of the gear train was not measured, and the recorded runs show a stick-slip limit cycle near upright (Section 5.3.4). Two observations follow. First, because the encoders are mounted on the motor shaft, this deadband is invisible to the measurement, so it appears to the estimator as an unmodeled input nonlinearity rather than as a measurable state discrepancy. Second, (7.7) is not smooth and therefore violates the smoothness assumption underlying Assumption 3.3. The networks can approximate it only in an averaged sense, and the residual appears in ϵ_N and in δ_{i0}. This is the explanation for the persistent oscillation in the uncertainty estimates reported in Chapter 8: the limit cycle is the signature of a nonsmooth input nonlinearity that no smooth approximator can cancel, not a tuning failure.

7.7 DISCRETE KALMAN FILTER

The discrete Kalman filter is used for comparison against the DMSO. The Kalman filter, developed by Kalman in 1960, provides the statistically optimal state estimate for a linear system with measurements corrupted by Gaussian, zero-mean, uncorrelated, white noise

[57]. The filter processes measurements together with knowledge of the system dynamics to produce a more accurate state estimate. The version used here is the discrete-time filter and follows Simon [58].

Assume the dynamic system is given by

$$x_{k+1} = Fx_k + Gu_k + w_k, \tag{7.8}$$

with measurements

$$y_k = Hx_k + v_k, \tag{7.9}$$

where w_k and v_k are zero-mean, uncorrelated, white Gaussian sequences with

$$w_k \sim \mathcal{N}\big(0, Q_{\mathrm{K}}\big), \qquad v_k \sim \mathcal{N}\big(0, R_{\mathrm{K}}\big). \tag{7.10}$$

Q_{K} is the process noise covariance and R_{K} the measurement noise covariance; both may be adjusted for performance. States with greater uncertainty in the dynamics can be assigned larger values in Q_{K}, which causes the filter to weight the measurements more heavily. R_{K} can be obtained accurately by statistical analysis of the available measurements.

The filter is given by the update equations

$$P_{k+1}^{-} = FP_k^{+}F^T + Q_{\mathrm{K}}, \tag{7.11}$$

$$L_{k+1} = P_{k+1}^{-}H^T\big(HP_{k+1}^{-}H^T + R_{\mathrm{K}}\big)^{-1}, \tag{7.12}$$

$$\hat{x}_{k+1}^{-} = F\hat{x}_k^{+} + Gu_k, \tag{7.13}$$

$$\hat{x}_{k+1}^{+} = \hat{x}_{k+1}^{-} + L_{k+1}\big(y_{k+1} - H\hat{x}_{k+1}^{-}\big), \tag{7.14}$$

$$P_{k+1}^{+} = \big(I - L_{k+1}H\big)P_{k+1}^{-}\big(I - L_{k+1}H\big)^T + L_{k+1}R_{\mathrm{K}}L_{k+1}^T, \tag{7.15}$$

where the superscripts $-$ and $+$ denote a priori and a posteriori quantities, P is the esti-

mation error covariance, L is the Kalman gain, and $\hat{x}$ the state estimate. The equations are processed sequentially from (7.11) to (7.15). Equation (7.15) is the Joseph form, which preserves symmetry and positive definiteness of the covariance under finite-precision arithmetic. Measurements need not arrive at the same rate as the dynamics are propagated; (7.14) may be processed only at intervals where measurements are available. In the simulations here the measurements are assumed available at the dynamics rate.

7.7.1 Three Baselines

A comparison between the DMSO and a Kalman filter must be constructed carefully to be meaningful. The Kalman filter is optimal for a linear system driven by zero-mean white process noise of known covariance. It is not designed to reject a deterministic modeling error, and demonstrating that it fails to do so establishes little about the DMSO. Three baselines are therefore used in Chapter 8:

(a) a Kalman filter with Q_{K} tuned to the sensor noise alone, which is the naive baseline;

(b) a Kalman filter with Q_{K} inflated to account for the parameter error, which is what a competent practitioner would actually do;

(c) an augmented-state Kalman filter in which the uncertainty is modeled as a random-walk state and estimated alongside the dynamic states, which is the closest linear analogue of the DMSO and the fairest comparison.

For baseline (c) the state is augmented as $x^{\mathrm{aug}} = \begin{bmatrix} x^T & f^T \end{bmatrix}^T$ with

$$F^{\mathrm{aug}} = \begin{bmatrix} F & B_{\mathrm{MSO}} \\ 0 & I_p \end{bmatrix}, \qquad G^{\mathrm{aug}} = \begin{bmatrix} G \\ 0 \end{bmatrix}, \qquad H^{\mathrm{aug}} = \begin{bmatrix} H & 0 \end{bmatrix}, \tag{7.16}$$

and the process noise covariance on the appended block set to the assumed random-walk

intensity of the uncertainty. Claims of improvement are stated against (c).

Chapter 8

RESULTS

All simulations and hardware replays in this chapter are produced by the open-source Python implementation accompanying this thesis (`twip-dmso`; script `scripts/thesis_rev8.py`), which also records every number quoted below.

8.1 TEST MATRIX AND METRICS

8.1.1 Cases

Each case is run for both the observer study and the control study.

Table 8.1: Simulation test matrix

Case	Description	Noise	Param. error	Nonlinearity
O1	Nominal	—	—	—
O2	Parameter uncertainty	—	✓	—
O3	Noise and uncertainty	✓	✓	—
O4	Unmodeled dynamics	✓	✓	—
C1	Control, nominal	✓	—	—
C2	Control, uncertainty	✓	✓	—
C3	Control, deadzone	✓	✓	deadzone
C4	Control, deadzone + backlash	✓	✓	both

8.1.2 Simulation setup

The truth model is the nonlinear model of Section 7.2, integrated with fourth-order Runge–Kutta at one tenth of the $\Delta t = 10\,\mathrm{ms}$ loop period of the hardware. The estimators and controllers use the nominal linearization discretized by (7.1).

Parameter error. Cases O2–O4 and C2–C4 propagate the truth with $M_p \times 1.1$, $M_w \times 1.5$, $k_m \times 1.25$, $k_e \times 0.95$, and $R \times 1.2$; the observer and controller retain the nominal values. The torque-constant error is kept moderate deliberately. At $\times 2.5$ all four controllers of Section 8.3 still balance (case C2, five seeds), but the input-gain error $(B^{\mathrm{true}} - B)u$, which is not a function of the state, is large during the recovery from the initial tilt: the largest residual of the basis fit in case O2 grows from $\epsilon_N = 0.013$ to 0.42, which makes the bound of Theorem 3.8 vacuous, although the RMS fit residual stays below 6% of the RMS uncertainty. The smaller error keeps the observer comparison within the theorem.

Unmodeled dynamics. Case O4 adds the unmodeled terms $0.5\sin(x_1)x_2$ to $\dot{x}_2$ and $0.25x_2^2$ to $\dot{x}_4$.

Observer study. For each case a single truth trajectory is generated by the LQR of Section 6.1 acting on the *true* state, tracking a position sinusoid of amplitude 0.3 m and period 8 s from an initial tilt of 5°, for 30 s. Every estimator then processes the same measurement and input sequences, so all are compared on identical information. The measurement y_k is the full state as the robot constructs it: the encoder position and velocity corrected for body tilt, the complementary-filtered tilt of (4.5), and the gyroscope rate. Without noise, $y_k = x_k$. Each estimator is scored by its one-step prediction of x_{k+1} after processing y_k and u_k, which is the native output of the DMSO and is formed for the Kalman filters as $F\hat{x}_{k|k} + Gu_k$. The uncertainty truth is $f_k = B_{\mathrm{MSO}}^T\big(x_{k+1} - Fx_k - Gu_k\big)$ computed from the true state.

Control study. Each controller closes the loop itself for 20 s, from 5° of tilt at rest, tracking a constant velocity of 0.1 m/s. The state is supplied to every controller by the same Kalman filter, which models the accelerometer kinematics as described in Section 7.2.

The LQR includes the nominal steady-state feedforward voltage for the commanded velocity. Case C3 uses a 0.5 V deadzone and C4 adds an input backlash of half-width $\Delta_b = 0.25\,\mathrm{V}$. The command filters have damping $\varsigma_i = 0.9$ and are initialized at the initial state. Each network has ten first-layer units with a fixed random input layer of unit-normal entries and eight hidden units; the first-layer weights start small and random, so that both regressors in (6.26)–(6.27) are nonzero, and the output weights start at zero.

8.1.3 Metrics

Scalar summaries are reported per case rather than averaged across cases. An average of percentage improvements taken over heterogeneous scenarios is not a meaningful statistic, since the scenarios have different error scales and the mean is dominated by whichever case happens to have the smallest baseline. The reported metrics are, for each state and each case:

(i) RMS estimation error over the run, after excluding the first 5 s as initial transient;

(ii) time to first entry into the ultimate bound predicted by (3.18), together with that predicted bound;

(iii) RMS regulation or tracking error for the control cases;

(iv) RMS and peak control effort, to confirm that improvements are not bought with unreported actuator authority.

8.1.4 Verification of the theoretical conditions

Each run records $\sigma_{\max}(A)$, $\phi_{\max}$ as realized along the trajectory, and the product $\Gamma\phi_{\max}^2$, so that satisfaction of (3.20) can be confirmed rather than assumed.

The DMSO is configured by the constructive procedure of Appendix A. The error-injection gain is $K_{\mathrm{MSO}} = F - \bar{\sigma}I$, so that $A = \bar{\sigma}I$ is normal and $\sigma_{\max}(A) = \bar{\sigma} = 0.5$

exactly. The basis is $\phi(y) = \big[1,\ \tanh(y_1/1),\ \tanh(y_2/1),\ \tanh(y_3/0.2),\ \tanh(y_4/1)\big]^T$, which is bounded by construction with $\phi^2_{\max} = 5$. The adaptation gain is one tenth of the bound (3.20), $\Gamma = 0.01$. Table 8.2 confirms that the conditions hold along every trajectory; the realized $\phi^2_{\max}$ never exceeds 1.92, well below its design value.

Table 8.2: DMSO stability conditions as realized in each observer case

Case	$\sigma_{\max}(A)$	$\phi^2_{\max}$ (design)	$\phi^2_{\max}$ (realized)	Γ	$\Gamma\phi^2_{\max}$	bound on Γ	(3.20) holds
O1	0.50	5	1.90	0.0100	0.0190	0.100	yes
O2	0.50	5	1.89	0.0100	0.0189	0.100	yes
O3	0.50	5	1.92	0.0100	0.0192	0.100	yes
O4	0.50	5	1.92	0.0100	0.0192	0.100	yes

8.2 OBSERVER RESULTS

Six estimators are compared: the three Kalman baselines of Section 7.7; the DMSO of Chapter 3 as configured above, with and without σ-modification ($\kappa = 0.1$); and, for reference, the same adaptation law at a larger gain, $\Gamma = 0.1$ with $K_{\mathrm{MSO}} = I$ and the basis evaluated on the estimate $\hat{x}_k$ rather than on the measurement, a configuration outside the proven region. A seventh row, marked †, uses the adaptation law with the steady-state Kalman gain as error injection, $K_{\mathrm{MSO}} = FL$, so that $A = F(I - L)$. That A is Schur but strongly non-normal ($\sigma_{\max}(A) = 1.03$ without noise and 6.5 with it), so Theorem 3.8 does not apply in the Euclidean norm and the row is an empirical extension; see Remark 3.12 and Section 8.2.3.

8.2.1 Nominal and parameter-uncertainty cases

In case O1 there is no model error and every estimator predicts the state to within numerical precision (Figure 8.1); the DMSO weights remain near zero because there is nothing to learn.

Case O2 is the setting the DMSO is designed for: a deterministic model error and a noise-free measurement. Figure 8.2 shows the norm of the estimation error against the ultimate

bound of Theorem 3.8. The bound requires ϵ_N; a valid value on the realized trajectory is the largest residual of a least-squares fit of the true uncertainty onto the basis, $\epsilon_N = 0.0133$, which gives $b_e = 0.0276$. The error stays inside the bound for the whole run, initial transient included; its largest value is 0.0259.

The DMSO identifies the uncertainty itself, not only the state (Figure 8.3). Against the augmented-state Kalman filter, the baseline against which improvement is claimed, the RMS error of $\hat{f}$ is lower by a factor of 1.6 in the velocity channel and 4.0 in the tilt-rate channel (Table 8.4). The one-step prediction error of the uncertain states is mixed (Table 8.3): 27% higher than the augmented filter's in velocity and half of it in tilt rate. σ-modification costs accuracy here, as the analysis predicts, because the leakage biases the weights toward zero while the uncertainty is persistently nonzero.

The large-gain configuration identifies the O2 uncertainty about four times more accurately than the DMSO at $\Gamma = 0.01$. The difference is the adaptation gain, not the structure. With $K_{\mathrm{MSO}} = I$, $\sigma_{\max}(F - K_{\mathrm{MSO}}) = 0.92$ and the bound of Corollary 3.9 is at most 0.085, so $\Gamma = 0.1$ runs outside the proven region, while the design value is one tenth of its bound. At half and nine tenths of the bound ($\Gamma = 0.05$ and 0.09), still inside the conditions, the errors are 0.11/0.74 and 0.06/0.42 ($\times 10^{-3}$), equal to and then better than the large-gain configuration's. Faster error injection is not the cause: with $K_{\mathrm{MSO}} = I$ and $\Gamma = 0.1$ the DMSO with the basis on y_k gives 0.12/0.80. With sensor noise (case O3) the gain makes no difference, 0.88–1.00 and 11.4–11.5 across the same range, because the error is then set by the measurements. The value $\Gamma = 0.01$ is kept below so that every reported run satisfies the conditions with a wide margin.

8.2.2 Cases with sensor noise

With sensor noise the picture is mixed, and it is reported as such (Figure 8.4 and Table 8.3).

In the velocity channel the DMSO remains better than both unaugmented Kalman filters,

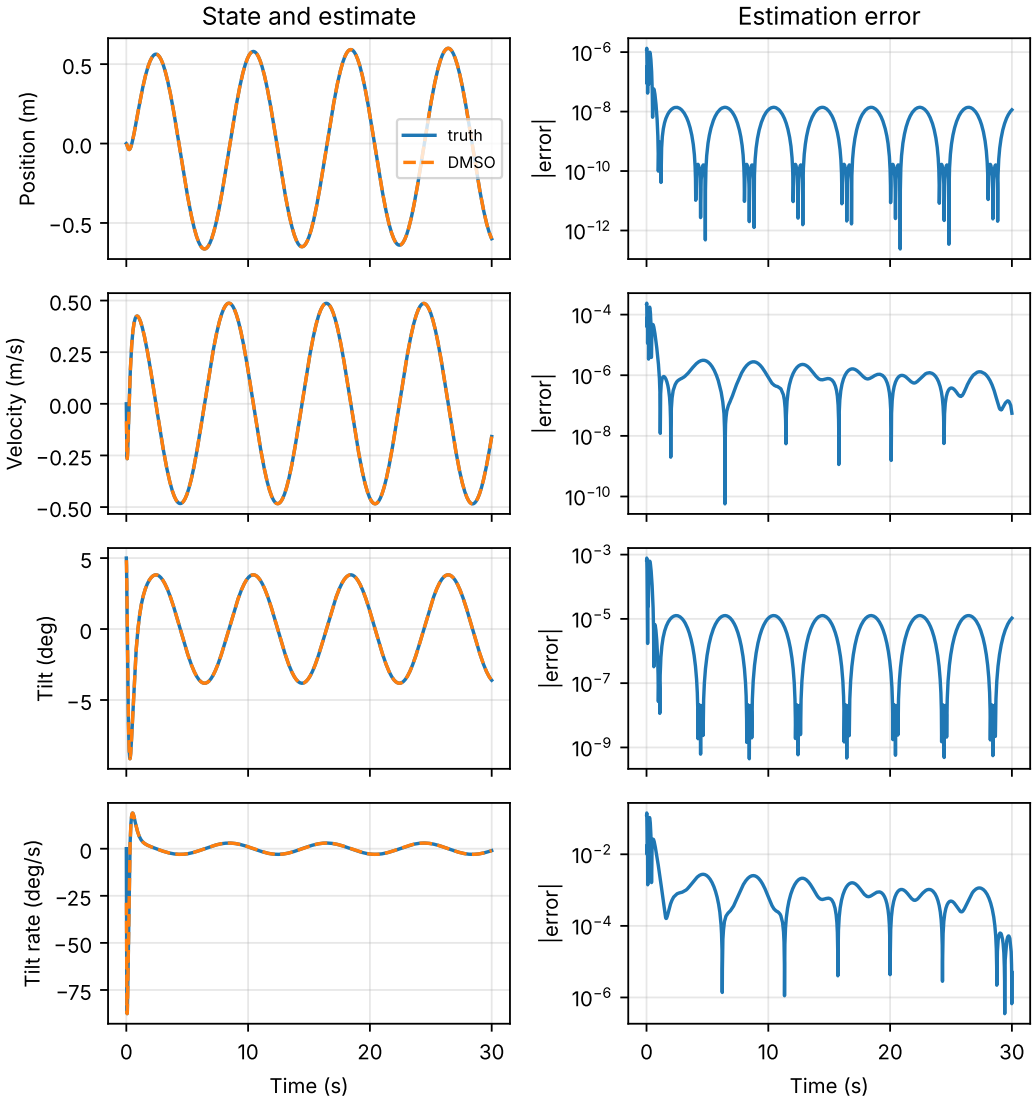


Figure 8.1: Case O1: states, DMSO one-step predictions, and prediction error.

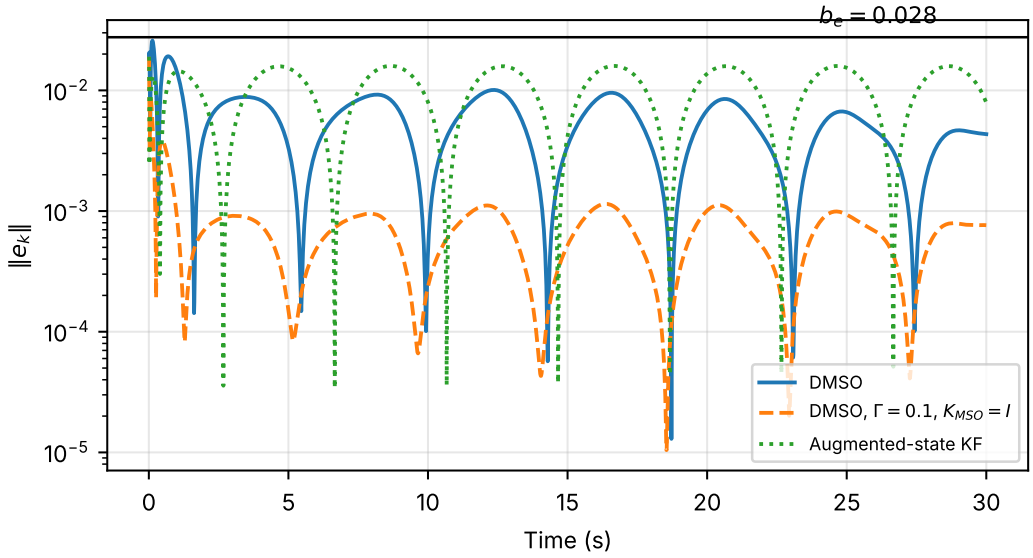


Figure 8.2: Case O2: norm of the estimation error with parameter uncertainty, against the predicted ultimate bound b_e of (3.18), with the large-gain configuration ($\Gamma = 0.1$, $K_{\mathrm{MSO}} = I$) and the augmented-state Kalman filter for comparison.

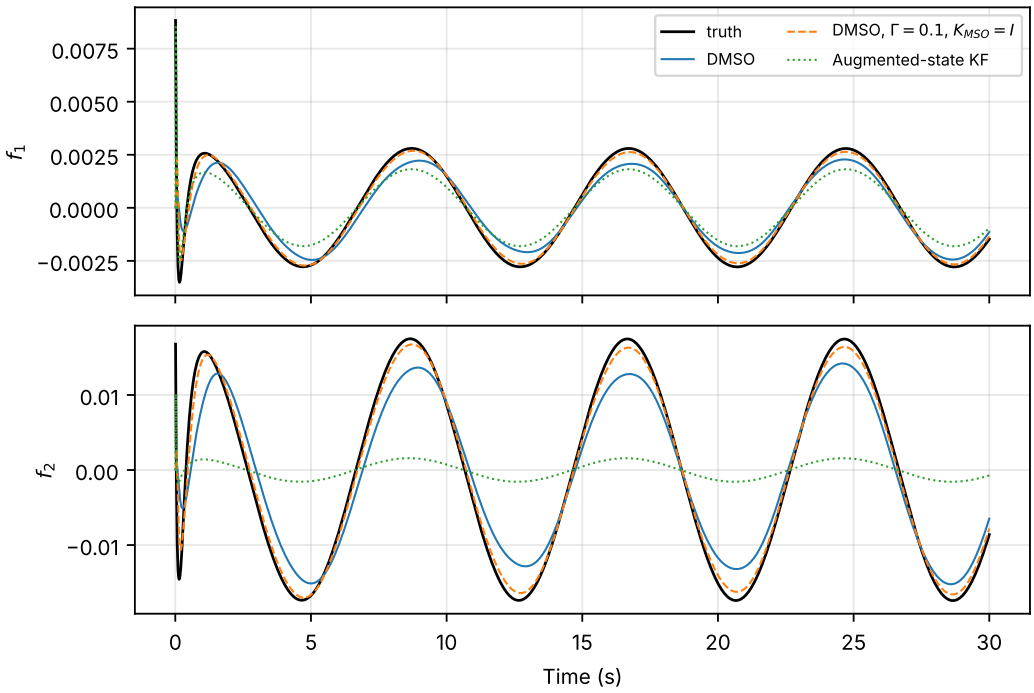


Figure 8.3: Case O2: DMSO uncertainty estimates against truth, with the large-gain configuration ($\Gamma = 0.1$, $K_{\mathrm{MSO}} = I$) and the augmented-state Kalman filter for comparison.

by a factor of 2.5 to 3 in case O3, and is about 40% worse than the augmented-state filter. In the tilt channel it is three to five times worse than the two unaugmented filters and 50% worse than the augmented one. The cause is the measurement rather than the adaptation. The complementary-filtered tilt carries a slowly varying error of about 1.1° RMS, because the accelerometer measures tilt relative to the apparent vertical, which is displaced by the robot's own acceleration. With $A = \bar{\sigma} I$ the DMSO corrects each channel only from its own measurement, and so it follows that error. The Kalman filters instead infer tilt through the model from the measured wheel acceleration, which the $A_2 x_3$ coupling makes available.

The Kalman-gain variant (†) combines the two mechanisms. Its velocity and tilt errors lie between those of the unaugmented and augmented filters, and its tilt-rate error is about 9% below the nominal filter's in O3 and O4. Its gain lies outside the conditions of Theorem 3.8, so these numbers are empirical.

In the noisy cases the augmented-state filter gives the best uncertainty estimate in both channels (Table 8.4). The Kalman-gain variant is 1.4 and 1.3 times worse in the velocity and tilt-rate channels of O3, and the DMSO with $A = \bar{\sigma} I$ about 1.6 times worse in both. In the tilt-rate channel every estimator's error is between 60% and 100% of the RMS of the true uncertainty itself, so at this noise level f_2 is barely identifiable from the available measurements; f_1 is identified to within 30% to 60%.

Table 8.3: Observer RMS one-step prediction error by case and estimator: velocity (mm/s) / tilt (mrad) / tilt rate (mrad/s). Position errors are not listed: in O3 and O4 they are 0.56 to 0.84 mm for every estimator, close to the 0.55 mm RMS error of the tilt-compensated position measurement itself.

Estimator	O1	O2	O3	O4
KF, nominal Q_{K}	<0.01 / <0.01 / 0.01	1.99 / 0.08 / 12.39	7.74 / 5.36 / 7.60	7.70 / 5.28 / 7.66
KF, inflated Q_{K}	<0.01 / <0.01 / 0.01	1.99 / 0.06 / 12.38	6.44 / 3.69 / 11.44	6.44 / 3.66 / 11.34
Augmented-state KF	<0.01 / <0.01 / 0.01	0.69 / 0.09 / 11.28	1.82 / 12.76 / 1.04	1.83 / 12.69 / 1.03
DMSO, $K_{\mathrm{MSO}} = I$, $\Gamma = 0.1$	<0.01 / <0.01 / 0.01	0.10 / 0.07 / 0.69	6.28 / 19.36 / 4.44	6.28 / 19.31 / 4.44
DMSO	<0.01 / <0.01 / 0.02	0.88 / 0.13 / 5.62	2.59 / 19.39 / 5.06	2.60 / 19.34 / 5.08
DMSO with σ-mod	<0.01 / <0.01 / 0.01	1.04 / 0.13 / 6.50	2.82 / 19.39 / 6.62	2.85 / 19.34 / 6.61
DMSO, Kalman-gain K_{MSO} †	<0.01 / <0.01 / 0.01	1.42 / 0.28 / 8.92	3.51 / 4.06 / 7.01	3.60 / 3.15 / 6.94

† outside the conditions of Theorem 3.8; empirical.

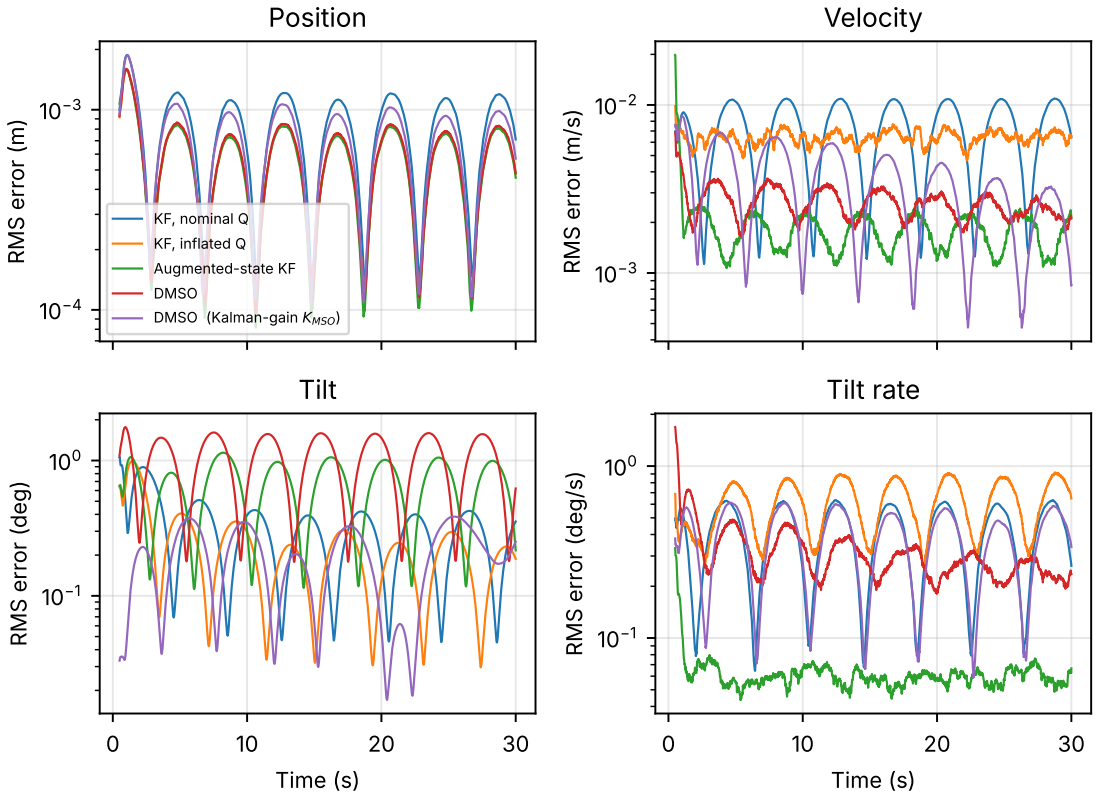


Figure 8.4: Case O3: estimation error (0.5 s moving RMS), DMSO against the three Kalman baselines of Section 7.7.

Table 8.4: RMS uncertainty estimation error $\hat{f} - f$ ($\times 10^{-3}$), velocity channel / tilt-rate channel. Only estimators that produce an uncertainty estimate are listed; O1 has no uncertainty.

Estimator	O2	O3	O4
Augmented-state KF	0.69 / 11.27	0.59 / 7.28	0.59 / 7.25
DMSO, $K_{\mathrm{MSO}} = I$, $\Gamma = 0.1$	0.11 / 0.75	1.22 / 11.65	1.22 / 11.61
DMSO	0.44 / 2.81	0.94 / 11.44	0.94 / 11.36
DMSO with σ-mod	0.52 / 3.25	1.12 / 12.57	1.14 / 12.51
DMSO, Kalman-gain K_{MSO} †	1.41 / 8.84	0.84 / 9.47	0.88 / 9.38
RMS of true f	1.99 / 12.38	1.99 / 12.38	1.98 / 12.24

8.2.3 Discussion

Three conclusions follow. First, the DMSO does what the analysis claims. Its conditions are satisfied with margin, its error stays inside the predicted bound, and under deterministic model error it identifies the uncertainty 1.6 to 4 times more accurately than the fairest Kalman baseline, with more to gain from a larger adaptation gain inside the proven region. Second, the choice $A = \bar{\sigma} I$, which makes the proof simplest, discards the cross-channel information that a model-based filter exploits. When one measurement carries a structured error, that choice costs more than the adaptation gains. Third, Theorem 3.8 admits any K_{MSO} with $\sigma_{\max}(F - K_{\mathrm{MSO}}) < 1$, and the empirical success of the Kalman-gain injection suggests extending the proof through the weighted norm of Remark 3.12. That extension is not immediate: the cancellations of Proposition 3.7 survive only if the adaptation law is weighted by the same matrix as the error, as Remark 3.12 sets out, and the Kalman-gain row above uses the unweighted law.

8.3 CONTROL RESULTS

8.3.1 Choice of the tracked output

The backstepping recursion of Section 6.5 is built on the flat output $y = x_1 - b\,x_3$ rather than on the position x_1, for the reason given in Remark 6.2. The numerical comparison is made here. Stage 3 of Appendix A was followed: the backstepping structure was run first with both networks disabled, on the nominal linear model, with perfect state knowledge and no saturation, for both choices of output.

With x_1 as the tracked output, so that $z_1 = x_1 - x_{1d}$, $z_2 = x_2 - x_2^c$, the virtual-control gain is A_2 and α_2 carries the term $-B_1 u$ that x_1-tracking requires, the closed loop diverges. Linearizing the continuous-time closed loop (plant and three command filters, networks disabled) gives a real eigenvalue between +5.0 and +5.9 rad/s for every gain set tried: 720

combinations of $c_1, \ldots, c_4$ and $\varpi_1, \varpi_2, \varpi_3$, all satisfying (6.28). As the filter bandwidths are raised, that eigenvalue converges to +5.95 rad/s (Figure 8.5a), which is the right-half-plane transmission zero of the transfer function $x_1(s)/u(s)$ from voltage to position. The recursion drives all four error coordinates to zero, which requires the closed loop to invert the plant from u to x_1; for a non-minimum-phase output, that inversion places a closed-loop pole at the zero.

The flat-output design, with the attitude coordinates weighted by $w = a_2^2$, has a linearized closed loop that is stable at every bandwidth in Figure 8.5a and at all 720 points of the grid above. In closed-loop simulation of case C2 (Figure 8.5b) the robot balances at every bandwidth tested, from $\varpi_2 = 20$ to 120 rad/s, including bandwidths beyond the one-fifth-of-sample-rate guideline of Appendix A. The filter errors do not decrease with bandwidth: χ_1 and χ_2 stay near 10^{-3}, and χ_3 grows from 0.005 to 0.18, because the virtual controls carry measurement noise (Section 6.8).

The results below use $\varpi = (20, 40, 60)$ rad/s and $c = (0.8, 1.5, 10, 12)$. A closed-loop sweep over 64 gain sets, $c_1 \in \{0.4, 0.8, 1.5, 3\}$, $c_2 \in \{1.2, 1.5, 3, 6\}$, $c_3 \in \{5, 10\}$, $c_4 \in \{6, 12\}$ (networks off, one seed), balanced in both C2 and C4 for all but $c = (3, 6, 5, 6)$, which fell in C4. Larger c_2 tracks better: $c = (1.5, 6, 10, 12)$ gives 6.2 mm RMS in C2 and 3.7 mm in C4, against 16 mm and 35 mm at the design gains. The design gains are conservative, chosen for robustness to input-gain error over the sweep rather than for tracking, so the errors reported below understate what the structure can achieve.

8.3.2 Tracking performance

Table 8.5 and Figures 8.6–8.7 compare the LQR alone, a two-step extra control, and the flat-output command-filtered design with the networks off and on (learning rates $\gamma_{i1} = 0.05$, $\gamma_{i2} = 0.2$, leakage $\kappa_{ij} = 0.01$). The two-step extra control is the virtual-control construction of Huang [56] without command filters: the virtual tilt is formed from the position and

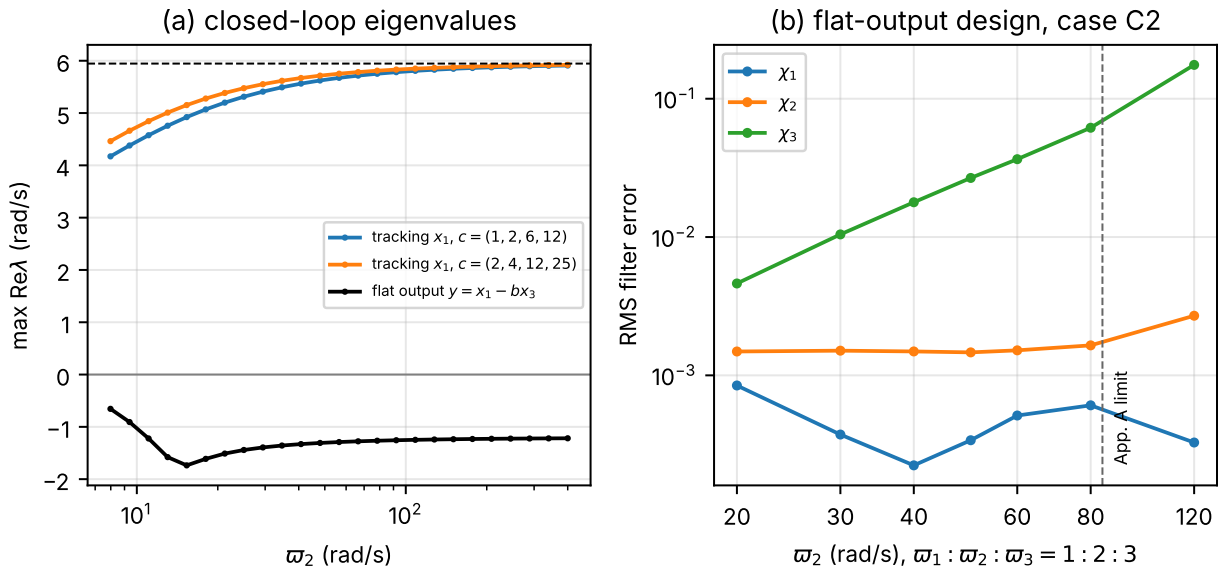


Figure 8.5: Command-filter bandwidth. (a) Largest real part of the closed-loop eigenvalues (networks disabled, nominal linear model) for the flat-output design of Section 6.5 and, for comparison, for the same recursion tracking x_1 directly. The dashed line is the right-half-plane zero of $x_1(s)/u(s)$. (b) RMS command-filter errors χ_i of the flat-output design in case C2 against the bandwidth; the robot balanced at every bandwidth shown (a fall would be marked by a dotted red line), and the dashed line is the sample-rate limit of Appendix A applied to ϖ_3.

velocity errors and the first network, the extra voltage from the tilt and tilt-rate errors and the second network, and the networks are driven by the velocity and tilt-rate errors; it carries no damping of its own on the position or tilt error and relies on the LQR term for them. Each entry is the mean over five noise seeds. All four controllers balanced in all twenty runs of every case.

The flat-output design reduces the RMS position tracking error relative to the LQR by a factor of 13 in the nominal case C1, 4.2 with parameter uncertainty (C2), 5.8 with the deadzone (C3), and 20 with deadzone and backlash (C4). The price is control effort: its RMS voltage is higher than the LQR's by 30%, 24%, 22%, and 136% in the four cases, and its peak voltage during the initial recovery is about 9.7 V, close to the 10 V limit, against 2.6 V for the LQR, which does not saturate.

The improvement comes from the structure, not the adaptation. With the networks enabled the error is within 0.3% of the networks-off value in C1–C3 and 2% above it in C4. The sign and size of that difference depend on the random initialization of the networks, that is, on the fixed input layer and the starting first-layer weights. Over ten initializations the mean change is -0.2%, -0.8%, -1.0%, and -1.4% in C1–C4, and in C4 it ranges from -5.8% to $+2.0\%$; the table reports one of the ten. At these learning rates the networks therefore make no measurable contribution to the tracking. That is consistent with Section 6.8: part of the uncertainty in these cases is the input-gain error, which the networks are not given the information to learn, and the deadzone and backlash act on the voltage and are not functions of the network inputs. A persistent small-amplitude oscillation remains in C4 (Figure 8.7), the signature of the nonsmooth input nonlinearity described in Remark 7.3.

The two-step extra control changes the error relative to the LQR by -7%, 0%, $+9\%$, and -5% in C1–C4, with at most 4% more RMS control. Both it and the LQR show a slow position ramp in C2–C4 (0.012 to 0.042 m/s over the last 10 s). That is the velocity error left by an incorrect steady-state feedforward under parameter error, integrated by the

weak position loop of the LQR (the weights used place a closed-loop pole at $|\lambda| = 0.9995$ at 10 ms). Because it appears equally without the extra control, it does not separate the position damping of the two designs (test T3 of Appendix A). The flat-output design, which damps z_1 explicitly, drifts two orders of magnitude more slowly, at less than 0.0005 m/s.

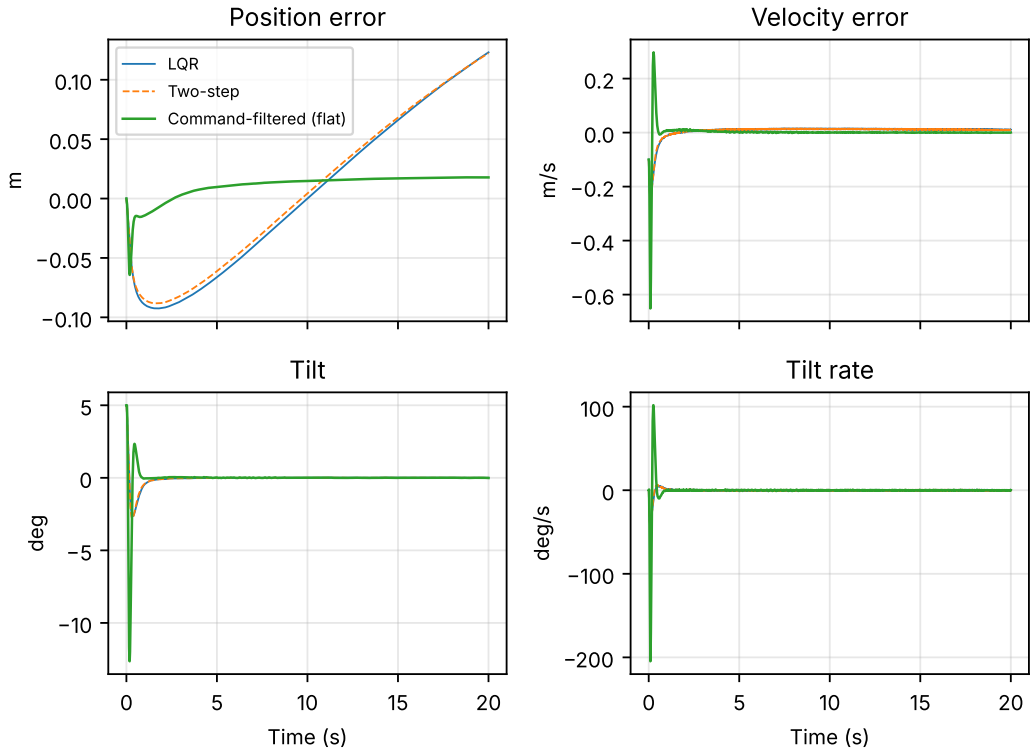


Figure 8.6: Case C2 (parameter uncertainty): position and velocity tracking errors, tilt, and tilt rate for the LQR, the two-step extra control, and the flat-output command-filtered design with networks.

8.4 IMPLEMENTATION RESULTS

The hardware results replay the LQR run logged in `implementationtest5.txt`, recorded with firmware v9 at a 10 ms loop. The replays use the encoder frame inferred from the switch-on transients of the same logs in Section 5.3.3: the logged position is taken as mirrored relative to the model, so the estimators receive $x = -\text{pos} - r\phi$, and the on-board estimates are

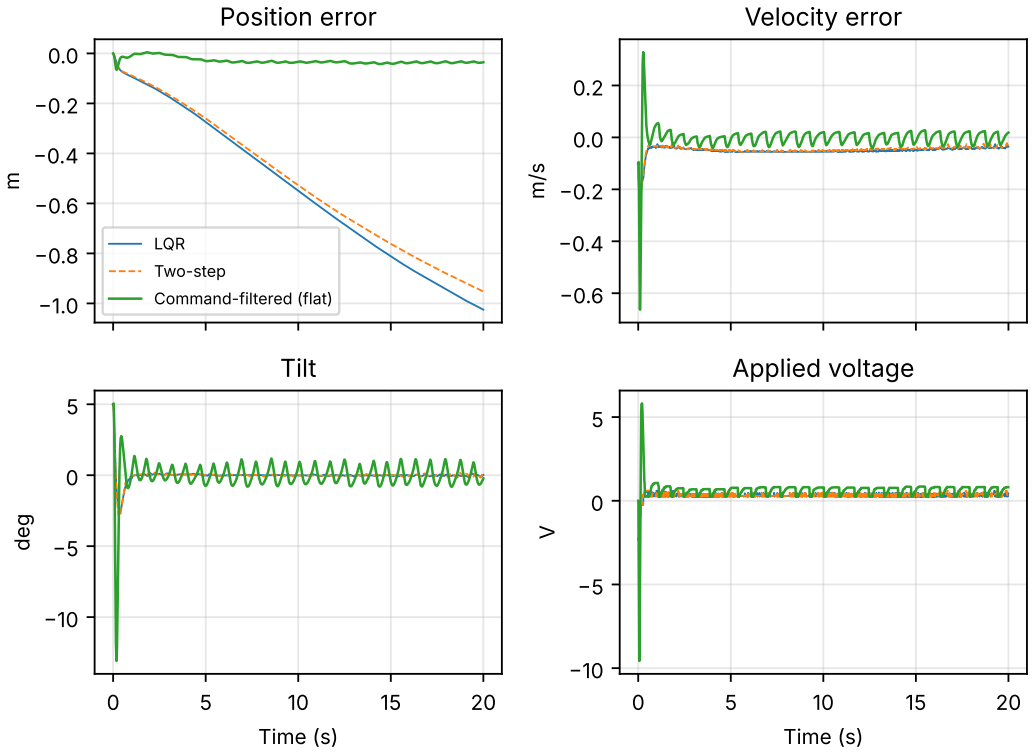


Figure 8.7: Case C4: response under deadzone and backlash.

Table 8.5: Control effectiveness by case, mean over five noise seeds: RMS position tracking error (m) after 5 s / RMS voltage (V) / peak voltage (V).

Controller	C1	C2	C3	C4
LQR alone	0.055 / 0.72 / 2.6	0.068 / 0.73 / 2.6	0.091 / 0.74 / 2.6	0.680 / 0.39 / 2.3
LQR + two-step extra control	0.051 / 0.72 / 2.6	0.068 / 0.73 / 2.6	0.100 / 0.75 / 2.6	0.647 / 0.41 / 2.3
Command-filtered, flat output, networks off	0.004 / 0.93 / 9.7	0.016 / 0.90 / 9.6	0.016 / 0.91 / 9.7	0.033 / 0.93 / 9.6
Command-filtered, flat output, networks on	0.004 / 0.93 / 9.7	0.016 / 0.90 / 9.6	0.016 / 0.91 / 9.7	0.034 / 0.93 / 9.6
Command-filtered, tracking x_1 directly	unstable for all gains tested (Section 8.3.1)			

mapped the same way for plotting. Figure 8.8 compares the measured states, the estimates computed on board by the firmware's DMSO (which the replay reproduces to print precision), and the DMSO of Chapter 3 run offline on the same data. After the first second the three agree to within the measurement quantization. There is no truth on hardware, so the replays establish consistency, not accuracy.

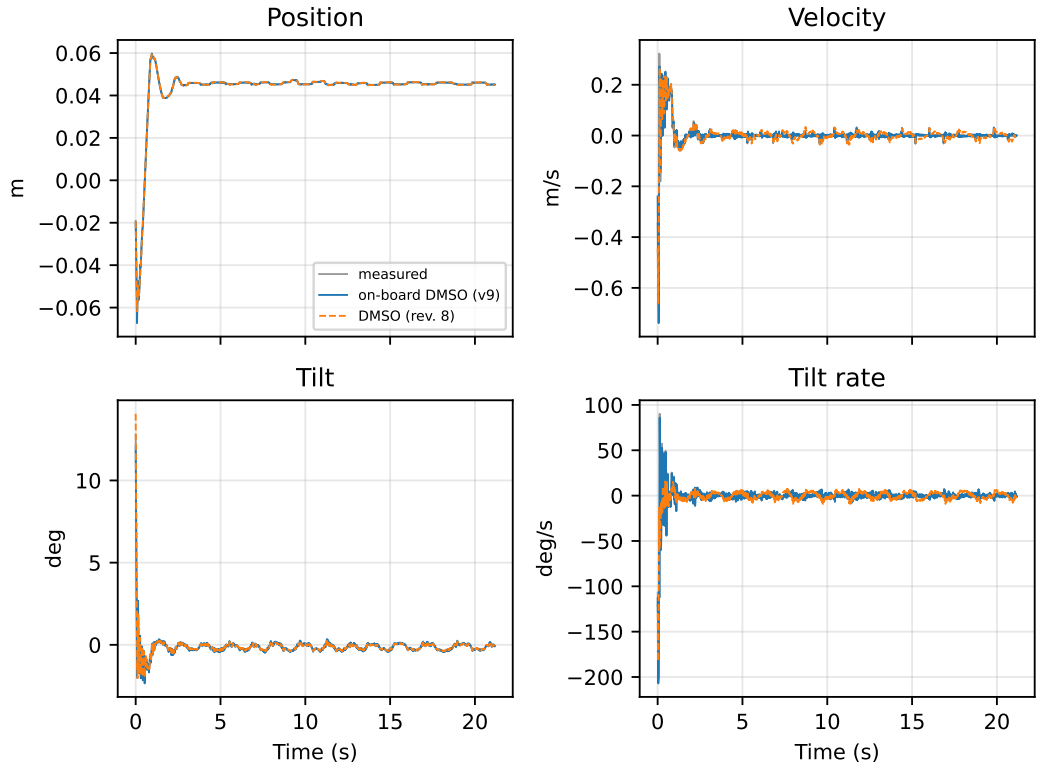


Figure 8.8: Hardware test: measured states, on-board (v9) DMSO estimates, and the DMSO of Chapter 3 replayed offline (legend: DMSO (rev. 8)).

Figure 8.9 compares the complementary filter with the model-based estimators fed raw accelerometer tilt. Without truth, each tilt estimate is scored by its consistency with the gyroscope, the RMS of $\mathrm{d}\hat{\theta}/\mathrm{d}t - g_y$. This measures high-frequency error and is blind to slow bias. On that score the raw accelerometer tilt is at 476 °/s, the complementary filter at 9.9 °/s, and the DMSO fed raw accelerometer tilt at 294 °/s. The Kalman filter fed the same raw tilt, with a measurement variance that reflects its actual error, reaches 4.1 °/s. Fed the

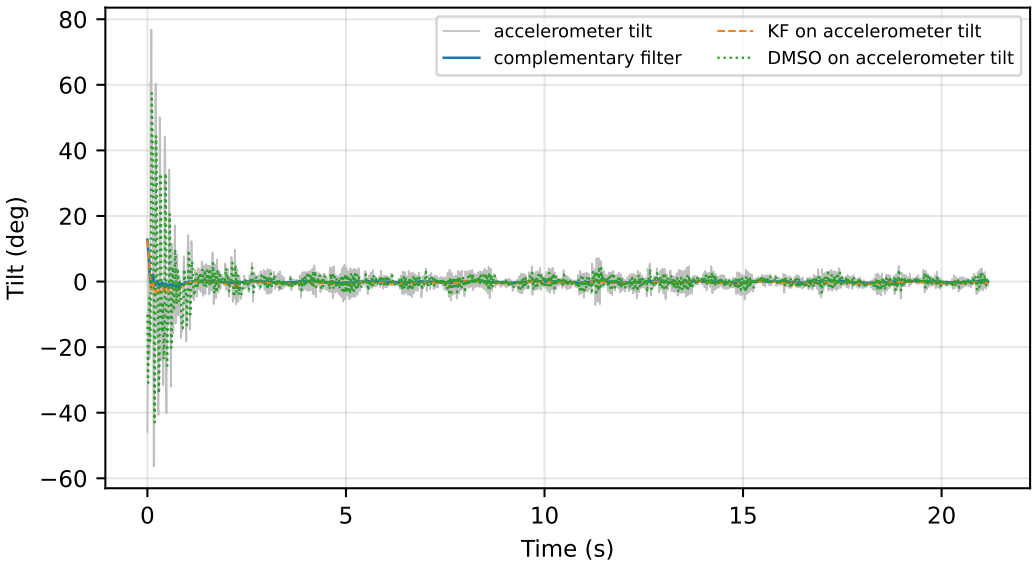


Figure 8.9: Hardware test: complementary filter against the model-based estimators using raw accelerometer tilt.

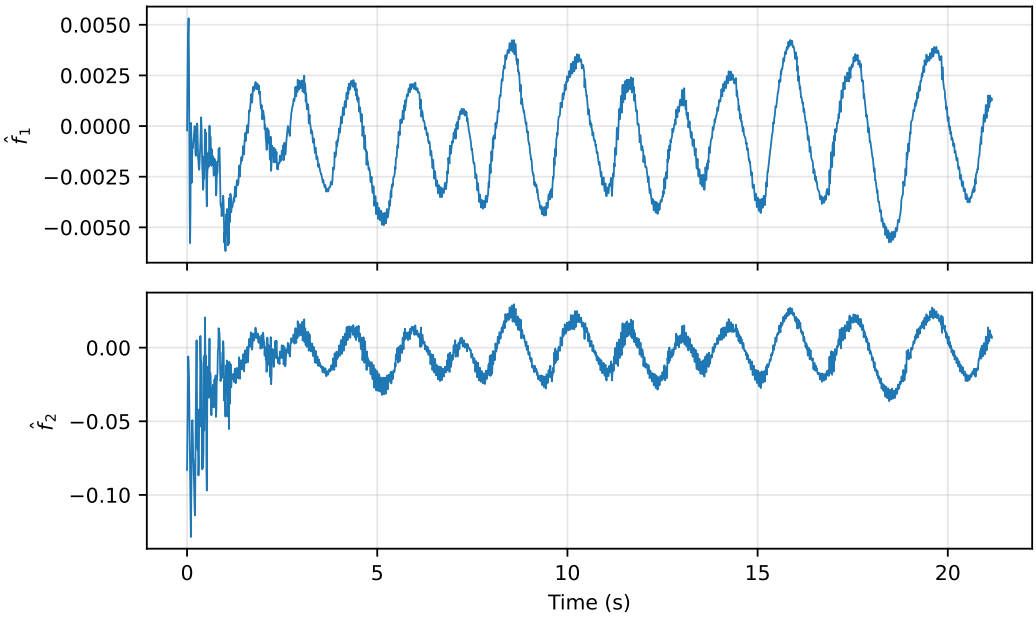


Figure 8.10: Hardware test: DMSO uncertainty estimates.

complementary-filter tilt instead, the Kalman filter and the DMSO reach 2.9 °/s and 5.9 °/s.

The complementary filter therefore outperforms the DMSO fed raw accelerometer tilt, but not a correctly weighted Kalman filter. The explanation needs no gearbox transients: an accelerometer measures tilt relative to the apparent vertical, which is displaced by $\ddot{x}/g$ whenever the robot accelerates. An estimator that corrects each channel from its own measurement follows that displacement, and one that weights the measurement by its true error does not. The comparison is a statement about measurement construction and error-injection design, not about the adaptation.

Figure 8.10 shows the uncertainty estimated on hardware. The tilt-rate channel has RMS 1.8×10^{-2}, against 1.24×10^{-2} for the true uncertainty of the simulated case O2, and the velocity channel 2.5×10^{-3}. The model error on the physical robot is therefore somewhat larger than the simulated perturbation. That is consistent with what Section 5.3 finds near upright: gear backlash and stick-slip, an external force (or an error in the integrated tilt or in l) that the steady-balancing coefficients call for, and possibly a motor deadzone (about 2 V on the earlier gearmotors, not measured on the present ones); none of these is in the nominal model.

Chapter 9

CONCLUSIONS

The Modified State Observer has been extended to discrete time and its stability established by Lyapunov analysis. The proof evaluates the first difference of the Lyapunov function exactly, which exposes the cancellations that the adaptation law was constructed to produce. The resulting conditions are $\sigma_{\max}(F - K_{\mathrm{MSO}}) < 1$ together with an upper bound on the adaptation gain, both satisfiable over an open set of designs, and both reducing to the expected limiting behavior as the adaptation gain tends to zero. The analysis bounds the state estimation error and the functional approximation error $\tilde{W}_k^T \phi_k$ in mean square without further assumptions, and uniformly once the weight estimate is bounded, which σ-modification of the update law provides directly and persistency of excitation provides with an additional argument.

The robot model was tested against the robot's recorded balancing runs (Section 5.3). The runs fix the sign with which the motor's reaction torque enters the body's moment balance, independently of the angular-momentum argument of Remark 5.2. At switch-on, with the command saturated at $-10\,\mathrm{V}$, the body was turning back toward upright at about $100\,^\circ/\mathrm{s}$ within $50\,\mathrm{ms}$, as the model predicts, while a model with the reaction reversed, at the data-sheet motor constant, has the body still falling further into its lean. Per unit of body rotation the base moved three to five times less than the reversed-sign models predict,

in a relation that does not involve the motor, and the drive-away of an early run matched the model's rate. With the gains the robot ran, a simulated reversed-sign plant either falls within 5.3 s or drives the base metres away, while the robot balanced for 21 s with its base held. The same runs do not validate the model quantitatively, and their switch-on transients indicate, without confirming, that the firmware's encoder position is mirrored relative to the model. In simulation, the observer satisfies its conditions with margin, stays inside the predicted ultimate bound, and under deterministic parameter error identifies the uncertainty 1.6 to 4 times more accurately than an augmented-state Kalman filter, with more available from a larger adaptation gain inside the proven region. With realistic sensor noise the augmented-state filter identifies the uncertainty better, and the observer's advantage over the unaugmented filters is confined to the velocity channel. The diagonal error injection that makes the proof simplest cannot use the model to reject the structured error of the accelerometer-derived tilt, which a Kalman-gain injection recovers empirically.

A command-filtered neural backstepping controller has been developed for the two-wheeled inverted pendulum, an underactuated system with unmatched uncertainty. The design assigns damping to every error coordinate, retains and explicitly cancels the coupling between the position and attitude error subsystems, and uses command filtering so that each network approximates a function of measured state and reference alone. It tracks the flat output $y = x_1 - b\,x_3$ of the linearized robot rather than the position itself, because the transfer function from voltage to position has a right-half-plane zero that a recursion on the position would have to invert; on the flat output the control enters only at the last step, and the attitude coordinates are weighted by a_2^2 in the Lyapunov function. In simulation it balanced in every run and reduced the RMS position tracking error relative to the LQR by factors of 4 to 20, at 22% to 136% more RMS control effort and with peak voltages near the 10 V limit. That gain comes from the structure. At the learning rates used, the networks changed the error by less than its spread over their random initialization, in part because the

parameter error acts on the input gain, which the networks are not given the information to learn.

Five directions follow naturally from this work.

Input-gain uncertainty. A parameter error in the motor constants changes B_2, so part of the residual the networks must absorb is proportional to the control. Giving the second network the previous control as an input, or estimating B_2 directly as in adaptive feedback linearization, would bring that error within the approximation hypothesis. At present, robustness to it comes from the feedback gains alone.

Output feedback. The present development assumes full state measurement, so the DMSO is properly a filter and uncertainty identifier. The structure of Proposition 3.7 carries over to $y_k = Cx_k$ with $A = F - K_{\mathrm{MSO}}C$, and the extension requires only a detectability assumption. This is the most direct route to a result that applies to systems where the tilt rate or the wheel velocity is not directly measured.

Nonsmooth input nonlinearities. Backlash violates the smoothness assumption underlying the universal approximation property, and the residual limit cycles observed both in simulation and on hardware are its signature. A network cannot cancel what it cannot approximate. Two remedies are worth pursuing: an explicit deadzone- and backlash-inverse placed ahead of the plant, or a discontinuous term in the control law that is robust to the deadband rather than attempting to estimate it.

Bench identification. The recorded runs fix the sign of the motor reaction but not the parameters. I_p, l, k_m, the deadzone and backlash of the present motors and the rolling resistance of the floor are unmeasured, and near upright the robot is dominated by stick-slip that no linear model contains. The bench tests of Table 5.7, with a check of the encoder

frame, would complete the validation, after which the LQR design could be tested on the robot as designed.

Deterministic timing. The achievable sample rate on the platform was set by telemetry rather than by the control computation, and the encoder velocity quantization scales inversely with the sample period. Both are implementation choices rather than properties of the algorithm, and both bound what any estimator can deliver. Appendix A treats the loop rate as a design variable and gives the trade.

Appendix A

IMPLEMENTATION AND VALIDATION GUIDANCE

This appendix gives a staged procedure for implementing and testing the observer of Chapter 3 and the controller of Chapter 6. The organizing principle is that each stage must have a falsifiable pass criterion that is checkable without the next stage being correct. A design that is assembled all at once and then tuned until it balances provides no evidence about which component is working.

A.1 STAGE 0: PARAMETER IDENTIFICATION

The detuning of the LQR gains reported in Section 4.4 is explained in part by the firmware's encoder frame (Section 5.3.6), but the parameters are still uncertain, and the two parameters most likely to be in error are the pendulum inertia I_p and the CG offset l. Both can be measured directly and should be, before any estimator is judged. The torque constant k_m is equally uncertain (Section 5.3.8); the motor tests below measure it. Table 5.7 gives each candidate model's prediction for the measurements below.

Procedure. Clamp the wheels to a fixture at the wheel axis, or remove them and suspend the chassis from a knife edge at the axis, so that the chassis swings as a physical pendulum. The clamped arrangement also allows the leads-shorted decay test of Table 5.7. Record the small-amplitude period T with the onboard gyroscope. Then

$$T = 2\pi\sqrt{\frac{I_{\text{pivot}}}{M_p g l}}, \qquad I_{\text{pivot}} = I_p + M_p l^2. \tag{A.1}$$

Measure l independently by balancing the chassis on a knife edge to locate the CG, then solve (A.1) for I_p. Repeat at several amplitudes to confirm the small-angle regime.

Motor constants. Measure R by four-wire resistance at a dozen shaft positions and average, since commutator position changes the reading. Obtain k_e by back-driving the motor at a measured rate and reading the open-circuit terminal voltage; k_e from the slope of V against $\dot{\vartheta}$ is far more reliable than k_e inferred from catalog free-run and stall figures. Obtain k_m from a locked-rotor test, as the torque at the wheel per ampere of armature current, in both directions of rotation. The two constants are tied by (4.1): referred to the output shaft, $k_m = \eta_g k_e$, with η_g the fraction of the armature torque that the gear train passes. A measured ratio k_m/k_e above one, or far below the efficiency expected of the gear train, indicates that one of the two is wrong.

Deadzone and backlash. Ramp the commanded voltage slowly in both directions with the wheels off the ground and record the breakaway voltages δ_+ and δ_- of (7.6). Measure backlash by holding the motor shaft and rotating the wheel through the deadband, reading the free rotation directly; this gives Δ_b in (7.7).

Sign conventions. The switch-on transients of the recorded runs indicate the frame of firmware v9 (Section 5.3.3): the logged tilt and command have the signs of ϕ and V, and the logged position is mirrored, $\text{pos} = -(x + r\phi)$, so the estimators and the replays of Section 8.4

use $x = -\text{pos} - r\phi$. The closed loop does not decide the frame (Section 5.3.6), so it should be confirmed by hand, and rechecked after any change to the firmware. Hold the wheels on the ground and rotate the body in the positive sense of ϕ (counterclockwise in Figure 5.2, center of gravity moving toward $-x$). The logged tilt should increase. Since the encoders read $x/r + \phi$, the logged position should then fall if it is mirrored and rise if it is not; rolling the robot toward $-x$ at a fixed lean gives the opposite change. With the wheels blocked, a small positive voltage should pitch the body backward (Table 5.7). The controller gains must be entered in the frame found; with the position mirrored, gains designed in the model frame change the sign of their position and velocity entries.

Pass criterion. The LQR gains computed from the measured parameters, entered in the firmware's frame, stabilize the hardware without manual detuning. If they do not, the model is still wrong and no amount of online estimation will make the comparison in Chapter 8 interpretable.

A.2 STAGE 1: OFFLINE ESTIMATOR REPLAY

Before the observer is allowed to influence any control signal, run it open loop on recorded data.

1. Drive the robot with the LQR controller alone and log, at the full loop rate, the raw gyroscope and accelerometer samples, the raw encoder counts, the commanded voltage, and a timestamp taken from a hardware timer rather than from a software counter.

2. Replay the log offline through the DMSO. Because $y_k = x_k$ here, the observer can be evaluated without closing any loop.

3. Verify the theoretical conditions numerically on the replayed data: compute $\sigma_{\max}(F - K_{\text{MSO}})$ for the Δt actually achieved, compute $\phi_{\max} = \max_k \|\phi(x_k)\|$ realized along the

recorded trajectory, and confirm (3.20). This is the step most often skipped. The $\phi_{\max}$ used to select Γ is a design assumption, and the trajectory is free to violate it.

Pass criterion. The estimation error settles inside the predicted ultimate bound b_e of (3.18), and $\Gamma\phi_{\max}^2$ computed from realized data satisfies (3.20). If the error settles but the bound is violated, the gains are outside the proven region and the result is empirical only.

A.3 STAGE 2: OBSERVER GAIN SELECTION

The conditions of Corollary 3.9 give a constructive selection order.

1. **Place A first.** Choose K_{MSO} so that $\sigma_{\max}(F - K_{\mathrm{MSO}})$ takes a target value $\bar{\sigma}$. Note that $\sigma_{\max}$, not the eigenvalues, is what matters; a gain that places the eigenvalues of A well inside the unit circle can still have $\sigma_{\max}(A) > 1$ if A is strongly non-normal. Compute the singular values explicitly. A practical starting point is $\bar{\sigma} \in [0.5, 0.8]$. Smaller values reject model error faster, with a response time that scales as $1/\log(1/\bar{\sigma})$, and lower the model-error floor $\epsilon_N/(1-\bar{\sigma})$ of (3.21). Their cost is measurement noise, which Theorem 3.8 does not model: as $\bar{\sigma}$ falls, $K_{\mathrm{MSO}} = F - \bar{\sigma} I$ approaches F and the prediction follows each noisy measurement more closely. That side of the trade has to be judged from data.

2. **Bound $\phi_{\max}$ by construction.** Rather than estimating $\phi_{\max}$ from a nominal trajectory, choose activation functions that are bounded by construction, for example a Gaussian radial basis or a tanh polynomial basis, so that $\phi_{\max}$ is known a priori and independent of how far the trajectory wanders.

3. **Select Γ from** (3.20)**.** With $\bar{\sigma}$ and $\phi_{\max}$ fixed,

$$\Gamma < \frac{1}{\phi_{\max}^2} \min\left\{\frac{1}{2},\ \frac{1-\bar{\sigma}^2}{2\bar{\sigma}^2}\right\}.$$

Begin at one tenth of this bound and increase. The speed of uncertainty identification is set by Γ, and the conditions permit any sufficiently small value, so there is no reason to operate near the boundary.

4. **Add σ-modification.** Use (3.23) with κ chosen so that $\Gamma\kappa \ll 1$; a decade below Γ is a reasonable start. This is strongly recommended over relying on persistency of excitation. During regulation about the upright equilibrium the regressor is nearly constant and (3.22) will not hold, which is exactly the condition under which weight drift occurs. Drift is insidious: the state estimate remains good while the weights wander, and the failure appears only when the trajectory eventually excites a direction the drifted weights misrepresent.

Diagnostic. Log $\left\|\hat{W}_k\right\|_F$ throughout every run. A weight norm that grows slowly and monotonically while the estimation error stays small is the signature of drift, and it is the reason to add leakage rather than to increase Γ.

A.4 STAGE 3: CONTROLLER BRING-UP

The controller of Section 6.5 has four damping gains, three filter bandwidths, four learning rates, and four leakage gains. Bring them up in this order, and only in this order.

1. **LQR alone.** Confirm stable regulation with the measured parameters from Stage 0. Record the baseline metrics of Section 8.1.

2. **Backstepping with networks disabled.** Set $\hat{F}_1 = \hat{F}_2 = 0$ and run the structure of Section 6.5 with u_e active. This tests the command filters, the error coordinate bookkeeping, and the sign conventions without any adaptation. Most implementation errors in a backstepping design are sign errors in the cross-term cancellations, and they are far easier to find with the adaptation off. Verify numerically that the implemented error dynamics

are the derived ones: difference the logged z_i to estimate $\dot{z}_i$, evaluate the right-hand sides of the error equations of Section 6.5 from the logged signals, and confirm that the two agree to within the discretization error. A sign error in a cancelling pair, such as $a_2 z_3$ in $\dot{z}_2$ against $-(a_2/w)z_2$ in $\dot{z}_3$, appears as a residual proportional to one of the coordinates. (Logging expressions such as $z_1 z_2 - z_2 z_1$ is not a test: they are zero for any numbers.) If the two do not agree, the implementation does not match the derivation.

3. **Filter bandwidths.** Raise ϖ_i until χ_i is small relative to the corresponding z_i, then stop. With noisy measurements χ_i stops shrinking, and can grow, as the bandwidth rises (Figure 8.5b), so raise ϖ_i only while χ_i falls. The upper limit is the lesser of one fifth of the sample rate and the lowest unmodeled structural mode of the chassis. The stiffening rods described in Section 4.4 exist because that mode was originally low enough to contaminate the inertial measurements; it should be measured, not assumed, by striking the chassis and recording the gyroscope response.

4. **One network at a time.** Enable network one with network two disabled, then the reverse, then both. Each network has its own error coordinate driving it, z_2 and z_4 respectively, so a network that is not helping shows up as a failure to reduce its own coordinate.

5. **Leakage last.** Raise κ_{ij} only as far as needed to hold the weight norms flat over a long run. Every increase adds $\kappa_{ij} W_{ijm}^2/2$ to the residual $\mathcal{C}$ in (6.35) and therefore to the steady-state error.

A.5 STAGE 4: HARDWARE ARCHITECTURE

Two limitations identified in Section 4.4 are architectural rather than algorithmic, and both bound what any estimator can achieve.

A.5.1 Separate the control path from the telemetry path

Serial transmission dominated the loop budget. The remedy is to make the control path deterministic and the telemetry path best-effort:

- Run the control law from a hardware timer interrupt at fixed period, and do no I/O inside it beyond sensor reads and the actuator write.
- Write telemetry into a ring buffer from the interrupt and drain the buffer from the main loop, so that a slow or blocked link cannot extend the control period.
- Log to onboard storage at full rate and stream a decimated subset, rather than streaming everything.
- Timestamp from a free-running hardware counter, not from a software accumulator, so that jitter is measurable after the fact. Report the achieved loop period distribution, not its nominal value. The discretization (7.1) assumes a fixed Δt, and jitter enters as an unmodeled disturbance that the networks will attempt, and fail, to explain.

A.5.2 The encoder placement limits the achievable result

The encoders are on the motor shaft, ahead of the gearbox, so backlash is structurally unobservable. Increasing the loop rate does not help: the measurement is not noisy, it is measuring the wrong thing. Two remedies are available, in increasing order of effort:

(a) **Load-side encoding.** Add a second encoder on the output shaft or the wheel. The difference between motor-side and load-side angle is then a direct measurement of the backlash state, which converts an unmodeled nonlinearity into an observable quantity and permits an explicit backlash inverse. This is the single highest-value hardware change available.

(b) **Deterministic timing in hardware.** Implementing the fixed-point control law in programmable logic removes scheduler jitter entirely and permits loop rates well beyond what the microcontroller supports. This is worth doing only after (a); a faster loop around a blind measurement does not improve the estimate.

A.5.3 The sample rate trade

There is an interior optimum in Δt and it should be found rather than assumed. Shortening Δt improves the fidelity of (7.1) and reduces the interval over which the zero-order hold is in error, but the encoder velocity quantization in Table 7.2 scales as $1.47 \times 10^{-4}/\Delta t$, so the velocity measurement degrades linearly as the loop speeds up. Sweep Δt in replay over the logged data from Stage 1, plot the RMS velocity estimation error against Δt, and operate at the minimum. If the minimum lies at the shortest Δt tested, the encoder resolution is not the binding constraint and the loop rate may be raised further; if it lies in the interior, raising the loop rate requires raising the encoder resolution first.

A.6 STAGE 5: FALSIFICATION TESTS

Each of the following is designed to fail if a specific claim is untrue.

T1. **Bound verification.** Run case O2 with a deliberately inflated parameter error and confirm that the estimation error settles inside b_e from (3.18) computed with the realized $\phi_{\max}$. A violation means either the conditions are not met or the implementation does not match the derivation.

T2. **Small-gain admissibility.** Run the observer at Γ one and two decades below the bound in (3.20). Stability must be retained, with slower identification, since the conditions of Corollary 3.9 admit arbitrarily small gains.

T3. **Position drift.** Run cases C2 and C4 for at least ten times the dominant closed-loop time constant with the two-step and command-filtered designs side by side, and plot z_1 over the full run. A design that assigns no damping to z_1 (requirement R2 of Chapter 6) permits a slow position ramp, since $\dot{z}_1 = z_2$ is a pure integrator. Either it appears, which confirms the analysis, or it does not, which means the LQR term is supplying the missing damping and should be credited explicitly in the design rather than absorbed into the approximation targets.

T4. **Algebraic loop.** Disable the command filters, replacing x_3^c with α_2 directly, and attempt to execute the ordering of Remark 6.3. The computation should fail to be well posed. This confirms that the filters are structural rather than cosmetic.

T5. **Weight drift.** Run a long regulation case with leakage disabled and log $\left\|\hat{W}_k\right\|_F$. Growth in the weight norm with a flat estimation error confirms that persistency of excitation does not hold in regulation and justifies σ-modification.

T6. **Nonsmoothness floor.** Run case C4 with the backlash amplitude Δ_b swept over a range including zero. The residual limit cycle amplitude should scale with Δ_b and vanish as $\Delta_b \to 0$. If it does not, the limit cycle has another cause and the explanation in Remark 7.3 is wrong.

A.7 SOFTWARE ORGANIZATION

A structure that supports the above is worth stating explicitly.

- **One model, three consumers.** The continuous model (5.34)–(5.35), the discretization (7.1), and the parameter set should exist in exactly one place and be consumed by the truth propagator, the estimator, and the controller. Divergence between a simulation model and a flight model is a classic and silent source of error.

- **Same source, both targets.** The estimator and controller should compile for both the desktop simulation and the embedded target from the same source, with the platform differences confined to sensor and actuator interfaces. This makes replay meaningful: the code that is tested is the code that runs.

- **Fixed-point discipline early.** If the eventual target is fixed-point, introduce the scaling in simulation and compare against the floating-point result, rather than discovering the scaling problems on hardware.

- **Record the conditions, not just the results.** Every run should emit $\sigma_{\max}(A)$, realized $\phi_{\max}$, $\Gamma\phi_{\max}^2$, the achieved loop period distribution, and the weight norms, alongside the state histories. These are what make a result checkable a year later.

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